

- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Chronic Kidney Disease

CKD: Guideline Recommendations and Supporting Information (2020)

Guideline recommendations are the essence of evidence-based guidelines and provide a course of action for the practitioner based on the evidence. Below, you will find a list of the **Chronic Kidney Disease (2020)** Recommendations with the Nutrition Care Process category. Download the [Summary of Changes Table](#) for an overview of the recommendation revisions and updates from the CKD 2010 guideline.

The Academy categorizes Recommendations in terms of either **Imperative** or **Conditional** statements. Conditional statements clearly define a specific situation or specific sub-population with the larger guideline population. Imperative statements are broadly applicable to the target population without restraints on their application. Each recommendation receives a rating, using the scale **Strong, Fair, Weak, Consensus, or Insufficient Evidence**. The rating for the recommendation is primarily based on the strength of the supporting evidence but also the balance between benefits or harms anticipated and the clinical practice implications when the statement is follows. [Learn more](#)

The Academy is transitioning to the GRADE (Grading of Recommendations, Assessment, Development and Evaluations) framework for developing clinical recommendations. The CKD recommendations were developed using both formats and were assigned ratings in both systems. [Learn more](#)

Resources Available with Each Recommendation

In addition to the recommendation statement and strength rating, you will find on each recommendation page:

- A brief narrative summary of the evidence analyzed to reach the recommendation
- A statement of justification, or reason for the strength of the recommendation
- Detailed information on the evidence supporting the recommendations and background narrative (available in the Supporting Evidence section toward the bottom of each recommendation page)
- A reference list at the end of each recommendation page that includes all the sources used in the evidence analysis for the particular recommendation (each reference is hyperlinked to a summary of the article analyzed in the evidence analysis).

Click on the [blue](#) recommendation title to view the full recommendations.

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Recommendations Summary

CKD: Nutrition Screening and Assessment: Usual-Care Statements (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Routine Nutrition Screening

In adults with CKD 3-5D or posttransplantation, it is reasonable to consider routine nutrition screening at least biannually with the intent of identifying those at risk of protein-energy wasting (OPINION).

Rating: Consensus
Conditional

CKD: Nutrition Screening Tools

In adults with CKD 3-5D or posttransplantation, there is limited evidence to suggest the use of one tool over others for identifying those at risk of protein-energy wasting (2D).

Rating: Weak
Conditional

CKD: Routine Nutrition Assessment

In adults with CKD 3-5D or posttransplantation, it is reasonable that a registered dietitian nutritionist (RDN) or an international equivalent conduct a comprehensive nutrition assessment (including but not limited to appetite, history of dietary intake, biochemical data, anthropometric measurements, and nutrition-focused physical findings) at least within the first 90 days of starting dialysis, annually, or when indicated by nutrition screening or provider referral (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms associated with these recommendations.

- [Conditions of Application](#)

Nutrition assessment methods should be conducted by individuals trained to perform the respective method. Medical Nutrition Therapy requires individualization of nutrition care based on patient's CKD stage, comorbidities, diet and lifestyle as well as patient preferences. Routine nutrition screening of adults diagnosed with CKD stages 1-5D should occur to allow for the identification and further assessment and treatment of nutritional concerns.

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with these recommendations.

- [Recommendation Narrative](#)

Optimal frequency of nutrition screening and assessment are not described in the literature, and, therefore, recommendations were based from expert opinion.

There are several nutrition screening mechanisms in clinical practice, but few are specific to CKD, and there are limited data on their validity and reliability. Most of the existing tools focus on identification of malnutrition risk; only one currently screens for protein energy wasting (PEW). Regardless of the mechanism used, the nutritional assessment conducted subsequent to the screening should be comprehensive and include the routine monitoring of nutrition care outcomes.

COMPOSITE NUTRITIONAL INDICES: SCREENING TOOLS

Geriatric Nutrition Risk Index (GNRI)

Three studies reported on the use of GNRI to assess nutritional status, including two validity/reliability studies (Beberashvili et al 2013, Yamada et al 2013) and one prediction study in MHD patients (de Roij van Zijldewijn et al 2015). In one study, GNRI had the greatest area under curve (using MIS as a reference) of the nutrition screening tools (Yamada et al 2013). GNRI showed a significantly negative correlation with the MIS ($r=-0.67$, $P=0.0001$), and the most accurate GNRI cutoff to identify a malnourished patient according to the MIS was 91.2. The GNRI's sensitivity, specificity, and accuracy of a score of 91.2 in predicting malnutrition according to the MIS were 73%, 82%, and 79% respectively. Another study reported that GNRI had high inter-observer agreement score ($k=0.98$) and high intra-observer reproducibility ($k=0.82$) (Beberashvili et al 2013).

In another study, GNRI was a significant predictor for mortality at 2.97 years ($p<0.001$) but had lower predictive value for all-cause mortality compared to MIS and albumin levels (de Roij van Zijldewijn et al 2015).

Malnutrition Universal Screening Tool/Malnutrition Screening Tool (MUST/MST)

Two validity/reliability study reported on the use of MUST and MST tools to assess nutritional status in MHD patients (Yamada et al 2013, Lawson et al 2012). A study by Lawson et al, reported on the validity and reliability of both MUST and MST tool in MHD patients. The sensitivity of both the MUST and MST tool was low (53.8% for MUST; 48.7% for MST), indicating that they are not particularly sensitive at identifying individuals with malnutrition in this group, compared to SGA. Both tools have a high specificity (MUST=78.3%; MST=85.5%), so they are good at excluding individuals who are not malnourished. Reliability assessed by kappa was 0.58 for MUST (95% CI, 0.20 to 0.80) and 0.33 for MST (95% CI, 0.03 to 0.54). Both tools had an NPV or 60% and PPV for MUST was 73.7% and for MST

was 78.7%. Though these tools are not sensitive enough to identify all malnourished renal in-patients, they are still fairly reliable and related to other nutrition status markers. In Yamada, et al., the authors compared results from various malnutrition assessment tools to the reference standard of MIS. MUST and MST scores were both significantly associated with MIS ($p < 0.0001$ for each). The ROC curves of the MUST and MST compared to MIS were the smallest of the tools measured, and sensitivity, specificity and accuracy to detect hypoalbuminemia were among the lowest of all tools considered, indicating these may not be the best tools to discriminate nutritional risk in patients on MHD.

Mini-Nutrition Assessment (MNA)

Four studies reported on the use of MNA to assess nutritional status in MHD patients, three were validity/reliability studies (Yamada et al 2013, Afsar et al 2006, Santin et al 2016) and one was a correlational study (Erdogan et al 2013). Afsar et al. reported on the reliability of MNA tool compared to SGA 3-point scale. The reliability coefficient (alpha) for MNA was 0.93 (good degree of reproducibility). MNA might underestimate the nutritional status of patients on MHD who are not in an inflammatory state. Hence, MNA may not be as reliable as SGA in detecting PEM in the MHD population. Erdogan et al. compared MNA to Bio-electrical Impedance Analysis (BIA), reported a significant correlation between MNA score and single frequency-BIA ($r = 0.2$, $p = 0.045$), muscle mass ($r = 0.382$; $p < 0.001$) and visceral fat ratio ($r = 0.270$; $p = 0.007$). Authors concluded BIA is not as sensitive as MNA to detect early effects of secondary causes for malnutrition. Santin et al 2015, compared SGA (7-point), MIS, MNA-Short Form (MNA-SF) to handgrip strength (HGS), albumin, c-reactive protein (CRP), and skinfolds. SGA and MNA-SF had fair agreement ($\kappa = 0.24$; $p < 0.001$). The worst agreement was found between MIS and MNA-SF ($\kappa = 0.14$, none to slight; $p < 0.004$). Again, both SGA and MIS had good concurrent and predictive validity for CKD population, whereas MNA-SF validity results were more comparable to non-CKD elderly individuals. Yamada et al. compared MNA to other nutritional tools and reported that MNA had lower area under curve (0.73) than GNRI and Nutritional Risk Score but higher than MUST and MST.

Nutrition Impact Symptoms (NIS)

One validity study reported on the use of NIS score for identifying those at risk of malnutrition in patients on HD and concluded that NIS score is a useful nutrition screening tool for identifying who are at risk of malnutrition (Campbell et al 2013). NIS score > 2 had the strongest predictive value for mortality and for predicting poor nutritional outcomes, behind the rating of malnourished by SGA. Concurrent validity indicated similar agreement between each of the malnutrition risk tools (patient-generated subjective global assessment (PG-SGA), an abbreviated PG-SGA and NIS). Serum albumin was negatively correlated with NIS (Spearman $Rho = -0.161$; $p = 0.018$).

Nutrition Screening Tool (NST)

One validity study reported on the use of NST to assess nutritional status in PD patients. In this study, NST had a sensitivity of 0.84 (range: 0.74 to 0.94; $p < 0.05$) and specificity of 0.9 (range: 0.82 to 0.99; $p < 0.05$) which is clinically acceptable (Bennett et al 2006).

Renal Nutrition Screening Tool (R-NST)

In another study by Xia et al in PD patients, the R-NST was compared to SGA-7 point scale (Xia et al 2016). Authors determined that the R-NST tool when compared to SGA-7 point scale is valid to detect risk of malnutrition (sensitivity=97.3% (95% CI 90.7-99.7), specificity=74.4% (95% CI 57.9-87.0), PPV=88.0% (95% CI 79.0-94.1), NPV=93.6% (95% CI 78.6-99.2). These results indicate that R-NST is a good tool for identifying renal in-patients at risk of undernutrition.

Protein Energy Wasting (PEW) score

Two predictive studies reported on the use of PEW score to assess nutritional status. Leining and colleagues identified that SGA and albumin were significant predictors of mortality, but BMI, mid-arm muscle circumference (MAMC) and PEW score did not predict mortality at 24 months in PD patients (Leinig et al 2011). However, Moreau-Gaudry et al, a study conducted in patients on MHD recorded that PEW predicts survival. Each unit decrease in score was related with a 5-7% reduction in survival ($p < 0.01$) (Moreau-Gaudry et al 2014). This score can be helpful in identifying subgroups of patients with a high mortality rate and recommend nutrition support.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on nutrition screening tools is based on Grade III/ Grade D evidence. The recommendations regarding frequency of nutrition screening and assessment are based on Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of the Geriatric Nutrition Risk Index \(GNRI\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Malnutrition Universal Screening Tool/Universal Screening Tool \(MUST/MST\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Mini Nutrition Assessment \(MNA\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Nutrition Impact Symptom \(NIS\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the PEW score for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Nutrition Screening Tool \(NST\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Renal Nutrition Screening Tool \(R-NST\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

- [References](#)

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- [References not graded in Academy of Nutrition and Dietetics Evidence Analysis Process](#)

No additional references.

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Quick Links

Recommendations Summary

CKD: Nutrition Assessment: Other Anthropometric Assessment Methods (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Skinfold Thickness

In adults with CKD 1-5D (1B) or posttransplantation (OPINION), in the absence of edema, we suggest using skinfold thickness measurements to assess body fat.

Rating: Strong
Conditional

CKD: Waist Circumference

In adults with CKD 5D, we suggest that waist circumference may be used to assess abdominal obesity, but its reliability in assessing changes over time is low (2C).

Rating: Weak
Conditional

CKD: Conicity Index

In adults with CKD 5D on MHD, we suggest that the conicity index may be used to assess nutritional status (OPINION) and as a predictor of mortality (2C).

Rating: Weak
Conditional

CKD: Creatinine Kinetics

In adults with CKD 5D, we suggest that creatinine kinetics may be used to estimate muscle mass, though very high or very low dietary intake of meat and/or creatine supplements will influence accuracy of this measurement (2C).

Rating: Weak
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms associated with these recommendations.

- [Conditions of Application](#)

Special Discussions

The recommendation for skinfold measurements requires specialized equipment. Good quality calipers are needed to obtain an accurate measurement of skinfold thickness. However, the measurer must be trained in order to obtain accurate results. To obtain waist circumference, only a measuring tape is required. Once again, the measurer must be trained on how to obtain this measure.

Implementation Considerations

Skinfold Measurements

- The guideline for skinfold measurements apply to all adult CKD patients, including post-transplant. However, for the measurements to be useful to the practitioner, longitudinal assessments must be done to provide meaningful information about changes in percent body fat for that patient.
- There are no potential risks or harms associated with the application of the guideline for skinfold measurements in all adult CKD patients.
- Skinfold measurements may not be accurate for obese patients, since calipers may have upper limits that do not accommodate high levels of adiposity.

Creatinine Kinetics

- The guideline for using creatinine kinetics to measure muscle mass applies to all adult CKD patients. However, the procedure requires the patient to collect his/her urine for a 24-hour period and, preferably, to keep the collection on ice, which may make the procedure inconvenient for some patients. Furthermore, intake of meat or protein supplements containing creatine may contribute to urine creatinine excretion and this must be considered when calculating creatinine kinetics. In MHD patients, creatinine kinetics is more useful for patients who are anuric.
- There are no potential risks or harms associated with the application of the guideline for creatinine kinetics in adult CKD patients.

Monitoring and Evaluation

- Anthropometric measurements for assessment of body composition should be done routinely in CKD patients; these include skinfold measurements, waist circumference and creatinine kinetics.
- [Potential Costs Associated with Application](#)

The recommendations for skinfold measurements and waist circumference require specialized equipment. Good quality calipers are needed to obtain an accurate measurement of skinfold thickness.

- [Recommendation Narrative](#)

Methods of assessing body composition, including anthropometric measurements, are components of the nutrition assessment in CKD. Anthropometric measurements are practical, inexpensive and non-invasive techniques that describe body mass, size, shape, and levels of fatness and leanness; they are the most basic and indirect methods of assessing body composition. These include height, weight, skinfolds, circumferences, bioelectrical impedance analysis (BIA) and creatinine kinetics.

Regardless of the method selected to assess body composition, none are perfect, and the errors surrounding them should not be ignored. Errors may have clinical relevance, especially if the individual is treated and observed over time (Chumlea et al 2008). Moreover, the results of the measures are only as useful as the availability of suitable reference data from a group of persons of at least the same age, race, gender and disease status.

Skinfold Measurements

Ten studies reported on the use of skinfold measurements to assess body composition, including four agreement/validity/reliability studies (Avesani et al 2004, Bross et al 2010, Kamimura et al 2003, Woodrow et al 1996), one prediction study (Araujo et al 2006) and six correlation studies Aatif et al 2013, Kalantar-Zadeh et al, Kamimura et al 2003b, Oe et al, Stall et al 1996).

Maintenance Hemodialysis Patients

Bross et al used DEXA as the reference test and showed that, triceps skinfold thickness (TSF), BIA (Kushner), and near-infrared interactance were most accurate of the index tests in estimating total BF%, although the BIA (Segal) and BIA (Lukaski) equations overestimated total BF%. These results were not affected by skin color. In Bross, et al, there were significant correlations (all $p < 0.001$) between DEXA measurements and triceps skinfold measures of body fat in MHD participants. Kamimura et al compared skinfold thickness with DEXA and BIA and found that body fat estimates using skinfold thickness (SKF) and BIA were not significantly different from those obtained by DEXA in the total group. There were significant intra-class correlations between DEXA with SKF ($r = 0.94$) and BIA ($r = 0.91$). DEXA showed relatively good agreement with both SKF [0.47 ± 2.8 (-5.0 to 6.0) kg] and BIA [0.39 ± 3.3 (-6.9 to 6.1) kg] in the total group, but BIA showed greater mean prediction error for both men and women. This study indicated that SKF was preferable over BIA, which showed gender-specific variability in the assessment of body fat.

A prediction study by Araujo et al 2006 showed that TSF $< 90\%$ was not associated with higher odds of mortality. Oe et al in MHD patients found a significant correlation in LBM ($r = 0.69$, $p < 0.025$) between four skinfold anthropometry and BIA. BF-FSA was positively correlated with BF-BIA ($r = 0.65$, $p < 0.005$). Both techniques are comparable for LBM and BF measurements; however, four site skinfold anthropometry (FSA) is less affected by change in fluid status. Malnutrition score was significantly correlated with bicep skinfolds ($r = -0.32$) in MHD patients in a study by Kalantar-Zadeh et al. Aatif et al showed that fat tissue index and triceps skinfold thickness had a positive significant correlation ($r = 0.61$, $p < 0.001$). Kamimura et al found a strong correlation between BIA and skinfold thickness ($r = 0.87$) and near-infrared interactance and skinfold thickness ($r = 0.78$). This study confirmed that the most simple, long-established, and inexpensive method of SFT is very useful for assessing body fat in patients on long-term MHD therapy.

Peritoneal Dialysis Patients:

Stall et al examined five different tools to assess BF%. BF% measurements were different between all methods ($p < 0.001$), although there were differences according to sex. For men, all techniques were significantly different from each other ($p < 0.05$) except BIA and DEXA, as well as the Steinkamp method (skinfold thickness) and total body potassium. For women, all techniques were significantly different from each other ($p < 0.05$) except DEXA and the two methods for measuring skinfold thickness (Durnin & Womersley and Steinkamp). Despite the differences between modalities, all techniques were found to correlate significantly with each other ($p < 0.01$ or better for men and $p < 0.001$ or better for women).

Hemodialysis and Peritoneal Dialysis Patients

Woodrow et al compared skinfold thickness with DEXA and BIA. Bland & Altman analysis demonstrated no observed differences in 95% levels of agreement for percent total body fat (TBF) and FFM from SF-BIA or skinfold anthropometry (SFA) compared with DEXA (%TBF BIA-DEXA -13.7 to +8.3; %TBF SFA-DEXA -13.0 to +9.4%; FFM BIA-DEXA -5.1 to +9.6 kg; FFM SFA-DEXA -5.6 to +9.1 kg). There were considerable variations in agreement between the measures.

Pre-Dialysis Patients

Avesani et al used a Bland-Altman plot analysis for body fat% and showed that the best agreement was between skinfold thickness and DEXA compared to other measures. Skinfold thickness also had significant intraclass correlations with body fat% and it significantly correlated with FFM as measured by DEXA ($r = 0.74$, $r = 0.85$) indicating moderate and good reproducibility, respectively. This study indicated that skinfold thickness may be a good method to determine body fat% in pre-dialysis, and mild to advanced CKD patients.

Serum Creatinine/Creatinine Kinetics

Seven studies examined the relationship between serum creatinine or creatinine kinetics and comparative measures of muscle mass in MHD, PD and pre-dialysis patients.

Maintenance Hemodialysis Patients

One study in MHD patients showed that creatinine kinetics correlated with creatinine levels, and other traditional measures of muscle mass (e.g. CT scan, anthropometric measurements) (Kaizu et al 2002). Three studies in MHD patients showed that pre-dialysis, interdialytic change, and weekly creatinine clearance levels predicted mortality (Kaizu et al 2002, Walther et al 2011, de Roij van Zijdewijn et al 2015).

Peritoneal Dialysis Patients

In PD patients, creatinine kinetics was correlated with other body composition measurements in one study (Borovnicar et al 1996); however, significant differences existed between creatinine and anthropometric measures for LBM/FFM in another (Szeto et al 2000). A study in PD examined creatinine clearance and relative risk of mortality (Churchill et al 1996). Evidence was limited in pre-dialysis patients to one study (Avesani et al 2004). CK was significantly correlated with BF% and FFM from DEXA ($r = 0.47$ and $r = 0.57$, respectively, indicating moderate reproducibility, though there were significant differences in adjusted means of BF% and FFM between CK and DEXA ($p < 0.05$) (Avesani et al 2004).

Waist Circumference

Two studies reported on the use of waist circumference to assess nutritional status in dialysis patients.

Maintenance Hemodialysis Patients

Cordeiro, et al. examined risk of PEW, inflammation and mortality according to waist circumference tertile in MHD patients. As waist circumference increased, indicating increased abdominal fat, patients had increased odds of PEW (assessed by SGA) and inflammation (assessed by IL-6). In the fully adjusted model, there was no increased risk of mortality according to waist circumference tertile.

Peritoneal Dialysis Patients

Bazanelli et al found a strong correlation between waist circumference and trunk fat ($r = 0.81$, $p < 0.001$) for both men and women, and a significant association with BMI ($r = 0.86$, $p < 0.001$). There was a moderate agreement between WC and trunk fat ($\kappa = 0.59$) and area under the curve was 0.90. In a prospective evaluation of the same study, changes in waist circumference was also correlated with changes in trunk fat ($r = 0.49$, $p < 0.001$) and κ of 0.48 indicated a moderate agreement between the tools. The authors concluded that waist circumference is a reliable marker of abdominal adiposity in PD patients.

Conicity Index

In one positive-quality study with HD patients, as Conicity Index tertile increased, indicating increased abdominal fat, patients had increased risk of PEW (assessed by SGA), increased fat BMI, waist circumference and inflammatory marker levels of CRP and IL-6 and had lower serum creatinine levels and handgrip strength ($p < 0.01$ for each measure). In the model adjusted for age, sex, comorbidities and dialysis vintage, the highest tertile of conicity index was associated with mortality [HR (95% CI) 1.93 (1.06–3.49)], but the results were not significant after adjustment for IL-6 and PEW (Cordeiro, et al. 2010). Based on one study, Conicity Index may be a useful measure of nutritional status, inflammation and mortality in HD patients, but evidence was limited.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendations are based on Grades II and III/ Grade B, C and D evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of creatinine/kinetics for assessing body composition in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of skinfold measurements for assessing body composition in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of waist circumference for assessing body composition in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of conicity index for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Nutrition Assessment: Body Mass Index (BMI) as a Predictor of Mortality (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Body Mass Index (BMI) as a Predictor of Mortality, Maintenance Hemodialysis (MHD)

In adults with CKD 5D on MHD, we suggest that overweight/obese status (based on BMI) can be used as a predictor of lower mortality, whereas, underweight status and morbid obesity (based on BMI) can be used as a predictor of higher mortality (2B).

Rating: Fair
Conditional

CKD: Body Mass Index (BMI) as a Predictor of Mortality, Peritoneal Dialysis (PD)

In adults with CKD 5D on PD, we suggest that underweight status (based on BMI) can be used as a predictor of higher mortality (2C).

Rating: Weak
Conditional

CKD: Body Mass Index (BMI) as a Predictor of Mortality, Non-Dialyzed

In adults with CKD 1-5, it is reasonable to consider using underweight status (based on BMI) as a predictor of higher mortality, though the mortality risk associated with overweight or obesity status (based on BMI) is not clear (OPINION).

Rating: Consensus
Conditional

CKD: Body Mass Index (BMI) as a Predictor of Mortality, Post-Transplant

In posttransplantation adults, it is reasonable to consider using underweight and overweight/obesity status (based on BMI) as a predictor of higher mortality (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms associated with these recommendations.

- [Conditions of Application](#)

- BMI is not an ideal marker of obesity, since it cannot differentiate between higher weights due to increased adiposity vs. muscularity and it cannot identify visceral adiposity, which has negative metabolic effects.
- To ensure accuracy of BMI, height should be measured periodically.
- The standard weight status categories associated with BMI ranges for adults should be used in the CKD population; these include <18.5 kg/m² for underweight; 18.5 to 24.9 kg/m² for normal weight; 25.0 to 29.9 kg/m² for overweight; and ≥30 kg/m² for obese.
- Limited evidence suggested that obesity (BMI ≥30 kg/m²) may be a risk factor for higher mortality in individuals who are on dialysis and under the age of 65. Therefore, practitioners should consider patient age when determining mortality risk according to BMI.
- In patients on dialysis, weight to calculate BMI should be measured following dialysis treatment to improve accuracy.

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with these recommendations.

- [Recommendation Narrative](#)

Methods of assessing body composition, including anthropometric measurements, are components of the nutrition assessment in CKD. Anthropometric measurements are practical, inexpensive and non-invasive techniques that describe body mass, size, shape, and levels of fatness and leanness; they are the most basic and indirect methods of assessing body composition. Regardless of the method selected to assess body composition, none are perfect, and the errors surrounding them should not be ignored. Errors may have clinical relevance, especially if the individual is treated and observed over time (Chumlea et al 2008). Moreover, the results of the measures are only as useful as the availability of suitable reference data from a group of persons of at least the same age, race, gender and disease status.

Twenty-four studies reported on the use of BMI to assess nutritional status, including 17 prediction studies (Molnar et al 2001, de Roij van Zuijewijn et al 2015, Araujo et al 2006, Kim et al 2014, Kalantar-Zadeh et al 2005, Yen et al 2010, Chatzot et al 2009, Wiesholzer et al 2003, Leavey et al 2001, Mathew et al 2015, Badve et al 2014, Hoogeveen et al 2012 and Lieveense et al 2012, Leinig et al 2001 and McDonald et al 2003) and nine correlation studies (Atif et al 2013, Beberashvili et al 2009, Bross et al 2010, Kadiri et al 2011, Kahraman et al 2007, Leinig et al 2008, Nakao et al 2007, Steiber et al 2007, Visser et al 1999). There were no studies examining validity or reliability of using BMI in this population to classify nutritional status.

Maintenance Hemodialysis Patients

Eight studies examined MHD patients only. Seven studies examined mortality risk according to BMI category. In three studies (Chatzot et al., Wiesholzer et al and Yen et al), the authors examined mortality risk according to traditional weight categories (underweight, normal weight, overweight and obese), although in a study with Taiwanese participants (Yen et al), these categories were defined differently. In five additional studies, the authors examined risk according to 5 to 11 BMI categories (Kalantar-Zadeh et al, Leavey et al. and Badve et al, Ricks et al 2011, Doshi et al 2016).

In one study that only compared two groups (< or >25 kg/m²), the authors found no association between BMI and mortality at 10 years (Araujo et al 2006). However, in the remaining studies in which BMI was examined according to traditional weight status groups or by 5 to 11 categories, there was consistently a higher risk of mortality for participants who were underweight, and lower risk for participants who were overweight or obese (Kalantar-Zadeh et al 2005, Yen et al 2010, Chatzot et al 2009, Wiesholzer et al 2003, Leavey et al 2001 and Badve et al 2014, Ricks et al 2011, Doshi et al 2016). Length of follow-up for these studies ranged from 1.34 to 10 years. There was an inverse relationship with mortality when BMI was measured as a continuous variable in three studies (Molnar et al 2010, Wiesholzer et al 2003 and Leavey et al 2001), but the χ^2 statistic was not significant in de Roij van Zuijewijn, et al.

Findings from correlation studies indicated that BMI was positively associated with albumin levels, fat and lean body mass measured by a variety of methods in HD patients. Beberashvili et al showed that serum albumin was significantly and positively correlated with BMI and FM in MHD patients (Beberashvili et al 2009). The higher BMI group had greater LBM ($p=0.001$) and FM ($p=0.0001$), and higher

phase angle (PA) and ECM/BCM ($p < 0.05$). MHD patients with elevated BMI demonstrate better nutritional status compared to normal BMI or overweight patients. Severity of inflammation was not related to BMI in MHD patients.

Bross et al indicated that BMI had a strong linear correlation with total body fat percentage measured by NIR and BIA (Segal) ($r = 0.85$) in MHD patient. Fat tissue index, as estimated by BIA, was significantly correlated with BMI in the study by Aatif et al. In another study, Kadiri et al showed that BMI was positively correlated with FM ($r = 0.493$, $p = 0.002$), serum albumin ($r = 0.340$, $p = 0.04$), and anemia in MHD patients. BMI was negatively correlated with CRP ($r = -0.065$, $p = 0.702$) but had no correlation with LBM ($r = 0.278$, $p = 0.085$). Kahraman et al 2007 studied the relationship between CRP and BMI status and found that CRP levels were significantly higher in obese and underweight MHD patients compared with normal and overweight patients ($p < 0.05$).

Steiber, et al. found that mean BMI was significantly different across the 5 categories of SGA ($p < 0.05$) in MHD patients. Visser et al demonstrated that there was a strong correlation between the 7-point SGA scale and BMI in MHD patients ($r = 0.79$, $p < 0.001$), % fat ($r = 0.77$, $p < 0.001$).

Maintenance Hemodialysis and Peritoneal Dialysis Patients

Three studies reported on the relationship between BMI and mortality in a combination of MHD and PD patients (Badve, et al. reported results for MHD and PD patients separately). In Mathew et al 2015, participants who survived had higher baseline BMIs compared to the group that did not survive, but BMI category was not a significant predictor. Hoogeveen et al 2012 demonstrated that underweight and obesity were risk factors in a combination of MHD/PD patients less than 65 years of age, but for those who were at least 65, there was no relationship between BMI and mortality. Lievense et al demonstrated that PD patients had lower mortality risk compared to MHD patients.

Leinig et al showed that there was a positive correlation between BMI and FM in predialysis ($r = 0.67$, $p = 0.0002$), in MHD ($r = 0.67$, $p = 0.0002$), and peritoneal dialysis ($r = 0.79$, $p < 0.0001$) patients. Nakao et al indicated that BMI was significantly correlated with BPI in MHD and PD patients (r values ranging from 0.778 to 0.886, $p < 0.0001$). Hoogeveen et al 2012 followed dialysis patients < 65 years of age for seven years. In the multivariable adjusted model, compared to those with normal weight status, those who were categorized as underweight (2.00 (1.30-3.07) and obese (1.57 (1.08-2.28) had a significantly higher hazard of mortality for those ≥ 65 years of age, but there was no significant relationship between weight status and mortality for those ≥ 65 years of age.

Peritoneal Dialysis Patients

Four studies reported on the relationship between BMI and mortality in PD patients. Badve et al. found that underweight increased mortality risk at 2.3 years, but results regarding higher BMI categories were not consistent. Leinig et al. found no difference in mortality risk according to whether PD patients had a BMI < 23 kg/m² at 2 years. McDonald et al. found that, in adjusted analysis, PD patients who were obese had higher risk of mortality (up to 10 years) compared to patients with normal weight status. In the study by Kim et al., the group with the lowest quartile of BMI had the highest mortality risk at 2 years, but there were no other significant associations. In a systematic review performed by Ahmadi et al 2016, authors confirmed an increased risk of 1 year mortality for people with CKD who were underweight, but this relationship did not persist for 2, 3 and 5 year mortality. Conversely, Ahmadi et al 2016 found that overweight/obesity status decreased mortality risk at 1, but not 2, 3 or 5 years (Ahmadi et al 2016).

Non-dialyzed Patients

Finally, two studies examined the relationship between BMI and mortality in non-dialyzed CKD patients. Madero, et al. examined risk according to BMI quartile and found no relationship. Hanks, et al. took a different approach and examined risk not only according to traditional BMI categories, but also according to whether participants were metabolically healthy. Of those who were metabolically healthy, there was decreased risk for overweight/obese participants compared to those with a normal BMI. However, there was no difference in mortality risk according to weight status in those who were metabolically unhealthy. These findings were consistent with a systematic review by Ahmadi et al 2015.

Post-transplant Patients

A systematic review by Ahmadi et al 2014 examined the relationship between BMI and mortality in 150,000+ adults with CKD with kidney transplant. Authors conclude that, compared to participants with normal weight status at baseline, those who were underweight [HR (95% CI): 1.09 (1.02, 1.20)] or overweight/obese [1.20 (1.14, 1.23)] were at increased hazard of mortality.

- [Recommendation Strength Rationale](#)

The evidence supporting these recommendations is based on Grade II/Grade B, Grade III/Grade C evidence as well as Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of BMI for assessing body composition in adults with CKD on hemodialysis?](#)

[Is there evidence to support the use of BMI for assessing body composition in adults with CKD, pre-dialysis and on peritoneal dialysis?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Recommendations Summary

CKD: Nutrition Assessment: Body Weight and Body Composition Usual-Care Statements (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Body Composition and Body Weight/Body Mass Index (BMI)

In adults with CKD 1-5D or posttransplantation, it is reasonable to consider assessing body composition in combination with body weight/BMI at the first visit and to monitor overall nutrition status periodically over time (OPINION).

Rating: Consensus

Conditional

CKD: Frequency of Body Weight/Body Mass Index (BMI) and Body Composition Assessment

In adults with CKD 1-5D or posttransplantation who are clinically stable, it is reasonable to measure body weight and BMI and to monitor for changes in body weight/BMI and body composition as needed (OPINION).

- At least monthly in MHD and PD patients
- At least every 3 months in patients with CKD 4-5 or posttransplantation
- At least every 6 months in patients with CKD 1-3

Rating: Consensus

Conditional

CKD: Assessment of Body Weight

In adults with CKD 1-5D or posttransplantation, it is reasonable for a registered dietitian nutritionist (RDN) or an international equivalent or physicians to use clinical judgement to determine the method for measuring body weight (e.g. actual measured weight, history of weight changes, serial weight measurements, adjustments for suspected impact of edema, ascites and polycystic organs) due to absence of standard reference norms (OPINION).

Rating: Consensus

Conditional

CKD: Body Mass Index (BMI) and Protein Energy Wasting (PEW)

In adults with CKD 1-5D or posttransplantation, BMI alone is not sufficient to establish a diagnosis of PEW unless the BMI is very low (<18 kg/m²) (OPINION).

Rating: Consensus

Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms associated with these recommendations.

- [Conditions of Application](#)

BMI

- The standard weight status categories that have been defined by the WHO according to BMI ranges for adults should be used in the CKD population; these include <18.5 kg/m² for underweight; 18.5 to 24.9 kg/m² for normal weight; 25.0 to 29.9 kg/m² for overweight; and ≥30 kg/m² for obese. Population-specific BMI cut-offs to define weight status may be lower for Asian populations.
- BMI is not an ideal marker of obesity, since it cannot differentiate between higher weights due to increased adiposity vs. muscularity and it cannot identify visceral adiposity, which has negative metabolic effects.
- To ensure accuracy of BMI, height should be measured periodically.
- Limited evidence suggested that obesity (BMI ≥30 kg/m²) may be a risk factor for higher mortality in individuals who are on dialysis and under the age of 65. Therefore, practitioners should consider patient age when determining mortality risk according to BMI.

Measuring Body Weight

When using published weight norms in the anthropometric assessment of adult CKD patients, caution must be used as each norm has significant drawbacks.

- Ideal body weight (IBW) is the body weight associated with the lowest mortality for a given height, age, sex and frame size and is based on the Metropolitan Life Insurance Height and Weight Tables. *[Caution: Not generalizable to the CKD population and data-gathering methods were not standardized.]*
- Hamwi method can be used to estimate ideal body weight. *[Caution: A quick and easy method for determining optimal body weight but has no scientific data to support its use.]*
- Standard Body Weight, NHANES II (SBW as per KDOQI Nutrition Practice Guidelines) describes the median body weight of average Americans from 1976 to 1980 for height, age, sex and frame size. *[Caution: Although data is validated and standardized and uses a large database of ethnically-diverse groups, data is provided only on what individuals weigh, not what they should weigh in order to reduce morbidity and mortality.]*
- BMI often defines generalized obesity and CKD research, specific to dialysis patients, has identified that patients at higher BMIs have a lower mortality risk. *[Caution: The researchers may not have statistically adjusted for all confounders related to comorbid conditions occurring in CKD on dialysis (diabetes, malignancy, etc.) and it is unclear how it may relate to CKD patients not on dialysis.]*
- Adjusted Body Weight (ABW) is based on the theory that 25% of the excess body weight (adipose tissue) in obese patients is metabolically active tissue. *[Caution: This has not been validated for use in CKD and may either overestimate or underestimate energy and protein requirements.]*

Monitoring and Evaluation

BMI should be used routinely to assess weight status in CKD patients since it is useful in predicting mortality. However, in isolation, BMI is not sufficient to establish a diagnosis of PEW unless it is very low ($<18 \text{ kg/m}^2$)

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with these recommendations

- [Recommendation Narrative](#)

Methods of assessing body composition, including anthropometric measurements, are components of the nutrition assessment in CKD. Anthropometric measurements are practical, inexpensive and non-invasive techniques that describe body mass, size, shape, and levels of fatness and leanness; they are the most basic and indirect methods of assessing body composition. These include height, weight, skinfolds, circumferences, bioelectrical impedance analysis (BIA) and creatinine kinetics. Dual-energy X-ray absorptiometry (DXA) is a direct method that is considered the gold standard for assessing body composition in patients with CKD; however, this measure is labor intensive, invasive, expensive and can be influenced by a number of CKD related factors such as hydration status.

Timing of body composition assessments is important in CKD since assumptions of hydration are required for accurate interpretation of the results, and fluid/electrolyte balance is likely to be altered significantly in CKD patients. For these reasons, in adults undergoing dialysis, assessments are best obtained after treatment when body fluid compartments are in relatively normal interrelationship (Carrero et al 2015, Chumlea et al 2008).

Regardless of the method selected to assess body composition, none are perfect, and the errors surrounding them should not be ignored. Errors may have clinical relevance, especially if the individual is treated and observed over time (Chumlea et al 2008). Moreover, the results of the measures are only as useful as the availability of suitable reference data from a group of persons of at least the same age, race, gender and disease status.

BMI should be used routinely to assess weight status in CKD patients since it is useful in predicting mortality (see BMI and mortality section). However, in isolation, BMI is not sufficient to establish a diagnosis of PEW unless it is very low ($<18 \text{ kg/m}^2$).

- [Recommendation Strength Rationale](#)

The evidence supporting these recommendations is based on Consensus/expert opinion evidence..

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

- [References](#)
- [References not graded in Academy of Nutrition and Dietetics Evidence Analysis Process](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Nutrition Assessment: Body Composition with Bioelectrical Impedance (BIA) and Dual-Energy X-Ray Absorptiometry (DEXA) (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Bioelectrical Impedance for Patients on Maintenance Hemodialysis (MHD)

In adults with CKD 5D on MHD, we suggest using bioimpedance and preferably multi-frequency bioelectrical impedance (MF-BIA) to assess body composition when available. Bioimpedance assessments should ideally be performed a minimum of 30 minutes or more after the end of the hemodialysis session to allow for redistribution of body fluids (2C).

Rating: Weak
Conditional

CKD: Bioelectrical Impedance for Patients, Non-Dialyzed and on Peritoneal Dialysis (PD)

In adults with CKD 1-5 or CKD 5D on PD, there is insufficient evidence to suggest using bioelectrical impedance to assess body composition (2D).

Rating: Weak
Conditional

CKD: Dual-Energy X-Ray Absorptiometry (DXA) for Body Composition Assessment

In adults with CKD 1-5D or posttransplantation, it is reasonable to use DXA when feasible as it remains the gold standard for measuring body composition despite being influenced by volume status (OPINION).

Rating: Consensus Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms associated with these recommendations.

- [Conditions of Application](#)

Special Discussions

The guidelines for multi-frequency bio-electrical impedance analysis (MF-BIA), dual-energy x-ray absorptiometry (DEXA), skinfold measurements, and waist circumference require specialized equipment. Good quality calipers are needed to obtain an accurate measurement of skinfold thickness. However, the measurer must be trained in order to obtain accurate results. To obtain waist circumference, only a measuring tape is required. Once again, the measurer must be trained on how to obtain this measure. MF-BIA is becoming more widely available as the technology advances. However, training is needed to understand and to appropriately interpret the output from the device.

Implementation Considerations

MF-BIA

- The guideline for MF-BIA applies to all adult patients receiving MHD. The measure must be obtained post-dialysis on a non-conducting surface for an accurate assessment.
- When bioimpedance is performed in patients on PD, measurements should be done with an empty abdominal cavity (following PD fluid drainage) and bladder. For individuals on MHD with residual kidney function, bladder should be empty.
- There are no potential risks or harms associated with the application of the guideline for MF-BIA in adult patients receiving MHD.

Dual Energy X-ray Absorptiometry

- DEXA is a valid technique for measuring body composition in adult CKD patients, including post-transplant patients. In MHD and PD patients, this is despite the measurement being influenced by over-hydration.
- DEXA is associated with very small amounts of radiation and this should be considered when weighing benefits and risks of this method for a particular individual. Ten screenings with DEXA results in a similar amount of radiation exposure as one chest x-ray.

- [Potential Costs Associated with Application](#)

The guidelines for MF-BIA, DEXA, skinfold measurements, and waist circumference require specialized equipment. Because of the cost associated with some of these measures (e.g., MF-BIA, DEXA), there is insufficient evidence for the workgroup to suggest the use of these measurements on a routine basis in clinical practice.

- [Recommendation Narrative](#)

Body composition is an important factor in nutrition assessment in CKD. Dual-energy X-ray absorptiometry (DEXA) is a direct method that is considered the gold standard for assessing body composition in patients with CKD; however, this measure is labor intensive, invasive, expensive and can be influenced by a number of CKD related factors such as hydration status.

Timing of body composition assessments is important in CKD since assumptions of hydration are required for accurate interpretation of the results, and fluid/electrolyte balance is likely to be altered significantly in CKD patients. For these reasons, in adults undergoing dialysis, assessments are best obtained after treatment when body fluid levels are balanced (Carrero et al 2015 and Chumlea et al 2008).

Regardless of the method selected to assess body composition, none are perfect, and the errors surrounding them should not be ignored. Errors may have clinical relevance, especially if the individual is treated and observed over time (Chumlea et al 2008). Moreover, the results of the measures are only as useful as the availability of suitable reference data from a group of persons of at least the same age, race, gender and disease status.

Multi-Frequency Bioelectrical Impedance Analysis (MF-BIA)

Twelve studies reported on the use of MF-BIA to assess fat mass (FM) and fat free mass (FFM) in MHD, PD and pre-dialysis patients. Four of these studies were validity/reliability studies: two in MHD patients (Donadio et al 2008, Furstenberg et al 2011); one in PD patients (Konings et al 2003); and one in pre-dialysis patients (Rigalleau et al 2004). Three were prediction studies: two in MHD patients, and one in MHD and PD patients (Abad et al 2011, Fiedler et al 2009, Rosenberger et al 2014). Eight were correlation studies: five in MHD patients (Donadio et al 2008, Furstenberg et al 2011, Konings et al 2003, Mancini et al 2003, Ohashi et al 2013, Rodrigues et al 2012, Cheng et al 2000); one in PD patients; one in MHD and PD patients (Nakao et al 2007); and one in pre-dialysis patients (Rigalleau et al 2004).

MHD Patients

FM and FFM measured by MF-BIA had good agreement with DEXA in two studies (Donadio et al 2008, Furstenberg et al 2011), had high correlations with several markers of nutritional status in four studies (Donadio et al 2008, Ohashi et al 2013, Rodrigues et al 2012, Nakao et al 2007) and predicted hard outcomes in three studies (Abad et al 2011, Fiedler et al 2009, Rosenberger et al 2014). Furstenberg, et al. concluded that MF-BIA was a more robust tool than DEXA for measuring body composition in MHD patients. Donadio, et al. found that MF-BIA yielded a smaller prediction error in MHD patients.

Body composition determined by MF-BIA was found to be predictive of hospitalization (Fiedler et al 2009) and survival (Abad et al 2011, Fiedler et al 2009, Rosenberger et al 2014). In Rodriguez et al., BIA underestimated FM and overestimated FFM when compared with air displacement plethysmography in MHD patients. PEW determined by MF-BIA was positively related to BMI and negatively associated with serum albumin level (Ohashi et al 2013). In Mancini, et al., bioimpedance vector analysis (BIVA) was predicted by normalized protein catabolic rate (nPCR) and albumin in MHD patients with normal nutritional status, but the predictive effects were not accurate in undernourished patients. In MHD patients, a body protein index score calculated from MF-BIA protein mass and height significantly correlated with blood protein levels in men on MHD, but there was no relationship in women on MHD (Nakao et al 2007).

PD Patients

FM and FFM measured by MF-BIA showed wide limits of agreement with DEXA in 1 study, which was affected by hydration status (Konings et al 2003), and was an independent risk factor for survival in another study Abad et al 2011). In CAPD patients, LBM measured by MF-BIA and creatinine kinetic method were highly correlated but there was no difference in LBM using BIA in patients with or without peritoneal dialysate (Cheng et al 2000). A body protein index score calculated from MF-BIA protein mass and height significantly correlated with blood protein levels in men on MHD, but there was no relationship in women on MHD or CAPD patients. The findings varied according to sex and dialysis treatment (Nakao et al 2007).

Pre-Dialysis Patients

In diabetic patients, % LBM measured by DEXA was greater than that predicted by BIA ($p < 0.05$). Bland & Altman analysis demonstrated biases by BIA, but the mean of the results obtained by combined anthropometry and BIA demonstrated no bias from DEXA measurements (Rigalleau et al 2004).

Near Infrared (NIR)

Evidence examining the validity of NIR as a measure of body composition was too limited to make recommendations.

- [Recommendation Strength Rationale](#)

The evidence supporting these recommendations is based on Grades III/Grade C, D evidence as well as Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of single-frequency bioelectrical impedance \(SF-BIA\) for assessing body composition in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of multi-frequency bioelectrical impedance \(MF-BIA\) for assessing body composition in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of bioimpedance spectroscopy \(BIS\) for assessing body composition in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of Near-Infrared Interactance for assessing body composition in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Nutrition Assessment with Laboratory Measurements (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Single Biomarker Measurements

In adults with CKD stages 1-5D or posttransplantation, biomarkers such as normalized protein catabolic rate (nPCR), serum albumin and/or serum prealbumin (if available) may be considered complementary tools to assess nutritional status. However, they should not be interpreted in isolation to assess nutritional status as they are influenced by non-nutritional factors (OPINION).

Rating: Consensus
Conditional

CKD: Serum Albumin Levels

In adults with CKD 5D on MHD, serum albumin may be used as a predictor of hospitalization and mortality, with lower levels associated with higher risk (1A).

Rating: Strong
Imperative

- [Risks/Harms of Implementing This Recommendation](#)

There are no potential risks or harms associated with the application of the guideline for serum albumin in adult patients with CKD on maintenance dialysis.

- [Conditions of Application](#)

The guideline for serum albumin applies to all adult patients with CKD on maintenance dialysis. The biochemical markers must be obtained pre-dialysis for maintenance dialysis patients.

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with the application of this guideline.

- [Recommendation Narrative](#)

Assessments of nutritional status in patients with CKD have traditionally relied upon biochemical or other related calculated indices such as serum albumin, prealbumin, and nPCR (normalized protein catabolic rate)/nPNA (normalized protein nitrogen appearance) as diagnostic tools. Albumin is a major circulating protein that plays a number of biologic roles, such as maintaining osmotic pressure and transporting a variety of molecules. Serum prealbumin, also known as transthyretin, is another circulating protein produced by the liver with a shorter half-life than albumin, it is therefore more sensitive to rapid changes in nutritional status. nPCR (or the more recent nomenclature nPNA) is a common tool used to estimate protein intake and is calculated using the intradialytic rise in the blood urea nitrogen in MHD patients and from urinary urea from 24-hour urine collection in non-dialyzed CKD patients. The advantages of such markers include the fact that they are easily quantifiable and available for each patient. However, these markers are known to be

heavily influenced by inflammation, illness, liver failure, volume expansion and urinary or dialysate protein losses (or in the case of nPCR/nPNA, protein balance and other factors). In fact, serum albumin is one of the best predictors of illness or death in patients with ESRD. In light of this, their utility in assessing nutritional status has been re-evaluated in recent years. Existing data suggest that such markers are not sufficiently reliable or valid to use in isolation for assessing nutritional status. Instead, it should be used as part of a more comprehensive and inclusive evaluation as used for screening purposes.

Detailed Justification

Serum Albumin

Sixteen observational studies that compared serum albumin concentration to other methods used to assess nutritional status, including twelve studies with MHD patients, two studies with PD patients, and two studies with both MHD and PD patients were included in this review

Maintenance Hemodialysis Patients:

Among the maintenance hemodialysis (MHD) studies, one was a prospective cohort study (de Roij et al 2015), two were retrospective cohort studies (Araujo et al 2006, Campbell et al 2010), seven were cross-sectional studies (Aatif et al 2013, Beberashvili et al 2009, Kadiri et al 2011, Jones et al 2002, Malgorzewicz et al 2008, Molino et al 2013, Yelken et al 2010). Two studies were diagnostic validity or reliability studies (Mancini et al 2003, Gurreebun et al 2007).

Gurreebun, et al. determined that serum albumin concentration was a sensitive method for identifying patients at risk of PEW defined by the 7-point SGA score. In a study by Mancini, et al., albumin independently predicted bioimpedance vector analysis (BIVA) in patients with normal values of other nutritional indexes, but the association was not significant in with patients with worse nutritional values. Araujo, et al. demonstrated that serum albumin concentration <3.5 g/dL were associated with higher odds of mortality over 10 years [OR (95%CI) = 2.34 (1.33-4.10); p=0.002]. Campbell, et al. found that low albumin concentration (<38 g/L) were significantly associated with higher mortality and morbidity (length of hospital stay), but there was no adjustment for comorbidities. De Roij van Zuidewijn, et al. determined that albumin concentration predicted all-cause mortality and was the most predictive of 8 other nutrition measures.

In Yelken, et al., serum albumin concentration were significantly correlated with high sensitivity C-reactive protein (hsCRP), tricep skinfold, mid arm circumference (MAC), and mid-arm muscle circumference (MAMC). Serum albumin concentration were associated with nPCR and inflammatory markers (Jones et al 2002, Molino et al 2013); BMI (Kadiri et al 2011); 7-point SGA score (Malgorzewicz et al 2008); and lean tissue index, but not fat tissue index from bioimpedance spectroscopy (Aatif et al 2013) BMI and FM (Beberashvili et al 2009).

Peritoneal Dialysis Patients

Of the two studies in PD, one was a prospective cohort study (Churchill et al 1996) and the other was a retrospective cohort study (Leinig et al 2011). Leinig, et al. demonstrated that hypoalbuminemia was a significant independent predictor of mortality [HR (95% CI): 2.3 (1.1-5.0)] after 24 months of follow-up. Churchill, et al. described that for every g/L increase in serum albumin, there was a 2-year relative mortality risk (95% CI) of 0.94 (0.90, 0.97).

Maintenance Hemodialysis and Peritoneal Dialysis Patients

Both MHD and PD patients were evaluated in two prospective cohort studies (Matthew et al 2015, de Mutsert et al 2009). Mathew, et al. found that serum albumin concentration did not predict mortality and was not correlated with lean tissue index. De Mutsert, et al demonstrated a 1g/dL decrease in serum albumin was associated with an increased mortality risk of 47% in MHD patients and 38% in PD patients (de Mutsert et al 2009). After adjusting for systemic inflammation, or for SGA and nPNA, these mortality risk ratios were not statistically significant indicating potential confounding effects of systemic inflammation.

In summary, one study showed that serum albumin concentration was a sensitive measure of nutritional status defined by 7-point SGA scores in MHD patients. Seven studies indicated that serum albumin was associated with other common markers of nutritional status in MHD patients. The preponderance of evidence suggested that lower serum albumin concentration predicts mortality in both MHD and PD patients.

Inflammatory Markers

There were no studies examining the validity and/or reliability of utilizing inflammatory markers to measure nutritional status. Thirteen studies examined correlations between inflammatory markers and other nutrition indices, including seven studies in MHD patients, one study in PD patients, two studies in both MHD and PD patients, one study in patients with kidney transplant, and two studies in pre-dialysis patients.

Maintenance Hemodialysis Patients:

Among the MHD studies, all seven were cross-sectional studies (Beberashvili et al 2009, Kadiri et al 2011, Kahraman et al 2005, Jones et al 2002, Molino et al 2013, Yelken et al 2010, Vannini et al 2009). hsCRP levels were positively associated with fat mass (Vannini et al 2009); and negatively associated with lean body mass (Vannini et al 2009), serum albumin (Jones et al 2002, Molino et al 2013, Yelken et al 2010, DiSilvestro et al 1997) and serum prealbumin (Molino et al 2013) concentrations. hsCRP was not associated with SGA score, PNA, anthropometric indices, or BIA measurements (Vannini et al 2009). While CRP was not associated with BMI in Vannini, et al., there was a negative correlation in Kadiri, et al. Kahraman, et al. found that CRP levels were highest in obese and underweight participants compared to their counterparts. Beberashvili, et al. found no relationship between proinflammatory cytokine level and BMI.

Peritoneal Dialysis Patients

de Araujo Antunes, et al. conducted a cross-sectional study in PD patients. Compared to patients with CRP level <1 mg/dL, those with CRP level ≥ 1 mg/dL had higher BMI (29.4 ± 6.1 vs. 24.4 ± 4.5 kg/m²; p=0.009), % standard body weight (124.5 ± 25.4 vs. 106.8 ± 17.9 %; p=0.012), and % BF measured by SF-BIA (38.9 ± 6.3 vs. 26.2 ± 12.6 %; p<0.001).

Maintenance Hemodialysis and Peritoneal Dialysis Patients:

Isoyama, et al. demonstrated that low handgrip strength, rather than low muscle mass measured with DEXA, was associated inflammatory markers including hsCRP, IL-6 and TNF- α (Isoyama et al 2014). In addition, CRP levels were negatively associated with BIA phase angle (Abad et al 2011).

Post-Transplant Patients

Only one cross-sectional study was identified for kidney transplant recipients. In this study, malnutrition inflammation score (MIS) was positively correlated with IL-6 (p=0.231; p<0.001), TNF- α (p=0.102; p<0.001), and CRP levels (p=0.094; p=0.003) (Molnar et al 2010).

Non-dialyzed Patients

Both studies in pre-dialysis patients were cross-sectional in nature (Cigarran et al 2013, Wing et al 2014). In a study by Wing, et al., hsCRP levels were higher in the highest BMI quartile, but results with other cytokines were mixed. In Stages 2-4 CKD men, CRP levels were negatively associated with testosterone distribution (Cigarran et al 2013).

In summary, many studies found correlations between higher inflammatory markers and suboptimal nutritional status, findings varied according to comparison measure. The relationship between BMI and inflammatory marker levels was unclear, and a U-shaped

relationship may exist. MIS was associated with inflammation inflammatory in kidney transplant patients.

- [Recommendation Strength Rationale](#)

The evidence supporting the above statements is based on Grade I/Grade A and Consensus/expert opinion evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of albumin levels for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of inflammatory markers for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of PNA/PCR for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of pre-albumin levels for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of testosterone levels for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

- [References](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Nutrition Assessment: Handgrip Strength (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Handgrip Strength

In adults with CKD 1-5D, we suggest that handgrip strength may be used as an indicator of protein-energy status and functional status when baseline data (prior measures) are available for comparison (2B).

Rating: Fair
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

The potential risk or harm associated with the application of the guideline for handgrip strength (HGS) in maintenance hemodialysis (MHD) patients involves the side of the body assessed. The measurement should be obtained on the opposite side of the vascular access. In all other patients (i.e. peritoneal dialysis and pre-dialysis), there are no potential risks or harms. Staff need to be properly trained on performing the measurement and interpreting the results.

- [Conditions of Application](#)

Implementation Considerations

- The guideline for handgrip strength (HGS) applies to all adult MHD, PD and non-dialyzed patients.

- The potential risk or harm associated with the application of the guideline for HGS in MHD patients involves the side of the body assessed. The measurement should be obtained on the opposite side of the vascular access. In all other patients (i.e. PD and pre-dialysis), there are no potential risks or harms. Staff need to be properly trained on performing the measurement and interpreting the results.
- Many individuals with CKD also have type 2 diabetes, a consequence of which may include peripheral neuropathy. Practitioners should account for potential loss in HGS due to peripheral neuropathy in patients with type 2 diabetes when comparing measurements over time (Gundmi et al 2018).

Measuring HGS is simple; however, it is not routinely used in clinical practice.

- [Potential Costs Associated with Application](#)

There is a cost associated with purchasing the equipment to measure Handgrip Strength.

- [Recommendation Narrative](#)

Handgrip strength (HGS) is a simple and reliable method to evaluate muscle function in patients with CKD. In addition, it can be used as an indirect measure of nutritional status in maintenance dialysis and non-dialyzed patients.

Detailed Justification

Five studies examined relationships between HGS and comparative measures in patients with CKD, including one study with non-dialyzed patients (Amparo et al 2013), one study with incident dialysis patients (Isoyama et al 2014), two studies with MHD patients (Hasheminejad et al 2016, Silva et al 2011), and one study with PD patients (Konings et al 2003). Overall, HGS was a valid measure of nutritional status compared to malnutrition inflammation score (MIS) in MHD patients (sensitivity=70-87%, specificity=43-66%) (Silva et al 2011) and was negatively associated with MIS in non-dialyzed patients ($r=0.42$; $p<0.001$) (Amparo et al 2013), but results may vary according to confounding variables. HGS was correlated with lean body mass (LBM) assessed by other methods, but there was no correlation with other markers of body composition or nutritional status in PD patients (Konings et al 2003). In incident dialysis patients, HGS had higher correlations with nutritional status and inflammatory markers, and was more predictive of mortality than muscle mass measured by dual energy X-ray absorptiometry (DEXA) (Isoyama et al 2014).

- [Recommendation Strength Rationale](#)

Evidence supporting this recommendation is based on Grade II/Grade B evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of the hand-grip strength for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

- [References](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Nutrition Assessment: Energy Requirements (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Assessment of Resting Energy Expenditure

In adults with CKD 1-5D or posttransplantation, it is reasonable to use indirect calorimetry to measure resting energy expenditure when feasible and indicated, as it remains the gold standard for determining resting energy expenditure (OPINION).

Rating: Consensus
Conditional

CKD: Resting Energy Expenditure Equations

In adults with CKD 5D who are metabolically stable, we suggest that in the absence of indirect calorimetry, disease-specific predictive energy equations may be used to estimate resting energy expenditure as they include factors that may influence the metabolic rate in this population (2C).

Rating: Weak
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no potential risks or harms related to these recommendations.

Patients should be monitored routinely to assess whether energy requirements are being met satisfactorily. Changes in nutritional status should be treated and the energy prescription modified accordingly.

- [Conditions of Application](#)

Special Discussions

Among patients with stage 5 CKD on MHD or PD, there are several factors that may influence energy expenditure beyond the traditional determinants (age, sex, and fat-free mass), such as hyperparathyroidism, hyperglycemia, and chronic inflammation that should be considered into the overall energy prescription. Energy needs will be variable depending on the health status of the patient (e.g., acutely ill versus chronically managed) as well as overall health goals (e.g., weight maintenance, repletion or loss). Energy needs may be different depending on the stage of CKD and its respective treatment (dialysis versus transplantation). In the context of these recommendations, “metabolically stable” indicates absence of any active inflammatory or infectious diseases; no hospitalization within two weeks; absence of poorly controlled diabetes and consumptive diseases such as cancer; absence of antibiotics or immunosuppressive medications; and no significant short-term loss of body weight.

Implementation Considerations

- The RDN should consider a number of factors when determining the energy requirements for adults diagnosed with CKD, and these include the patient’s overall health status, CKD diagnosis and associated therapies, level of physical activity, age, gender, weight status, disease-specific determinants, metabolic stressors, and treatment goals.
- Disease specific equations, such as the Maintenance Hemodialysis Equation, should be used when estimating energy requirements for the CKD population.
- Thermal effects of food may be decreased in individuals who are non-dialyzed compared to dialyzed due to lower protein intake.

Monitoring and Evaluation

Patients should be monitored routinely to assess whether energy requirements are being met satisfactorily. Changes in nutritional status should be treated and the energy prescription modified accordingly.

- [Potential Costs Associated with Application](#)

There are no costs associated with these recommendations.

- [Recommendation Narrative](#)

Achieving energy balance is critical in persons diagnosed with CKD so that protein-energy malnutrition and wasting can be prevented or treated in susceptible persons. Thus, obtaining reliable data regarding dietary energy intake as well as having a valid measure for energy expenditure is paramount.

Indirect calorimetry remains as the best practice measure for determining resting energy expenditure (REE) in adults diagnosed with CKD stages 1-5, including those receiving renal replacement therapies such as MHD, PD or post-transplant. More research is needed to demonstrate whether handheld indirect calorimetric devices may be a suitable alternative in this population.

In the absence of indirect calorimetry, there are over 200 predictive energy equations available that may be able to estimate REE in patients diagnosed with CKD. Several have been shown to either over- or under-estimate REE in earlier stages of CKD as well as those patients treated with maintenance dialysis. There have been several cross-sectional studies that suggest that the energy requirements of patients with earlier stages of CKD may not be substantially different than healthy adults, but the evidence is limited. Recent research has shown that predictive energy equations specifically designed for patients with CKD on maintenance dialysis have lower bias and greater precision.

Even the best predictive models designed for CKD do not account for the contribution of physical activity or structured exercise. Reliance on current estimates for physical activity may not determine total energy requirements accurately in this population.

Detailed Justification

There were six studies which tested REE equations in CKD patients and compared them to a reference standard of indirect calorimetry (Byham-Gray et al 2014, Dias Rodrigues et al 2014, Kamimura et al 2011, Lee et al 2008, Neyra et al 2003, Vilar et al 2014). Two of the six studies used indirect calorimetry data to derive a disease-specific equation (Byham-Gray et al 2014, Vilar et al 2014). The Harris-Benedict equation over-estimated REE in four studies across the spectrum of CKD; e.g., Dias Rodrigues, et al (MHD), Kamimura, et al (non-dialyzed, MHD and PD), Lee, et al. (CAPD) and Neyra, et al (CRF, MHD and PD), but the Harris-Benedict equation underestimated REE in MHD participants in Vilar, et al (MHD). Similarly, the Schofield equation over-estimated REE in Dias Rodrigues, et al. (MHD) and Kamimura, et al (non-dialyzed, MHD and PD), but under-estimated REE in Vilar, et al (MHD). Byham-Gray, et al demonstrated that the Maintenance Hemodialysis Equation more accurately predicted REE than the Mifflin-St. Joer equation. In Vilar, et al, they too found that their created equation for REE was best predictor of REE when compared to traditional predictive energy equations. Generally, agreement between equations and methods was low-to-moderate.

- [Recommendation Strength Rationale](#)

The evidence supporting these recommendation statements is based on Grade III/Grade C evidence and Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of REE equations for assessing energy requirements in adults with CKD 1-5D and post-transplant?](#)

- [References](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Nutrition Assessment: Composite Nutritional Indices (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: 7-Point Subjective Global Assessment (SGA)

In adults with CKD 5D, we recommend the use of the 7-point Subjective Global Assessment as a valid and reliable tool for assessing nutritional status (1B).

Rating: Strong
Imperative

CKD: Malnutrition Inflammation Score (MIS)

In adults with CKD 5D on MHD or posttransplantation, Malnutrition Inflammation Score may be used to assess nutritional status (2C).

Rating: Weak
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no risks of harms associated with the implementation of these recommendations.

- [Conditions of Application](#)

The large body of literature on nutritional assessment and composite nutritional indices have been completed in CKD 5D. While some of these tools may be relevant and can be translated to earlier stages (1-4) CKD, there is a need for the practitioner to conduct a comprehensive nutritional assessment comprising the main domains of the Nutrition Care Process.

Protein energy wasting (PEW), a term supported by the International Society of Renal Nutrition and Metabolism, describes the complexity of nutritional and metabolic alterations that exist in CKD. While PEW definition is useful in identify patients with overt nutritional abnormalities, its sensitivity is low given its strict criteria. While comprehensive nutritional indices have been validated for the recognition of a poor nutritional status (e.g., malnutrition), it is unclear how well some of these same tools may be applied in the early identification of PEW.

Implementation Considerations

Routine nutrition screening of adults diagnosed with CKD stages 1-5D should occur to allow for the identification and further assessment and treatment of nutritional concerns.

A comprehensive nutrition assessment, using a composite nutritional index, should be conducted at the initial visit and completed whenever there is a change in health status or as per institutional or regulatory policies.

Monitoring and Evaluation

The comprehensive nutrition assessment will guide the nutrition intervention prescribed. The clinician should monitor key nutrition care outcomes based on the treatment plan prescribed and re-assess and change the plan accordingly to achieve the goals established.

- [Potential Costs Associated with Application](#)

There are no significant costs associated with implementation of these recommendations.

- [Recommendation Narrative](#)

Assessment of nutritional status in adults diagnosed with CKD stages 1-5 must occur on a routine basis in order to prevent and/or treat malnutrition and wasting. The Nutrition Care Process (NCP) begins with a nutrition screening, whereby key nutritional indicators may trigger further assessment and intervention. There are several nutrition screening mechanisms in clinical practice, but few are specific to CKD, and there are limited data on their validity and reliability. Most of the existing tools focus on identification of malnutrition risk; only one currently screens for PEW. Regardless of the mechanism used, the nutritional assessment conducted subsequent to the screening should be comprehensive and include the routine monitoring of nutrition care outcomes. The main components of the comprehensive nutrition assessment comprise anthropometric measurements, biomarkers, clinical symptoms exhibited on physical exam, dietary intake assessment, and medical/psychosocial history. The availability of composite nutritional indices [e.g., the Subjective Global Assessment (SGA) or Malnutrition Inflammation Score (MIS)], collect such data and therefore assist the clinician in deciding about the individual's nutritional status and eventual plan of care, and are specific to the unique nutritional requirements of this patient population.

Detailed Justification

Composite Nutritional Indices: Screening Tools

Geriatric Nutrition Risk Index (GNRI)

Three studies reported on the use of GNRI to assess nutritional status, including two validity/reliability studies (Beberashvili et al 2013, Yamada et al 2013) and one prediction study in MHD patients. In one study, GNRI had the greatest area under curve (using MIS as a reference) of the nutrition screening tools (Yamada et al 2013). GNRI showed a significantly negative correlation with the MIS ($r=-0.67$, $P=0.0001$), and the most accurate GNRI cutoff to identify a malnourished patient according to the MIS was 91.2. The GNRI's sensitivity, specificity, and accuracy of a score of 91.2 in predicting malnutrition according to the MIS were 73%, 82%, and 79% respectively. Another study reported that GNRI had high inter-observer agreement score ($k=0.98$) and high intra-observer reproducibility ($k=0.82$) (Beberashvili et al 2013).

In another study, GNRI was a significant predictor for mortality at 2.97 years ($p<0.001$) but had lower predictive value for all-cause mortality compared to MIS and albumin levels.

Malnutrition Universal Screening Tool/Malnutrition Screening Tool (MUST/MST)

Two validity/reliability study reported on the use of MUST and MST tools to assess nutritional status in MHD patients (Yamada et al 2013, Lawson et al 2012). A study by Lawson et al, reported on the validity and reliability of both MUST and MST tool in MHD patients (Lawson et al 2012). The sensitivity of both the MUST and MST tool was low (53.8% for MUST; 48.7% for MST), indicating that they are not particularly sensitive at identifying individuals with malnutrition in this group, compared to SGA. Both tools have a high specificity (MUST=78.3%; MST=85.5%), so they are good at excluding individuals who are not malnourished. Reliability assessed by kappa was 0.58 for MUST (95% CI, 0.20 to 0.80) and 0.33 for MST (95% CI, 0.03 to 0.54). Both tools had an NPV or 60% and PPV for MUST was 73.7% and for MST was 78.7%. Though these tools are not sensitive enough to identify all malnourished renal in-patients, they are still fairly reliable and related to other nutrition status markers. In Yamada, et al., the authors compared results from various malnutrition assessment tools to the reference standard of MIS. MUST and MST scores were both significantly associated with MIS ($p<0.0001$ for each). The ROC curves of the MUST and MST compared to MIS were the smallest of the tools measured, and sensitivity, specificity and accuracy to detect hypoalbuminemia were among the lowest of all tools considered, indicating these may not be the best tools to discriminate nutritional risk in patients on MHD.

Mini-Nutrition Assessment (MNA)

Four studies reported on the use of MNA to assess nutritional status in MHD patients, three were validity/reliability studies (Yamada et al 2013, Afsar et al 2006, Santin et al 2016) and one was a correlational study (Yamada et al 2013, Afsar et al 2006, Santin et al 2016, Erdogan et al 2013). Afsar et al. reported on the reliability of MNA tool compared to SGA 3-point scale. The reliability coefficients (alpha) for MNA was 0.93 (good degree of reproducibility). MNA might underestimate the nutritional status of patients on MHD who are not in an inflammatory state. Hence, MNA may not be as reliable as SGA in detecting PEM in the MHD population. Erdogan et al. compared MNA to Bio-electrical Impedance Analysis (BIA), reported a significant correlation between MNA score and single frequency-BIA ($r=0.2$, $p=0.045$), muscle mass ($r=0.382$; $p<0.001$) and visceral fat ratio ($r=0.270$; $p=0.007$). Authors concluded BIA is not as sensitive as MNA to detect early effects of secondary causes for malnutrition. Santin et al. 2015, compared SGA (7-point), MIS, MNA-Short Form (MNA-SF) to handgrip strength (HGS), albumin, c-reactive protein (CRP), and skinfolds. SGA and MNA-SF had fair agreement ($kappa=0.24$; $p<0.001$). The worst agreement was found between MIS and MNA-SF ($kappa=0.14$, none to slight; $p<0.004$). Again, both SGA and MIS had good concurrent and predictive validity for CKD population, whereas MNA-SF validity results were more comparable to non-CKD elderly individuals. Yamada et al, compared MNA to other nutritional tools and reported that MNA had lower area under curve (0.73) than GNRI and Nutritional Risk Score but higher than MUST and MST.

Nutrition Impact Symptoms (NIS)

One validity study reported on the use of NIS score for identifying those at risk of malnutrition in patients on HD and concluded that NIS score is a useful nutrition screening tool for identifying who are at risk of malnutrition (Campbell et al. 2013). NIS score >2 had the strongest predictive value for mortality and for predicting poor nutritional outcomes, behind the rating of malnourished by SGA. Concurrent validity indicated similar agreement between each of the malnutrition risk tools (patient-generated subjective global assessment (PG-SGA), an abbreviated PG-SGA and NIS). Serum albumin was negatively correlated with NIS (Spearman $Rho=-0.161$; $p=0.018$).

Nutrition Screening Tool (NST)

One validity study reported on the use of NST to assess nutritional status in PD patients. In this study, NST had a sensitivity of 0.84 (range: 0.74 to 0.94; $p<0.05$) and specificity of 0.9 (range: 0.82 to 0.99; $p<0.05$) which is clinically acceptable (Bennett et al 2006).

Renal Nutrition Screening Tool (R-NST)

In another study by Xia et al in PD patients, the R-NST was compared to SGA-7 point scale. Authors determined that the R-NST tool

when compared to SGA- 7 point scale is valid to detect risk of malnutrition (sensitivity=97.3% (95% CI 90.7-99.7), specificity=74.4% (95% CI 57.9-87.0), PPV=88.0% (95% CI 79.0-94.1), NPV=93.6% (95% CI 78.6-99.2). These results indicate that R-NST is a good tool for identifying renal in-patients at risk of undernutrition.

Protein Energy Wasting (PEW) Score

Two predictive studies reported on the use of PEW score to assess nutritional status. Leining and colleagues identified that SGA and albumin were significant predictors of mortality, but BMI, mid-arm muscle circumference (MAMC) and PEW score did not predict mortality at 24 months in PD patients. However, Moreau-Gaudry et al, a study conducted in patients on MHD recorded that PEW predicts survival. Each unit decrease in score was related with a 5-7% reduction in survival ($p<0.01$). This score can be helpful in identifying subgroups of patients with a high mortality rate and recommend nutrition support.

Composite Nutritional Indices: Assessment Tools

Subjective Global Assessment (SGA)

Eleven studies examined the relationship between the 7-point SGA score and comparative measures, including three validity/reliability studies (Steiber et al 2007, Visser et al 1999, Santin et al 2016) and six additional prediction and/or correlation studies (de Roij van Zuijdewijn et al 2015, Malgorzewicz et al 2008, Vannini et al 2009, Jones et al 2004, Perez et al 2016, Tapiawala et al 2006).

Three studies examined the validity and/or reliability of the 7-point SGA score in MHD patients. In Visser, et al, 7-point SGA score demonstrated fair inter-observer reliability [intra-class correlation (ICC) = 0.72] and good intra-observer reliability (ICC=0.88) in MHD patients. In Santin, et al, 7-point SGA score had good agreement with MIS ($\kappa=0.43$; $p<0.001$) and MNA-SF ($\kappa=0.24$; $p<0.001$). In a study by Steiber, et al, SGA had fair interrater reliability ($\kappa=0.5$, Spearman's $\rho=0.7$), substantial intra-rater reliability ($\kappa=0.7$, Spearman's $\rho=0.8$) ($p<0.001$).

Three cohort studies examined whether the 7-point SGA score was predictive of hard outcomes in patients on MHD. In Perez, et al., SGA was a significant predictor of mortality at 2 years after adjustments for significant confounders (Perez et al 2016). In a study by de Roij van Zuijdewijn, et al., SGA was a significant predictor ($p<0.001$) for mortality at 2.97 years, but had lower predictive value for all-cause mortality compared to MIS and albumin levels. de Mutsert and others reported that hazard of mortality increased with SGA in a dose-dependent manner among patients on dialysis. Compared to normal nutritional status, persons who had a SGA of 4-5 had an increased HR (95% CI) at 7 year mortality of 1.6 (1.3, 1.9) and SGA of 1-3 had an HR of 2.1 (1.5, 2.8) at 7-year mortality. The strength of association increased in time-dependent models. Finally, in a study with PD patients, every one unit increase in the 7-point SGA adapted for end-stage renal disease (ESRD)/continuous ambulatory PD patients, there was a 25% decreased 2 year mortality risk ($p<0.05$) (Churchill et al 1996).

Six studies examined correlations between the 7-point SGA score and other measures of nutritional status. In Visser, et al., there was a strong correlation between the 7-point SGA score and BMI ($r=0.79$), % fat ($r=0.77$), and mid arm circumference ($r=0.71$) (all $p<0.001$) in MHD patients. In a study by Steiber, et al, there were statistically significant differences in mean BMI and serum albumin according to SGA score in MHD patients ($p<0.05$). Tapiawala, et al. assessed the 7-point SGA score in patients with CKD, ESRD and those on all types of dialysis. SGA scores were not correlated with dietary protein and energy intake or serum albumin levels, but anthropometric measures correlated with the SGA scores (skinfolts $r=0.2$, MAC $r=0.5$ and MAMC $r=0.5$). Authors concluded 7-point SGA is a reliable method of assessing nutritional status. Malgorzewicz, et al. compared near-infrared measurements and albumin levels to the SGA 7-point score in MHD patients. LBM measured by near-infrared was significantly decreased in malnourished patients ($p<0.05$) and there was a correlation between SGA score and LBM ($r=0.5$; $p<0.05$) as well as SGA score and albumin concentration ($r=0.7$; $p<0.05$). In Vannini, et al, SGA were associated with traditional nutritional markers, reinforcing validity for use among patients on MHD. SGA score was not associated with CRP level. Jones, et al examined the relationship between 3-point SGA score and a composite nutritional score that included SGA (3 point and 7 point), BMI, % reference weight, skinfold and MAMC measurements and albumin levels in patients treated by MHD. Compared to the composite score, the SGA score misclassified a large number of subjects and score was not associated with many nutrition parameters such as dietary intake, BMI or albumin levels.

In one study (Garagarza et al 2013), the authors utilized a version of the SGA that was adapted for patients on MHD, and in two studies (Leinig et al 2011, Passadakis et al 1999), the version of the SGA tool used was unclear. Garagarza et al, compared bioimpedance spectroscopy measurements to SGA scores from a version modified for MHD that included a 5-point score comprising weight changes, eating habits, gastrointestinal symptoms, functional activity and comorbidities. PEW measured by BIS extracellular weight (ECW)/body weight (BW) was positively associated with CRP ($p=0.009$) and SGA score ($p=0.03$). Leinig, et al. examined the relationship between SGA score and mortality risk at 24 months in PD patients, but version of the SGA employed was unclear. SGA score was a significant predictor of mortality in PD patients (Leinig et al 2011). Passadakis, et al. compared BIA measurements to SGA score in CAPD patients, but the version of SGA utilized was uncertain. SGA score was significantly correlated with impedance index ($r=0.48$; $p=0.0038$) and phase angle ($r=0.43$; $p=0.0048$).

Malnutrition Inflammation Score (MIS)

Eight studies reported on the use of MIS to assess nutritional status, including two validity/reliability studies (Beberashvili et al 2013, Santin et al 2016), four prediction studies (Fiedler et al 2009, de Roij van Zuijdewijn et al 2015, Perez et al 2016) and three correlation studies (Molnar et al 2010, Amparo et al 2013, Hou et al 2012).

One study by Beberashvili et al reported that MIS had moderate inter-observer agreement ($\kappa=0.62$) and inter-observer reproducibility ($\kappa=0.77$) and is a valid tool for longitudinal assessment of nutritional status of patients on MHD. Another study by Santin et al, indicated that MIS had good agreement with SGA ($\kappa=0.43$, $p<0.001$) and worse agreement with MNA-SF ($\kappa=0.14$, $p<0.004$). MIS also had good concurrent and predictive validity for the MHD population.

Four studies reported on the use of MIS as a predictor of mortality (Fiedler et al 2009, de Roij van Zuijdewijn et al 2015, Perez et al 2016). All three studies reported that in patients on MHD, MIS is a significant predictor of mortality. In one study, MIS was a significant predictor for mortality at 2.97 years ($p<0.001$), and best predictive tool for all-cause mortality and secondary end-points like cardiovascular events in patients on MHD (de Roij van Zuijdewijn et al 2015). Another study by Fiedler et al also reported that MIS was predictive of both mortality and hospitalizations in patients treated by MHD with survival analysis indicated that MIS was one of the best predictors of mortality [HR 6.25 (2.82 - 13.87), $p<0.001$]. Perez et al, also indicated that MIS was a significant predictor for 2 year mortality in MHD patients. Finally, in Santin, et al., while mild MIS did not predict mortality, severe MIS was a significant predictor of mortality in adjusted analysis [HR (95% CI): 5.13 (1.19, 13.7)].

Three studies reported on the use of MIS and correlation with other tools. Amparo et al, indicated that there was a significant negative correlation between hand grip strength and MIS ($r=-0.42$, $p<0.001$) in predialysis subjects. Hou et al, indicated that MIS was strongly correlated with modified quantitative subjective global assessment ($r=0.924$) and inversely correlated with BIA ($r=-0.213$) in MHD patients. Molnar et al, reported that MIS showed significant negative correlations with abdominal circumference ($p=-0.144$; $p<0.001$) and pre-albumin level ($p=-0.165$; $p<0.001$), whereas significant positive correlation was seen with IL-6 ($p=0.231$; $p<0.001$), TNF- α ($p=0.102$; $p<0.001$), and CRP levels ($p=0.094$; $p=0.003$) in kidney transplant recipients. All studies show that MIS is a useful tool to assess nutritional status in CKD patients.

Other Composite Nutritional Indices

Nutrition Risk Score (NRS)

A prediction study reported that NRS was a good predictor of mortality (HR 4.24 (1.92-9.38), $p < 0.001$) in patients on MHD and was superior when compared to lab markers and BIA in predicting mortality (Fiedler et al 2009).

Protein Nutrition Index (PNI)

A reliability study investigated PNI as a predictor of survival in PD patients. Compared to the reference standard (nPNA (nPCR) $\hat{=}$ ± 0.91 as malnutrition), the sensitivity, specificity, positive and negative predictive value of PNI were 0.4, 0.978, 0.901 and 0.783, respectively (Chen et al 2010). This study indicated that PNI is a good predictor of mortality (even after adjusting for age and comorbidities). An increase in PNI score by 1 led to a 16% decrease in mortality risk.

Composite Score of Protein Energy Nutrition Status (cPENS)

de Roij van Zijdewin et al studied eight nutrition assessment tools used to predict all-cause mortality. cPENS had a Harrell's C statistics of 0.63 (0.61 $\hat{=}$ 0.66) for predicting mortality. However, the study indicated that it had inadequate discrimination and calibration or a lower predictive value for mortality.

Other Measures

Blumberg et al, compared the integrative score with the SGA-7-point scale in HD patients. Integrative clinical nutrition dialysis score is based on biochemical measures of albumin, creatinine, urea, cholesterol, CRP, dialysis adequacy, and weight change. With every unit increase in integrative score, the odds of death were significantly decreased (HR=0.929, 95% CI 0.885-0.974, $p < 0.002$). SGA and integrative score were significantly correlated ($n=69$, $r=0.853$, $p < 0.01$) and according to the author this is a useful prognostic tool to detect early nutrition deterioration.

A prediction study investigated which nutritional composed scoring system best predicts all-cause mortality in MHD patients (Perez et al 2016). This study indicated that SGA and MIS are better predictors of all-cause mortality at 15.5 months in this study and International Society of Renal Nutrition and Metabolism criteria was not able to predict mortality in this sample.

One correlation study investigated the relationship between body adiposity index (BAI), BIA, anthropometrics, and DEXA (Silva et al 2013). The correlation coefficient was higher between DEXA vs. anthropometric measurements ($r=0.76$) and BAI ($r=0.61$) when compared to BIA ($r=0.57$) in the adjusted analysis ($p < 0.0001$). Results suggest BIA estimates body fat with high accuracy in non-dialyzed CKD patients.

- [Recommendation Strength Rationale](#)

The evidence supporting the above recommendations is based on Grade II/Grade B and Grade III/Grade C evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of the 7-point SGA for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the 3-point SGA for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Geriatric Nutrition Risk Index \(GNRI\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Malnutrition Inflammation Score \(MIS\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Malnutrition Universal Screening Tool/Universal Screening Tool \(MUST/MST\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Mini Nutrition Assessment \(MNA\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Nutrition Impact Symptom \(NIS\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Nutrition Risk Score \(NRS\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Malnutrition Score for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the PEW score for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Nutrition Screening Tool \(NST\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Renal Nutrition Screening Tool \(R-NST\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Integrative Score for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Composite Score of Protein Energy Nutrition Status \(cPENS\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the International Society of Renal Nutrition and Metabolism \(ISRNM\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of the Body Adiposity Index \(BAI\) for assessing nutritional status in adults with CKD 1-5D and post-transplant?](#)

Is there evidence to support the use of the Protein Nutrition Index (PNI) for assessing nutritional status in adults with CKD 1-5D and post-transplant?

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Nutrition Assessment: Dietary Intake (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Considerations when Assessing Dietary Intake

In adults with CKD 3-5D or posttransplantation, it is reasonable to assess factors beyond dietary intake (e.g. medication use, knowledge, beliefs, attitudes, behavior and access to food, depression, cognitive function etc.) to effectively plan nutrition interventions. (OPINION).

Rating: Consensus
Conditional

CKD: 3-Day Food Records to Assess Dietary Intake

In adults with CKD 3-5D, we suggest the use of a 3-day food record, conducted during both dialysis and non-dialysis treatment days (when applicable), as a preferred method to assess dietary intake (2C).

Rating: Weak
Conditional

CKD: Alternative Methods of Assessing Dietary Intake in CKD 3-5

In adults with CKD 3-5 (OPINION), or CKD 5D (2D), 24-hour food recalls, food frequency questionnaires and normalized protein catabolic rate (nPCR) may be considered as alternative methods of assessing dietary energy and protein intake (2D).

Rating: Consensus
Conditional

CKD: Alternative Methods of Assessing Dietary Intake in CKD 5D

In adults with CKD 5D (2D), 24-hour food recalls, food frequency questionnaires and normalized protein catabolic rate (nPCR) may be considered as alternative methods of assessing dietary energy and protein intake (2D).

Rating: Weak
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no potential risks or harms associated with these recommendation statements.

- [Conditions of Application](#)

Special Discussions

Despite the food record/diary being the most reliable and valid measure of dietary intake among patients diagnosed with CKD, it does rely on accurate reporting inclusive of portion sizes. The food record may be seen as cumbersome to complete for several days and is limited to individuals that are able to read and record intake reliably. With the generation of smartphone applications, there has been a burgeoning interest in recording dietary intake using technology, with limited success in its adoption among certain subgroups (e.g., elderly). Dietary intake methods may need to be simplified, modified, or be combined with a few strategies in order to obtain reliable dietary intake data, with emphasis on them being culturally appropriate.

Implementation Considerations

- Routine dietary assessment among adults diagnosed with CKD stages 1-5D should occur to allow for the identification and treatment of nutritional concerns related to nutrient intake.
- Assessing dietary intake using multiple, complementary methods, such as FFQ and 24-hr uring collection to measure urine urea nitrogen, sodium and potassium, may be useful to confirm accuracy of dietary intake estimates.
- Dietary assessment should be conducted at the initial visit and completed whenever there is a change in health status or as per institutional or regulatory policies.

Monitoring and Evaluation

A thorough assessment of dietary intake will guide the nutrition intervention prescribed. The clinician should monitor key nutrition care outcomes based on the treatment plan and re-assess and change the plan accordingly to achieve the goals established.

- [Potential Costs Associated with Application](#)

There are no costs associated with implementation of these recommendation statements.

- [Recommendation Narrative](#)

Poor nutritional intake and obesity are prevalent among patients diagnosed with CKD and therefore, it is important to monitor dietary intake that provides information on total energy, macro- and micro-nutrients as well as overall food/liquid servings and eating patterns. In this context, it is important to identify reliable methods for estimating dietary intake in diverse care settings. Under- and over-reporting of intake are a concern in this population.

Detailed Justification

A total of seven studies reported on use of methods to assess protein and energy intake in CKD subjects (Griffiths et al 1999, Kloppenburg et al 2001, Laxton et al 1991, Avesani et al 2005, Bazanelli et al 2010, Kai et al 2016, Shapiro et al 2015).

Food Records/Diary

Based on the findings of included studies, food records/diary for assessing dietary intake of protein and calories were reliable and correlated with reference standards. Food records can provide accurate information if patients are instructed and trained, and food intake is recorded for at least 7 days (Griffiths et al 1999, Kloppenburg et al 2001, Kai et al 2016). Two studies used food diary/3-day food records to determine underreporting of energy intake in non-dialyzed and PD patients (Avesani et al 2005, Bazanelli et al 2010). Underreporting was noticed in 72.5% of non-dialyzed CKD patients and 52.5% PD patients. Both the studies indicated that underreporting was more pronounced in overweight patients. Shapiro et al 2015 compared energy intake measured by 3-day food record (dietitian interview-assisted) and REE measured by indirect calorimetry. Energy intake reported by interview-assisted food records were lower than measured REE (Shapiro et al 2015).

Food Frequency Questionnaires

Delgado et al conducted a validation study comparing Block Brief 2000 food frequency questionnaire (BFFQ) against 3-day food diary records (Delgado et al 2014) and found BFFQ under-estimated energy and macronutrient intake in patients on hemodialysis. However, the use of simple calibration equations can be used to obtain intake similar to 3-day food diary records.

Protein Catabolic Rate

Three studies examined the use of protein catabolic rate (PCR) to assess protein intake in CKD patients (Laxton et al 1991, Lorenzo et al 1995, Virga et al 1996), and found significant correlations with reference standards for measuring dietary intake (ex: food records). However, PCR overestimated protein intake when daily protein intake was <1 g/kg and when daily protein intake was >1 g/kg it was underestimated by PCR. In PD patients, PNA (PCR) normalized to desirable body weight was correlated better with BUN ($r=0.702$) and Kt/V ($r=0.348$) (Virga et al 1996).

- [Recommendation Strength Rationale](#)

The evidence supporting these recommendations are based on Grade III/Grade C, D and Consensus/expert opinion.

- [Minority Opinions](#)

.Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[Is there evidence to support the use of food records or 24-hr recall for assessing dietary protein or energy intake in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of FFQ for assessing protein or energy intake in adults with CKD 1-5D and post-transplant?](#)

[Is there evidence to support the use of PNA/PCR for assessing dietary protein intake in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Medical Nutrition Therapy (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: MNT to Improve Outcomes

In adults with CKD 1-5D, we recommend that a registered dietitian nutritionist (RDN) or an international equivalent, in close collaboration with a physician, or other provider (nurse practitioner or physician assistant), provide medical nutrition therapy (MNT). Goals are to optimize nutritional status, and to minimize risks imposed by co-morbidities and alterations in metabolism on the progression of kidney disease (1C) and on adverse clinical outcomes (OPINION).

Rating: Fair
Imperative

CKD: MNT Content

In adults with CKD 1-5D or posttransplantation, it is reasonable to prescribe MNT that is tailored to the individuals'™ needs, nutritional status and co-morbid conditions (OPINION).

Rating: Consensus
Imperative

CKD: MNT Monitoring and Evaluation

In adults with CKD 3-5D or posttransplantation, it is reasonable for the registered dietitian nutritionist (RDN) or an international equivalent to monitor and evaluate appetite, dietary intake, body weight changes, biochemical data, anthropometric measurements, and nutrition-focused physical findings to assess the effectiveness of medical nutrition therapy (OPINION).

Rating: Consensus
Imperative

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms association with Medical Nutrition Therapy.

- [Conditions of Application](#)

Special Discussions

The full utility and value of MNT provided by the RDN on both nutrition outcomes and risk of morbidity, mortality and hospitalizations has not yet been fully identified. The impact of the RDN in many disease states and the value of repeated contacts with an RDN on specific nutrition parameters has been documented in the literature (Evidence Analysis Library 2015, Sikand et al, Muscaritoli et al). This is particularly true for CKD patients as well as in other disease states and metabolic phenotypes such as obesity that affect CKD risk and exacerbation of CKD progression. While MNT outcomes research is still in its infancy, the studies that do exist exhibit important relationships on nutrition parameters and other outcomes. An MNT database that monitors MNT intervention effectiveness on nutrition and overall outcome parameters would enable the formalization of this analysis. Studies that prove causality or significant association between MNT application and patient outcomes is currently in progress. In addition, the strength of the evidence in studies reviewed prohibits strong recommendations due to the variability in study populations, protocols and analyses. Therefore, this section included recommendations that are opinion based.

MNT facilitates the delivery of Nutrition Practice Guidelines through a systemic approach of delivery that is based on scientific evidence and expert opinion. The education, content and practice expertise for the provision of MNT individualized care is found within the scope of practice of the RDN with expertise in nephrology.

Implementation Considerations

- Evidence based protocols are inherent to MNT but do not replace individualized modification.
- Implementation of MNT for CKD patients requires the formation of a fiscal structure that will support the integration of MNT into routine medical management of CKD patients. The interest level to integrate MNT into clinical practice exists by many nephrology and general medicine clinics, however, the lack of adequate reimbursement for RDN services precludes the opportunity to pursue implementation.
- Demand for MNT is growing as the prevalence of CKD in the USA increases. Reimbursement policies for disease prevention need to include MNT. Legislation awareness is needed to disseminate the value of MNT.
- MNT may be delivered through telehealth options, in order to improve patient education and successful maintenance of nutrition interventions and adherence.

Monitoring and Evaluation

Monitoring and evaluation of MNT on patient's nutritional parameters is an essential component of treatment and includes assessment of patient's labs, nutritional status, etiology of kidney disease, lifestyle (stress, exercise, evaluation of smoking and alcohol use, etc.), and patient identified nutrition goals.

- [Potential Costs Associated with Application](#)

There are no obvious costs in implementing these recommendations. Implementation of MNT Therapy for CKD patients requires the formation of a fiscal structure that will support the integration of MNT therapy into routine medical management of CKD patients. The interest level to integrate MNT into clinical practice exists by many nephrology and general medicine clinics however the lack of adequate reimbursement for RDN services precludes the opportunity to pursue implementation.

Demand for MNT is growing as the prevalence of CKD in the USA increases. Reimbursement policies for disease prevention need to include MNT. Legislation awareness is needed to disseminate the value of MNT.

- [Recommendation Narrative](#)

Individualized management of nutritional intake is a crucial aspect of care for individuals diagnosed with any stage of CKD, including those on dialysis and those who have received a kidney transplant. These patients are vulnerable for malnutrition, which is associated with higher risk for morbidity, mortality, and length of hospital stay. Nutritional needs change throughout the disease course, from the earlier stages of CKD to the post-transplant period. The metabolic abnormalities and co-morbid diseases that often accompany CKD further emphasize the need for specialized nutrition health care. Therefore, it is essential that such individuals receive tailored nutrition assessment and counseling in the form of MNT. MNT is a collaborative approach that typically requires the medical expertise and prescription of MNT by a physician or other provider (nurse practitioner, physician assistant) and implementation by an RDN, or international equivalent). These roles are not mutually exclusive and involve ongoing team-patient analysis and discussion. Participating providers and RDN are recommended to have received specialized education and training in nutrition and CKD in accordance with the requirements set forth by local regulations.

In 2002, the then American Dietetic Association published a nutrition care model that provided evidence-based, high-quality standardized care for patients with CKD, non-dialyzed and post-transplant. The document was later revised in 2010, which reported that nutrition care provided by an RD up to twice monthly over a one-year period can have a valuable role in the medical care of the CKD patients by:

- Providing nutrition assessment and interventions to delay kidney disease progression in addition to co-morbid conditions such as diabetes mellitus, cardiovascular disease, dyslipidemia, gout, nephrolithiasis;
- Utilizing behavioral methods to individualize the approach and minimize barriers to individualized goals;
- Providing individualized meal plans and follow up on adherence and successful implementation. Interventions include but are not limited to weight management and maintenance/repletion of patient nutritional status;
- Addressing inflammation, supporting obtaining a euvolemic state, contributing to correction of electrolyte abnormalities, assisting in anemia management and managing bone disease through nutrition assessment and dietary interventions including individualized meal plans;
- Assisting identifying medication errors and need for adjustment- in collaboration with nephrology Provider (Medical Doctor, Nurse Practitioner, Physician Assistant);
- Providing and update nutrition therapy as new knowledge indicates such as incorporating plant proteins and addressing the acid nature of dietary intake.

Detailed Justification

MNT requires nutrition screening and assessment of nutritional status to provide individualized treatment for specific disease states. CKD patients are on a dynamic nutrition trajectory according to their disease stage and MNT is needed at each stage of CKD. Metabolic abnormalities, acid base, fluid and electrolyte balances often change as CKD progresses. For example, a patient can be hypokalemic during Stage 2 CKD requiring potassium supplementation and a high potassium diet. Months or years later, this same patient during Stage 4 CKD might become hyperkalemic, requiring medication adjustment and dietary potassium restriction rather than supplementation. Should this same patient receive a kidney transplant, they might stabilize potassium balance and have no need for

potassium supplementation or dietary potassium restriction. This type of complicated CKD patients requires specialized nutrition health care and ongoing monitoring by a nephrology RDN.

Sixteen RCTs examining the effect of MNT on nutrition-related outcomes were identified in the systematic review. However, these studies were heterogeneous in terms of the populations (five studies included patients who were non-dialyzed, nine included patients on MHD, one included patients on CAPD, and one included patients post-transplant); interventions (ex: RDNs utilized various methods of nutritional counseling among the studies); and outcomes (ex: protein intake, serum phosphate, serum albumin, BMI, and dyslipidemia). Intervention durations ranged from four weeks to two years.

CKD Progression

In four of the studies ranging from 4 weeks to 4 months, authors found no effect of MNT on CKD progression in non-dialyzed patients compared to participants receiving standard nutrition education for CKD, which may or may not have also been provided by an RDN. Interventions ranged from one in-person contact plus phone contacts with the RDN for 12 weeks (Stage 4 CKD) (Campbell et al 2008) to a multi-disciplinary intervention including 4 weeks of weekly counselling with an RDN (Stages 3-4 CKD) (Howden et al 2013) to 2, two-hour cooking classes and a shopping tour (Stages 2-4 CKD) (Fletcher et al 2011) to nutrition counselling plus nutrition education for four months (Stages 3-5 CKD) (Paes-Barreto et al 2013).

SGA Scores

Three RCTs, including two study populations, reported on the effect of MNT on SGA scores. Campbell et al. demonstrated that malnourished Stage 4 CKD patients' SGA scores significantly improved in the intervention group compared to the control group, for whom malnutrition by SGA score increased (Campbell et al 2008). The intervention consisted of nutritional counselling from an RDN for 12 weeks, with an emphasis on self-management techniques, face-to-face consultation at baseline, and telephone consultation every two weeks for the first month, and then monthly for the next 2 months. In Leon, et al., MHD participants received monthly consultation by RDN for 12 months (Leon et al 2001). Intervention RDNs were trained to determine potential barriers to achieving normal albumin levels for each patient, to attempt to overcome the barriers, and to monitor for improvements in barriers. There was no difference in the percentage of participants that had improved or decreased SGA scores between groups.

BMI

Four RCTs examined the effect of MNT interventions on BMI, including two studies with non-dialyzed patients (Stages 3-5 CKD) (Paes-Barreto et al 2013 and Howden et al 2013), one study with MHD participants (Leon et al 2001) and one with post-transplant patients (Orazio et al 2011). Howden, et al., examined the effect of a 12-month multi-disciplinary lifestyle intervention on BMI in patients with stages 3-4 CKD (Howden et al 2013). The intervention group received 4 weeks of group behavioral and lifestyle modification sessions provided by an RDN and a psychologist. Mean BMI significantly decreased in the intervention group compared to the standard care group ($p < 0.01$). Paes-Barreto, et al., examined the effect of MNT on BMI in participants with stages 3-5 CKD who received individualized dietary counselling monthly for four months. In addition to the routine counselling, an intervention group received intensive counselling, which included nutrition education materials emphasizing a low-protein and low-sodium diet. There was a significantly greater decrease in BMI in the intervention group compared to the standard care group ($p < 0.01$). In Leon, et al., MHD participants received monthly consultation by an RDN to determine and address barriers to reaching normal serum albumin levels for 12 months. There was no effect on BMI, though this was not the objective of the intervention. Finally, in Orazio, et al., intervention participants received RDN counselling using a Mediterranean-style diet, which consisted of a low glycemic index and moderate energy deficit. MNT counselling was based the Stages of Change Model. There was no difference in change in BMI between groups after 2 years.

In a pooled analysis of two studies (Paes Barreto et al 2013 and Howden et al 2013), participants who received MNT had a greater mean (95% CI) decrease in BMI compared to the control groups $[-0.89 (-1.52, -0.25) \text{ kg/m}^2]$. Results regarding effect of MNT on arm and waist circumference as well as body composition were limited and unclear.

Phosphate Levels

Eight studies examined the effect of MNT on phosphorus/phosphate levels in MHD patients for durations ranging from 8 weeks to 6 months. In Ashurst, et al. and Lou, et al., phosphorus-focused education, provided once and monthly for 6 months, respectively, significantly improved (decreased) mean serum phosphate levels. In Karavetian, et al. weekly education nutrition counselling for 2 months also decreased phosphate levels ($p < 0.01$). However, Morey et al. also used phosphorus-focused RDN counselling and education, monthly for 6 months, and found no difference in change in phosphate levels between groups at 6 months.

Participants receiving a multi-disciplinary nutrition education program did not have any changes in phosphate levels compared to participants receiving an oral nutrition supplement (ONS) (Hernandez et al 2014). In Reese, et al., participants who were coached by a trained RDN about dietary and medication adherence (≈ 3 times a week) for 10 weeks were compared to patients receiving a financial incentive or usual care. There were no between-group differences in change in phosphate levels. There was no effect of MNT in the form of dietary counselling in CAPD patients (Sutton et al 2007) or in the form of RDN counselling plus low-protein and low-sodium diet education in non-dialyzed patients (Paes-Barreto et al 2013) on phosphate levels, but the objectives of these studies were to improve energy, protein and sodium intake.

Meta-analysis of four studies with comparable data revealed that, mean (95% CI) phosphorus/phosphate levels were decreased $-0.715 (-1.395, -0.034) \text{ mg/dL}$, however heterogeneity is high ($I^2 = 67.71\%$, $p = 0.015$). Thus, there was evidence that MNT decreased phosphorus/phosphate levels in MHD patients (Hernandez-Morante et al 2014, Morey et al 2008, Karavetian et al 2013), but effect on phosphorus/phosphate levels as well as the effect on calcium or potassium levels in non-dialyzed patients (Paes-Barreto et al 2013), was unclear.

Lipid Profile

Three RCTs examined the effect of MNT from an RDN on lipid profile (Hernandez-Morante et al 2014, Howden et al 2013, Fletcher et al 2011). In Hernandez-Morante, et al., MHD participants in the intervention group received a 12-session multi-disciplinary Nutrition Education Program over four months, including group and individual therapy, while control participants received an oral nutrition supplement three days/week. Within group analysis showed no significant changes in mean triglycerides and total cholesterol levels over 4 months. There was a significant increase in mean low-density lipoprotein cholesterol (LDL-C) and a significant decrease in mean high-density lipoprotein cholesterol (HDL-C) in both groups over the 4-month study period ($p < 0.001$ for each measure). Between-group analysis was not reported.

Both Howden, et al. and Fletcher, et al. examined the effect of MNT in Stages 3-4 CKD participants. In Howden, et al., intervention participants received a multi-disciplinary lifestyle intervention for 12 months. It included 4 weeks of group behavioral and lifestyle modification by an RDN and a psychologist. No significant changes were observed in triglyceride or total, HDL or LDL cholesterol levels between the 2 groups. In Fletcher, et al., in addition to the standard nutrition care for CKD, the intervention group received cooking classes over 4 weeks for 2 hours per session and a shopping tour led by an RDN. No significant difference was observed in mean total cholesterol level between the 2 groups. Pooled analysis confirmed no effect of MNT on total cholesterol and triglyceride levels. However, in pooled analysis, LDL levels were decreased by MNT (Mean (95% CI): $-6.022 (-7.754, -4.290) \text{ mg/dL}$). There was no clear effect of MNT on blood pressure (BP).

Protein intake

Six RCTs examined the effect of MNT on protein intake in CKD patients (Paes-Barreto et al 2013, Leon et al 2011, Howden et al, Campbell et al 2008, Sutton et al 2007 and Orazio et al 2011). Two of those studies targeted protein intake as their primary outcome of the MNT provided to the participants. Paes-Barreto, et al. educated non-dialysis patients on eating a low protein diet, while Leon, et al. counseled MHD participants on following a high protein diet. Both studies showed high compliance of recommended protein intake among the participants in the intervention group as compared to the control group. The other four studies did not show any significant differences in protein intake between the intervention and control groups, but protein intake was not the primary outcome.

The utilization of MNT protocols has the potential to preserve nutritional status, modify risk factors for progression of kidney disease, as well as assist with living with CKD from a diet and lifestyle perspective through teaching patients healthy food choices in an individualized manner.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on improved outcomes is based on Grade III /Grade C evidence and Consensus/expert opinion. The remaining MNT recommendations are based on Consensus/expert opinion.

The full utility and value of the MNT intervention by the RDN on both nutrition outcomes and risk of morbidity, mortality and hospitalizations has not yet been fully identified. The impact of the RDN in many disease states and the value of repeated contact visits of RDN with patient on specific nutrition parameters has been documented in the literature. This is particularly true for CKD patients as well as in other disease states and metabolic phenotypes such as obesity that affect CKD risk and exacerbation of CKD progression. In consideration that MNT outcomes research is still in its infancy, the studies that do exist exhibit important associations on nutrition parameters and outcomes as reviewed in this document. An MNT database that monitors MNT intervention effectiveness on nutrition and overall outcome parameters would enable the formalization of this analysis. Studies that prove causality or significant association between MNT application and patient outcomes is currently a work in progress. In addition, the strength of the evidence in studies reviewed prohibits strong grading due to the variability in study population sizes, protocol and statistical structuring. Retrospective cohort analysis trials were not included here. Therefore, suggestions from this section are opinion based.

MNT facilitates the delivery of Nutrition Practice Guidelines through a systemic approach of delivery that is based on scientific evidence and expert opinion. The education, content and practice expertise for the provision of MNT individualized care is found within the scope of practice of the RDN with expertise in nephrology.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of MNT provided by a dietitian on lipid profile in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on calcium, potassium or sodium levels or calcium-phosphorus product in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on phosphorus/phosphate levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on BMI in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on body weight in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on CRP levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on nutritional status indicators in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on protein and energy intake in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on quality of life in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on blood pressure in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on blood glucose in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on eGFR in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on serum creatinine levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on anemia indices in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on body composition in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on waist circumference in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of MNT provided by a dietitian on arm circumference in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Macronutrients: Dietary Patterns (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Dietary Patterns: Mediterranean Diet

In adults with CKD 1-5 not on dialysis or posttransplantation, with or without dyslipidemia, we suggest that prescribing a Mediterranean Diet may improve lipid profiles (2C).

Rating: Weak
Conditional

CKD: Dietary Patterns: Fruits and Vegetables

In adults with CKD 1-4, we suggest that prescribing increased fruit and vegetable intake may decrease body weight, blood pressure and net acid production (NEAP) (2C).

Rating: Weak
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

- Safety of various dietary patterns, including the DASH and Mediterranean with high intakes of fruit and vegetables in advanced stages of kidney disease.

- [Conditions of Application](#)

Implementation Considerations

- Safety of various dietary patterns, including the DASH and Mediterranean with high intakes of fruit and vegetables in advanced stages of kidney disease, especially in regard to serum potassium control and adequacy of protein intake.
- Individualized support and follow-up may be required to support patients to implement complex dietary changes.

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with these interventions.

- [Recommendation Narrative](#)

Dietary patterns reflect the variety of foods which represent habitual dietary intake (Leech, et al. 2015). Particular dietary patterns, including the Mediterranean Diet, the Dietary Approach to Stop Hypertension (DASH), plant-based and diets high in fruits and vegetables (including vegetarian diets) are examples of healthy dietary patterns which have been the subject of interest much in nutritional epidemiology (Hu, et al. 2002, Kelly et al 2017). By taking a whole-of-diet approach, this considers the synergistic effects of nutrients resulting in cumulative effects on health and disease (Hu, et al. 2002).

Chronic Kidney Disease (CKD) presents many challenges for nutrition management, including increased risk of death and appreciable cardiovascular disease (CVD) burden among affected persons. Traditionally, nutrition education has focused on individual nutrients, such as protein, phosphorus, potassium and sodium. Recent evidence has linked healthy dietary patterns with reduced chronic CVD and mortality risk in the healthy population (Estruch, et al. 2013; Salehi-Abargouei, et al. 2013; Sofi, et al. 2014). However, these relationships have not been explored conclusively with the CKD population.

This systematic review focused on controlled trials of dietary patterns. While various dietary patterns were investigated (Fruits and Vegetables, Mediterranean Diet, Low Fructose Diet, Hypolipidemic, Carbohydrate restricted- low iron, polyphenol-enriched diet (CR-LIPE), High-protein/Low-carbohydrate), there was little evidence examining the efficacy of most of these patterns in controlled trials. Hence, only the Mediterranean and High Fruit and Vegetable dietary patterns had sufficient evidence to create recommendations.

Mediterranean Dietary Pattern

eGFR

One RCT reported on the effect of Mediterranean dietary pattern on eGFR (Mekki, et al. 2010). Mekki et al 2010 indicated an increase in eGFR in both the groups, but stayed unchanged at 90-days (NS) (SMD 0.23, 95% CI -0.39, 0.85). Mediterranean dietary pattern had no clear effect on eGFR at 90 days post intervention in adults with CKD stage 2. Additional research on effect of Mediterranean dietary pattern intake is needed.

Lipid Profile

Limited evidence from three studies, two of which examined non-dialyzed patients and one of which examined post-transplant patients, demonstrated that the Mediterranean diet improved lipid panel by decreasing total cholesterol (TC), low-density lipoprotein (LDL-C) and triglyceride (TG) level compared to control groups.

Two controlled trials reported on the effect of Mediterranean dietary pattern on lipid profile in non-dialyzed patients (Mekki, et al. 2010; Di Daniele et al. 2014). In the RCT, Mekki et al 2010 (stage 2) reported a 35% reduction in TC ($p < 0.05$) in the Mediterranean diet group, whereas, no change in TC was observed in the control group. LDL-C levels and TG levels were also reduced compared to standard care (Mekki, et al. 2010). In an NRCT, Daniele et al 2014 reported a significant reduction in TC in both Mediterranean diet group and organic Mediterranean diet group. However, most reduction was noted in the organic Mediterranean diet group.

In post-transplant patients, one RCT reported that Mediterranean diet led to significant reduction in TC, TG and LDL-C levels compared with a low fat diet (Stachowska, et al. 2005).

Other Outcomes

Compared to a control group, the Mediterranean Diet had no clear effect on BP in transplant recipients (Stachowska, et al. 2005) or on CRP levels in stage 2 patients (Mekki, et al. 2010).

However, one NRCT (Daniele et al 2014) reported on the effect of Mediterranean dietary pattern on albuminuria in stage 2 and 3 CKD adults, and both Mediterranean diet groups (normal and organic) had significant reductions in albuminuria values compared to low protein group (Daniele, et al. 2014).

High Fruit and Vegetable Dietary Pattern

CKD Progression

In adults with stages 3-4 CKD, fruits and vegetables dietary pattern has mixed effects/uncertain effects on eGFR compared with oral bicarbonate supplementation (Goraya, et al. 2014 and 2013).

Body Weight

Two RCTs reported on the effect of a fruit and vegetable dietary pattern on body weight in adults with CKD. Goraya et al 2014, reported that the group following fruit and vegetable dietary pattern had greater net body weight loss than both oral bicarbonate group and standard care group (p -value < 0.05), also the control group had greater net body weight loss than HCO₃ group (p -value < 0.05). Goraya et al 2013, reported lower body weight in adults with CKD stages 3-4 following a fruit and vegetable dietary pattern compared to oral bicarbonate supplementation group at 1-year follow up ($p < 0.01$). (MD = -5.09 kg, 95% CI -7.73, 2.44; $I^2 = 56\%$).

Blood Pressure

Three studies (2 RCT, 1 NRCT) reported on the effect of increased fruit and vegetable intake on BP in adults with CKD. All three studies indicated that increased intake of fruit and vegetable had a significant effect on lowering systolic BP compared to oral bicarbonate

supplement intake group or standard care group. Goraya et al 2014 indicated reductions in systolic BPs in all groups, however, the 3-year value for FV was lower than those in HCO₃ and control. Goraya et al 2013, showed that compared to HCO₃ group, the fruits and vegetable group had lower systolic BP at 1-year follow up (p-value < 0.01). Goraya et al 2012 (NRCT), showed that fruit and vegetable intake, but not control or HCO₃, significantly decreased systolic BP in individuals with CKD Stages 1 and 2 (p-values < 0.001). Pooled analysis of data from Goraya et al 2013 and Goraya et al 2014 indicated a MD (95% CI) of -5.6, CI: -8.3, -2.8 mmHg. Increased intake of fruits and vegetable dietary pattern lowered systolic BP compared to oral bicarbonate supplement intake or standard care group in adults with CKD stages 1-4.

Comparison with Recent Research

A recent systematic review (SR) examined the effect of dietary patterns on CKD outcomes using cohort studies (Kelly, et al. 2017). In agreement with the current analysis of controlled trials, Kelly, et al. found no effect of dietary pattern on CKD progression in studies with follow-ups ranging from 4 to 6.4 years. However, unlike the current SR, Kelly, et al. was able to demonstrate a relationship between a dietary pattern rich in vegetables, fruit, fish, cereals, whole grains, fiber, legumes, and nuts and seeds, and lower in red meat, sodium, and refined sugars in studies reporting outcomes from 4 to 13 years of follow up [RR (95% CI): 0.73 (95% CI, 0.63 to 0.83)] (Kelly, et al. 2017).

A recent Cochrane review of dietary interventions in CKD found from 6 RCTs evaluated on dietary patterns (one study (n=191) of a carbohydrate-restricted low-iron, polyphenol enriched diet, two studies (n=355) of a Mediterranean diet, two studies (n=181) of increased fruit and vegetable intake and one study (n=12) of a high protein/low carbohydrate diet. From this review, dietary interventions had uncertain effects on all-cause mortality and cardiovascular events. However, with low quality evidence, there was reduced systolic and diastolic BP, and higher GFR and albumin levels following dietary interventions (Palmer, et al. 2017).

Although the intervention studies examining dietary patterns in CKD are limited, there is consistent evidence from observational analyses on dietary patterns containing fruits, vegetables, whole grains, lean meats, low fat dairy and low added salt, and improved clinical outcome (notably mortality) in CKD (Kelly, et al. 2017). Another recent study confirmed that nuts, low-fat dairy products, and legumes are protective against development of CKD (Joshi et al 2019). There is therefore a need to undertake future trials to further investigate more holistic dietary interventions over single nutrient approaches in these patients. Dietary pattern may improve additional outcomes not reported in the systematic review, including constipation.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation is based on Grade III /Grade C or D evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of fruit and vegetable dietary patterns on blood pressure in adults with CKD 1-4?](#)

[What is the effect of fruit and vegetable dietary patterns on body weight in adults with CKD 3-4?](#)

[What is the effect of the Mediterranean dietary pattern on lipid profile in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on acid-base levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of fruit and vegetable dietary patterns on eGFR in adults with CKD 3-4?](#)

[What is the effect of fruit and vegetable dietary patterns on albuminuria in adults with CKD 1-4?](#)

[What is the effect of the Mediterranean dietary pattern on CRP levels in adults with CKD 2?](#)

[What is the effect of the Mediterranean dietary pattern on albuminuria in adults with CKD2-3?](#)

[What is the effect of the Mediterranean dietary pattern on blood pressure in adults with CKD post-transplant?](#)

[What is the effect of the Mediterranean dietary pattern on lipid profile in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of the Mediterranean dietary pattern on eGFR in non-dialyzed adults with CKD?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Macronutrients: LC n-3 PUFA Nutritional Supplements for Lipid Profile (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: LC n-3 PUFA Nutritional Supplements for Lipid Profile, Maintenance Hemodialysis

In adults with CKD 5D on MHD, we suggest that 1.3-4 g/d LC n-3 PUFA may be prescribed to reduce triglycerides and LDL cholesterol (2C) and raise HDL levels (2D).

Rating: Weak
Conditional

CKD: LC n-3 PUFA Nutritional Supplements for Lipid Profile, Peritoneal Dialysis

In adults with CKD 5D on PD, it is reasonable to consider prescribing 1.3-4 g/d LC n-3 PUFA to improve the lipid profile (OPINION).

Rating: Consensus
Conditional

CKD: LC n-3 PUFA Nutritional Supplements for Lipid Profile, Non-dialyzed

In adults with CKD 3-5, we suggest prescribing ~2g/d LC n-3 PUFA to lower serum triglyceride levels (2C).

Rating: Weak
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

- Adverse effects of capsule-based supplementation may lead to gastrointestinal side effects like stomach upset and eructation (though the latter can be masked by different formulations). Theoretical risks like bleeding have not been borne out in clinical trials.
- There is no need to routinely monitor dietary LC n-3 PUFA intake other than in the context of general dietary counselling. An exception would be if the patient is specifically instructed to consume greater dietary quantities of LC n-3 PUFA.

- [Conditions of Application](#)

The clinical impact of LC n-3 PUFA supplementation in patients with CKD was challenging to assess due to short study durations, modest sample sizes, and wide heterogeneity in terms of the absolute and relative amounts of n-3 PUFA supplemented, the type of placebo used, and the study duration, making it more difficult to provide conclusive evidence for or against the use of LC n-3 PUFA as a treatment option. Furthermore, baseline LC n-3 PUFA levels (either in blood or tissues) were not typically used to target populations that would most benefit. This is an important but often overlooked point because the putative benefits of LC n-3 PUFA supplementation may be inversely related to baseline blood or tissue concentrations.^{178, 211}

Implementation considerations

- LC n-3 PUFA supplementation considerations will differ depending on whether the intervention is diet-based or capsule-based.
- For dietary interventions the goal of supplementation must be clearly defined. If it is to raise blood levels of $\pm$ -linolenic acid then supplementation should focus on soybean, flaxseed, and other oils as well as meat and dairy products. If it is to raise EPA or DHA blood/tissue levels then the primary dietary sources must be sardine, mackerel, salmon and other high-content marine-based foods (Kris-Ehrtenton et al 2002). Potential limitations to dietary supplementation include their relatively high cost and difficulty in achieving high daily intake. In addition, the source and processing method will influence LC n-3 PUFA foodstuff

content. For example, farmed fish typically (but not always) has lower LC n-3 PUFA compared to wild fish, while frying fish could alter the n-3/n-6 ratio which may be of clinical significance.²¹²

- Capsule-based supplementation involves a set of different considerations. While dozens of commercial LC n-3 PUFA supplements are available, quality control is often lacking.²¹³ This makes precise dosing recommendations difficult. An alternative route is to have the patient obtain supplements via physician prescription (e.g. icosapent ethyl, omega-3 ethyl esters). For either option cost could be an issue. Achieving high dose supplementation will be easier with capsules than through dietary consumption. Adverse effects of capsule-based supplementation may lead to gastrointestinal side effects like stomach upset and eructation (though the latter can be masked by different formulations). Theoretical risks like bleeding have not been borne out in clinical trials.
- LC n-3 PUFA content is listed on the website of the National Institutes of Health (<https://ods.od.nih.gov/factsheets/Omega3FattyAcids-HealthProfessional/>).

- **Potential Costs Associated with Application**

- LC n-3 PUFA supplementation considerations will differ depending on whether the intervention is diet-based or capsule-based.
- Potential limitations to dietary supplementation include their relatively high cost and difficulty in achieving high daily intake.

- **Recommendation Narrative**

Rationale/Background

Long chain omega-3 polyunsaturated fatty acids (LC n-3 PUFA) include eicosapentaenoic (EPA), docosapentaenoic and docosahexaenoic acids (DHA), all of which are obtained primarily from dietary sources like cold-water fish (i.e. fish oil), or linoleic acid, which is derived from flaxseed or certain other vegetable oils. In recent decades LC n-3 PUFA have demonstrated protean biologic effects that mediate eicosanoid production, cell membrane physiology, signal transduction, metabolism, apoptosis, oxidation, and inflammation. Accordingly, they have been tested in a variety of medical conditions. Of particular interest has been their putative effects on cardiac membrane stabilization leading to possible reduction of malignant arrhythmias and sudden cardiac death. Patients with CKD have been documented to have some of the lowest record blood levels of LC n-3 PUFA on record (Friedman, et al. 2012) thus making them potentially very suitable candidates for supplementation interventions. In fact, LC n-3 PUFA supplementation has been studied as possible therapy for a number of conditions commonly observed in patients with CKD including dyslipidemia, hemodialysis access failure, cardiovascular disease and death, as well as for their immunomodulatory effects in patients with kidney allografts.

Lipid Profiles

Aside from with HDL, there were no signals suggesting that LC n-3 PUFA improved circulating plasma lipids. Nineteen separate RCTs (though one without a true control group; Guebre-Egziabher et al 2005) addressed the impact of LC n-3 PUFA supplementation on lipid levels. Thirteen studied patients on MHD (Tayebi-Khosroshahi et al 2013; Saifullah et al 2007; Kooshki et al 2011; Lemos et al 2012; Sorensen et al 2015; Bowden et al 2009 and Daud et al 2012; Pouliat et al 2011; Ewers et al 2009; An et al 2012 (included patients on PD also); Khajehdehi et al 2000; Taziki et al 2007 and Khalatbari et al 2013), four studies patients with CKD 2-5 (Svensson et al 2004, Mori et al 2009, Bouzidi et al 2010, Guebre-Egziabher et al 2013), and two studied patients with CKD with kidney allografts (Bennett et al 1995; Ramezani et al 2011). The studies ranged greatly in terms of the type of supplement (seventeen with fish oil, two with flaxseed oil or ground flaxseed (2 g/d oil in Lemos et al 2012 and 40g/d seed in Khalatbari et al 2013) and duration (3-6 months). Additionally, amount and reporting of dosing was inconsistent. Studies reporting amount of specific LC n-3 PUFAs described doses ranging from 0.42-1.8g/d EPA and 0.25-0.82g/d DHA. The specific amounts of EPA and DHA were not clear in several studies (Bennett et al, Ewers et al, Khajehdehi et al, Sorrenson et al, Taybei-Khosroshahi et al, Guebre-Egziabher et al).

Triglycerides

Eighteen RCTs studied the impact of LC n-3 PUFA on serum triglycerides. Seven of the thirteen trials studying patients with MHD found no effect (An et al 2012; Bowden et al 2009; Daud et al 2012; Ewers et al 2009; Pouliat et al 2011; Saifullah et al 2007; Tayebi-Khosroshahi, et al. 2013) and six reported a reduction in levels (Lemos et al 2012; Khalatbari et al 2013; Khajehdehi et al 2000; Kooshki et al 2011; Sorensen et al 2015; Taziki et al 2007). In a pooled analysis of twelve of these studies, LC n-3 PUFA supplementation lowered triglyceride levels by an average (95% CI) of -33.78 (-63.21, -4.36) mg/dL as compared to placebo/controls, though heterogeneity was high ($I^2=92.36\%$, $p<0.001$). While outcomes did not appear to be related to study quality or duration, triglyceride lowering tended to be associated with using lower doses of LC n-3 PUFA (0.42-0.96 g/d EPA and 0.24-0.6 g/d DHA daily), flaxseed oil (2g/day) or ground flaxseed (40g/day), a counterintuitive finding. Interestingly, positive results were more consistent in pre-dialysis patients (Svensson et al 2004; Mori et al 2009; Bouzidi et al 2010; Guebre-Egziabher et al 2005) where fish oil supplementation (1.8 g/d total or 0.46-1.44 g/d EPA with 0.25-0.96 g/d DHA) for 8-12 weeks consistently lowered triglycerides.

Total Cholesterol

The literature did not suggest a beneficial effect of LC n-3 PUFA supplementation on total cholesterol. Eleven of thirteen studies in patients on MHD reported no effect (0.42-1.8 g/d EPA and 0.24-1.14 g/d DHA for 4 weeks to 6 months) (Lemos et al 2012, Khajehdehi et al 2000; Bowden et al 2009; Daud et al 2012; Ewers et al 2009, Kooshki et al 2011, Pouliat et al 2011, Saifullah et al 2007, Sorensen et al 2015, Tayebi-Khosroshahi et al 2013, Taziki et al 2007), while the two studies supplementing with flaxseed oil (2g/d for 120 days) (Lemos, et al. 2012) or ground flaxseed (40 g/d for 8 weeks) (Khalatbari, et al. 2013) noted a significant reduction in total cholesterol levels (though one study did not compare differences between groups; Khalatbari, et al. 2013). A pooled analysis of all 13 studies did not find any effect on mean total cholesterol, but did note a high level of heterogeneity in the data ($I^2=95.77\%$, $p<0.001$). Three of four supplementation studies in pre-dialysis patients reported no effect on total cholesterol levels (Svensson, et al 2006; Mori, et al. 2009; Guebre-Egziabher, et al. 2005) while the fourth demonstrated a reduction at three months ($p<0.05$) with no difference between arms (Bouzidi, et al. 2010). Results could not be pooled for these four studies. While Ramezani et al. 2011 reported lower cholesterol levels in CKD patients with kidney allografts compared to placebo after 6 months of supplementation with 1.76 g/d EPA with 0.96 g/d of DHA in fish oil, Schmitz et al found no such benefit in a similar population.

LDL Cholesterol

Eight of twelve studies in patients on MHD found no benefit of supplementation (Lemos et al 2012; Daud et al 2012; Ewers et al 2009; Kooshki et al 2011; Saifullah et al 2007; Sorensen et al 2015; Tayebi-Khosroshahi et al 2013; Taziki et al 2007) while four reported a reduction in LDL (Lemos et al 2012; Khalatbari et al 2013; Khajehdehi et al 2000; Bowden et al 2009). Two of the four positive studies supplemented with fish oil (1.5g total in Khajehdehi et al 2000 and 0.96 g/d EPA with 0.6 g/d DHA in Bowden et al) while the other two used flaxseed oil or ground flaxseed, with both latter studies observing a drop in LDL (Lemos, et al. 2012; Khalatbari, et al. 2013). Study quality or duration or the type of comparison group used did not influence the outcome. A pooled analysis of all twelve studies noted an improvement in LDL only when excluding the flaxseed-based supplement studies [mean difference (95% CI): -5.26 (-9.51, -1.00) mg/dL] and even then the result was clinically marginal. In pre-dialysis patients, four studies of 8-12 weeks length using fish oil found no effect on LDL (1.8g/d total in Guebre-Egziabher et al and 0.46-1.44g/d EPA with 0.25-0.96g/d DHA) (Svensson et al 2006; Mori et al 2009; Bouzidi et al 2010; Guebre-Egziabher et al 2005). In patients with CKD with kidney allografts, one study reported that EPA 6g capsules per day increased LDL levels (but a higher dose did not) while another study reported negative results (Bennett et al and Schmitz et al).

HDL Cholesterol

Seventeen RCTs included HDL as an outcome. Though HDL may be influenced by physical activity, smoking status, and gender, the preponderance of these studies did not control for these factors. Of the twelve studies in patients on MHD six reported negative results (An et al 2012; Daud et al 2012; Ewers et al 2009; Kooshki et al 2011; Saifullah et al 2007; Tayebi-Khosroshahi et al 2013) and six found that HDL levels were increased (An et al 2012; Khalatbari et al 2013; Khajehdehi et al 2000; Bowden et al 2009; Sorensen et al 2015; Taziki et al 2007). Effects were not clearly influenced by study quality or duration. However, the positive studies tended to use lower doses of LC n-3 PUFA (0.72-0.96 g/d EPA with 0.42-0.6 g/d DHA), flaxseed oil (2g/day), or ground flaxseed (40g/day). In a pooled analysis of all twelve studies LC n-3 PUFA supplementation was found to raise HDL by a mean (95% CI) of 7.1 (0.52, 13.63) mg/dL. However, the heterogeneity was high overall. Results were mixed in the four trials of pre-dialysis patients, with two showing a benefit (Svensson et al 2004; Guebre-Egziabher et al 2005) and two reporting no effect (Mori et al 2009; Bouzidi et al 2010). Again, the outcome was not clearly influenced by quality of study, study duration or dosage, and results could not be pooled. Finally, the only study in CKD patients with kidney allografts showed no benefit (Bennett et al 1995).

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendations is based on Grade III/Grade C, D evidence and consensus/expert opinion. The clinical impact of LC n-3 PUFA supplementation in patients with CKD was challenging to assess due to short study duration, modest sample size, and wide heterogeneity in terms of the absolute and relative amounts of n-3 PUFA supplemented, the type of placebo used, and the study duration. Furthermore, baseline LC n-3 PUFA levels (either in blood or tissues) were not typically used to target populations that would most benefit. This is an important but often overlooked point because the putative benefits of LC n-3 PUFA supplementation may be inversely related to baseline blood or tissue concentrations (Friedman et al 2012, Mozaffarian et al 2006).

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of omega 3 fatty acid supplementation on total, LDL and HDL cholesterol and triglyceride levels in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Macronutrients: LC n-3 PUFA Nutritional Supplements and Hard Outcomes (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: LC n-3 PUFA Nutritional Supplements for Mortality and Cardiovascular Disease, Maintenance Hemodialysis and Post-Transplant

In adults with CKD 5D on MHD or posttransplantation, we suggest not routinely prescribing LC n-3 PUFA, including those derived from fish or flaxseed and other oils, to lower risk of mortality (2C) or cardiovascular events (2B).

Rating: Fair
Imperative

CKD: LC n-3 PUFA Nutritional Supplements for Mortality and Cardiovascular Disease, Peritoneal Dialysis

In adults with CKD 5D on PD, it is reasonable to not routinely prescribe LC n-3 PUFA, including those derived from fish or flaxseed and other oils, to lower risk of mortality or cardiovascular events (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Adverse effects of capsule-based supplementation may lead to gastrointestinal side effects like stomach upset and eructation (though the latter can be masked by different formulations). Theoretical risks like bleeding have not been borne out in clinical trials.

- [Conditions of Application](#)

All treatment should be individualized according to a patient's preferences, health history and co-morbidities as well as clinical judgement.

- [Potential Costs Associated with Application](#)

- LC n-3 PUFA supplementation considerations will differ depending on whether the intervention is diet-based or capsule-based.

- For dietary interventions the goal of supplementation must be clearly defined. If it is to raise blood levels of α -linolenic acid then supplementation should focus on soybean, flaxseed, and other oils as well as meat and dairy products. If it is to raise EPA or DHA blood/tissue levels then the primary dietary sources must be sardine, mackerel, salmon and other high-content marine-based foods. Potential limitations to dietary supplementation include their relatively high cost and difficulty in achieving high daily intake. In addition, the source and processing method will influence LC n-3 PUFA foodstuff content. For example, farmed fish typically (but not always) has lower LC n-3 PUFA compared to wild fish, while frying fish could alter the n-3/n-6 ratio which may be of clinical significance (Strobel, et al. 2012).
- Capsule-based supplementation involves a set of different considerations. While dozens of commercial LC n-3 PUFA supplements are available, quality control is often lacking (Kleiner, et al. 2015). This makes precise dosing recommendations difficult. An alternative route is to have the patient obtain supplements via physician prescription (e.g. icosapent ethyl, omega-3 ethyl esters). For either option cost could be an issue. Achieving high dose supplementation will be easier with capsules than through dietary consumption. Adverse effects of capsule-based supplementation may lead to gastrointestinal side effects like stomach upset and eructation (though the latter can be masked by different formulations). Theoretical risks like bleeding have not been borne out in clinical trials.
- LC n-3 PUFA content is listed on the website of the National Institutes of Health (NIH 2018) (<https://ods.od.nih.gov/factsheets/Omega3FattyAcids-HealthProfessional/>).

◦ Recommendation Narrative

Rationale/Background

Long chain omega-3 polyunsaturated fatty acids (LC n-3 PUFA) include eicosapentaenoic (EPA), docosapentaenoic and docohexaenoic acids (DHA), all of which are obtained primarily from dietary sources like cold-water fish (i.e. fish oil), or linoleic acid, which is derived from flaxseed or certain other vegetable oils. In recent decades LC n-3 PUFA have demonstrated protean biologic effects that mediate eicosanoid production, cell membrane physiology, signal transduction, metabolism, apoptosis, oxidation, and inflammation. Accordingly, they have been tested in a variety of medical conditions. Of particular interest has been their putative effects on cardiac membrane stabilization leading to possible reduction of malignant arrhythmias and sudden cardiac death. Patients with CKD have been documented to have some of the lowest record blood levels of LC n-3 PUFA on record (Friedman, et al. 2012) thus making them potentially very suitable candidates for supplementation interventions. In fact, LC n-3 PUFA supplementation has been studied as possible therapy for a number of conditions commonly observed in patients with CKD including dyslipidemia, hemodialysis access failure, cardiovascular disease and death, as well as for their immunomodulatory effects in patients with kidney allografts.

Detailed Justification

All-Cause Mortality and Cardiovascular Events

Despite the putative overall benefits of LC n-3 PUFA and the elevated risk of death in patients with CKD, three RCTs demonstrated no improvement in mortality with supplementation. However, the studies were heterogeneous in terms of study population (Svensson et al 2006 in patients with MHD, two in patients with CKD with kidney allografts- Maachi et al and Berthoux et al), the dose of LC n-3 PUFA (Maachi et al 1.44g/d EPA + 0.96g/d DHA; Berthoux et al 1.62g/d EPA+1.08g/d DHA; Svensson et al 2006 0.77g/d EPA+ 0.64g/d DHA) and the study duration (1-2 years), with the combined study population being fairly modest in size (n=264).

Two well-designed but modestly sized (combined N=351) RCTs in patients on MHD reported mixed results on the effect of LC n-3 PUFA supplementation on cardiovascular events (Svensson et al 2006, Lok et al 2012). Lok et al. reported that 4 g/day fish oil (1.6 g/d EPA, 0.8 g/d DHA) for 12 months as compared to a corn oil-based placebo significantly lowered the cardiovascular event rate 0.41 (0.20 to 0.85) (p=0.02) and improved cardiovascular event-free survival 0.43 (0.19 to 0.96) (p=0.04) but did not influence the number of patients with one or more events 0.78 (0.55 to 1.09) (p=0.15). All were secondary outcomes in a trial designed to study MHD vascular access. In a secondary prevention trial Svensson et al. reported that 1.7 g/d fish oil (0.77g/d EPA and 0.64g/d DHA) for two years had no effect on the primary combined endpoint of cardiovascular events or death as compared to olive oil-based placebo (Svensson et al 2006) but did improve the secondary endpoints of myocardial infarctions (0.30 (0.10, 0.92) (p=0.036) and major coronary events (0.40 (0.17, 0.97) (p=0.043).

Glomerular Filtration Rate and CKD Progression

Based on six RCTs with widely differing study design and populations (CKD stage 3, MHD, kidney allografts) (Maachi et al 1995, Berthoux et al 1992, Mori et al 2009, Bennett et al 1995), fish oil supplementation was not found to influence estimated or measured GFR. In the study by Guebre Egziabher et al 2013, participants received 1.8g of n-3 fatty acids, but authors do not describe EPA or DHA amount. In the study by Bennett et al, participants received 9 or 18g/d EPA capsules. In the remaining studies, EPA dose ranged from 0.46-1.62g/d and DHA ranged from 0.25-1.08g/d.

Similarly, fish oil supplementation for 8-12 weeks did not influence serum creatinine levels in three studies of non-dialyzed patients who used placebo or non-placebo-based control groups. In the study by Guebre Egziabher et al 2013, participants received 1.8g/d of n-3 fatty acids, but authors do not describe EPA or DHA amount. In the remaining studies, EPA amount ranged from 0.69-1.44g/d and DHA amount ranged from 0.25-0.96g/d (Svensson et al 2004, Bouzidi et al 2010).

Blood Pressure

Five RCTs examined the effect of LC n-3 PUFA supplementation on BP, two in non-dialyzed patients (no stage reported) (Svensson et al 2004, Mori et al 2009), two in patients on MHD (Lok et al 2012, Khajehdehi et al 2000) and one in patients with CKD with kidney allografts (Bennett et al 1995). The results were mixed. In non-dialyzed patients Svensson et al reported that fish oil (0.96g/d from DHA and 1.44g/d EPA) for 8 weeks did not affect BP (Svensson et al 2004) while Mori et al found that fish oil (0.38g/d DHA and 0.46g/d EPA) for 8 weeks lowered both systolic (mean $\pm$ SEM, (-3.3 ($\pm$ 0.7)) and diastolic (-2.9 ($\pm$ 0.5)), (p<0.0001 for each change) blood pressures. A pooled analysis of these two trials did not find an overall beneficial effect. In patients on MHD, Lok et al reported an improvement in systolic BP in patients on MHD with fish oil (0.8g/d DHA and 1.6g/d EPA) for one year [Mean Difference (95% CI): -8.10 (-15.4, -0.85), p=0.014] and a reduction in the number of BP medications but no effect on diastolic BP. In contrast, Khajehdehi et al found no effect at all on BP of 1.5 g/day fish oil (DHA and EPA content not reported) as compared to placebo for two months. Data from these two trials could not be pooled. Bennett et al randomized patients with CKD with kidney allografts and reported no benefit of 9 or 18 g EPA capsules per day versus placebo for 26 weeks on systolic BP but did note a reduction in diastolic BP in both EPA arms (p<0.05 for each) only.

Inflammatory Markers

The putative anti-inflammatory effects of LC n-3 PUFA were tested on two established biomarkers of inflammation.

C-Reactive Protein

Fifteen RCTs studied the effect of fish oil supplementation on circulating CRP. In pre-dialysis patients throughout the stages of CKD, fish oil either compared to placebo (Mori et al 2009, Madsen et al 2007) or at varying doses (Guebre-Egziabher et al 2005) had no effect. The pattern in patients on MHD was similar. A pooled analysis of ten studies (Lemos et al 2012, Khalatbari et al 2013, Bowden et al 2009, Daud et al 2012, Kooshki et al 2011, Poulia et al 2011, Saifullah et al 2007, Gharekhani et al 2014, Hung et al 2015, Harving et al 2015) found no effect of LC n-3 PUFA supplementation (nine using fish oil containing 0.42-1.8 g/d EPA with 0.24-1.14 g/d DHA, one using 2g/d flaxseed oil) on circulating CRP as compared to placebo (MD -1.73 mg/L, 95% CI: -3.54, 0.09). Ewers et al found that fat

supplementation (which also included fats other than LC-n-3 PUFA and specific n-3 PUFAs were not described) was associated with a reduction in CRP ($p=0.01$) after 12 weeks as compared to non-supplemented patients.

Interleukin-6

Six RCTs studied the effect of LC n-3 PUFA on circulating IL-6. Neither of the two studies in predialysis patients comparing fish oil supplementation to placebo or at varying doses found a significant effect on IL-6 (one study reported 1.8g/d n-3 PUFAs total and one reported 1.4g/d EPA with 1.0g DHA) (Guebre-Egziabher et al 2005, Deike et al 2012), nor did a pooled analysis of four studies in patients on MHD (MD 5.32 pg/ml, 95% CI: -5.637, 16.275) in which participants received 0-1.93 g/d EPA with 0.72-0.97 g/d DHA) (Kooshki et al 2011, Gharekhani et al 2014, Hung et al 2015, Himmelfarb et al 2007).

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation statements is based on Grade III/ Grade C evidence for MHD or post-transplant; and for individuals on PD the supporting evidence is based on Consensus/expert opinion. The clinical impact of LC n-3 PUFA supplementation in patients with CKD was challenging to assess due to short study duration, modest sample size, and broad heterogeneity in terms of the absolute and relative amounts of n-3 PUFA supplemented, the type of placebo used, and the study duration, making it more difficult to provide conclusive evidence for or against the use of LC n-3 PUFA as a treatment option. Furthermore, baseline LC n-3 PUFA levels (either in blood or tissues) were not typically used to target populations that would most benefit. This is an important but often overlooked point because the putative benefits of LC n-3 PUFA supplementation may be inversely related to baseline blood or tissue concentrations (Friedman, et al. 2012; Mozaffarian, et al. 2006).

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of omega 3 fatty acid supplementation on all-cause mortality in adults with CKD on hemodialysis and post-transplant?](#)

[What is the effect of omega 3 fatty acid supplementation on cardiovascular disease events in adults with CKD on hemodialysis?](#)

[What is the effect of omega 3 fatty acid supplementation on homocysteine levels in adults with CKD 5D on hemodialysis?](#)

[What is the effect of omega 3 fatty acid supplementation on CRP levels in adults with CKD 5D on hemodialysis?](#)

[What is the effect of omega 3 fatty acid supplementation on IL-6 levels in adults with CKD on hemodialysis?](#)

[What is the effect of omega 3 fatty acid supplementation on inflammatory marker levels in non-dialyzed adults with CKD?](#)

[What is the effect of omega 3 fatty acid supplementation on blood pressure in adults with hemodialysis or renal transplant?](#)

[What is the effect of omega 3 fatty acid supplementation on blood pressure in non-dialyzed adults with CKD?](#)

[What is the effect of omega 3 fatty acid supplementation on eGFR in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Macronutrients: Protein Amount (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Protein Restriction, Non-Dialysis, Non-Diabetic

In adults with CKD 3-5 who are metabolically stable, we recommend, under close clinical supervision, protein restriction with or without keto acid analogs, to reduce risk for ESKD/death (1A) and improve quality of life (2C).

- a low protein diet providing 0.55 to 0.60 g dietary protein/kg body weight/day, OR
- a very-low protein diet providing 0.28 to 0.43 g dietary protein/kg body weight/day with additional keto acid/amino acid analogs to meet protein requirements (0.55 to 0.60 g /kg body weight/day)

Rating: Strong
Conditional

CKD: Dietary Protein Intake, Maintenance Hemodialysis, Non-Diabetic

In adults with CKD 5D on MHD (1C) who are metabolically stable, we recommend prescribing a dietary protein intake of 1.0 -1.2 g /kg body weight per day to maintain a stable nutritional status.

Rating: Fair
Conditional

CKD: Dietary Protein Intake, Peritoneal Dialysis, Non-Diabetic

In adults with CKD 5D on PD (OPINION) who are metabolically stable, we recommend prescribing a dietary protein intake of 1.0 -1.2 g /kg body weight per day to maintain a stable nutritional status.

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Compliance to diets should be monitored frequently during the first year of dietary intervention by dietary interviews (3 is optimal) and urine collection for urea output measures. Then yearly follow-up may be recommended until start of maintenance dialysis.

These diets should be progressively installed to allow a careful dietary counselling and adequate compliance. Although such diets are not associated with wasting in carefully monitored research studies, on a routine basis, attention should be focused on energy intake which may decrease over time and induce wasting.

- [Conditions of Application](#)

Are LPD/VLPD+ ketoanalogues indicated for malnourished CKD patients? This question cannot easily be answered since it may depend on the cause of patient wasting. For example, an acute catabolic state may induce malnutrition even if the nutrient intake is adequate. However, priority should be given to the correction of wasting and protein intake should be increased until the wasting state improves. An LPD/VLPD with KA diet should not be started during a wasting state in CKD patients.

Do LPD and VLPD with ketoanalogues have impact on the nutritional status? In a post-hoc analysis of the MDRD study, the authors compared the randomized groups (LPD versus VLPD with ketoanalogues) for various outcomes related to nutritional status. Overall, the results demonstrate the safety of dietary protein restriction over two to three years in patients with moderate to advanced CKD. However, there were small but significant changes from baseline in some nutritional indices, and minimal differences between the randomized groups in some of these changes. In both LPD and VLPD with ketoanalogues, both protein and energy intake declined. Serum albumin rose, while serum transferrin, body weight, percentage of body fat, arm muscle area and urine creatinine excretion declined. In a longitudinal study looking at body composition, a VLPD with ketoanalogues diet induced a small decline in lean body mass on the average of 1.2 kg, with concomitant increase in fat mass, mainly in the first 3 months; these parameters subsequently stabilized and even improve slightly thereafter. Other short-term studies did not show noticeable effects of LPD and VLPD with ketoanalogues on nutritional parameters. Nevertheless, small anthropometric measurements decline observed in some studies are of concern since, in routine practice, LPD and VLPD with ketoanalogues are used on the long term and because of the adverse effect of protein-energy wasting in patients with end-stage renal disease. This is why physicians who prescribe low-protein diets must regularly monitor patients' protein and energy intake and nutritional status.

In the context of these recommendations, "metabolically stable" indicates absence of any active inflammatory or infectious diseases; no hospitalization within two weeks; absence of poorly controlled diabetes and consumptive diseases such as cancer; absence of antibiotics or immunosuppressive medications; and absence of significant short-term loss of body weight.

Implementation considerations

- Increase the training and number of specialized renal dietitians worldwide who could effectively and safely implement low and very low protein diets;
 - Promote low protein products to simplify dietary counseling and help achieving low protein diet;
 - Be more aggressive with the dietary interventions to improve symptoms when chronic dialysis is not a treatment option or need to be postponed (vascular access maturation, organizing preemptive renal transplant);
 - The need for food information is important to obtain a good compliance to the restricted protein intake. However, therapeutic education can help patients to improve personal motivation, and can even become a personal goal to achieve. Getting more interested in food harvesting, preparation, and cooking may improve quality of life. In addition, postponing initiation of dialysis undoubtedly maintains a better quality of life rather than undergoing chronic dialysis (Neumann et al 2018).
- [Potential Costs Associated with Application](#)

Although costs of MNT sessions and reimbursements vary, MNT sessions are essential for improving outcomes. In international settings where ketoacid analogs are available, the cost of ketoacid analogs should be considered before recommending to a patient.

- [Recommendation Narrative](#)

Protein metabolism in the body is responsible for adequate growth in children and maintenance of body protein mass such as muscle mass in adults. Every day, approximately 250 g of protein are catabolized, leading to protein catabolic products, such as urea and many other known or unidentified compounds. Most of these degradation products are normally cleared by the kidneys and excreted in urine. When kidney function impairs, there will be an accumulation of these by-products into the blood, which will progressively impair organ function (Kalantar-Zadeh et al 2017). This has been clearly identified for compounds such as P-cresylsulphate, indoxyl-sulphate, trimethyl aminoxide, fibroblast-growth factor 23, which are now considered as uremic toxins. Secondly, protein intake is responsible for a major fraction of renal workload, and many experimental and clinical research have confirmed the renal effects of a protein load and a deleterious role of the renal hyperfiltration response associated with protein intake. Therefore, in a situation of nephron reduction, such as CKD, reducing protein intake will reduce hyperfiltration, with an additive effect to those of angiotensin-reducing drugs (Kalantar-Zadeh et al 2017). As a consequence of both actions, reducing uremia on one hand and improving renal hemodynamics on the other hand, a reduction in protein intake may reduce the body poisoning and postpone the need to start maintenance dialysis treatment.

Detailed Justification

Reducing protein intake may impair nutritional status. However, it is a well-known fact that adults in western countries eat too much protein (1.35 g protein/kg/d) as compared with their optimal daily needs, estimated to be 0.8 g protein/kg/day. Further, metabolic balances in healthy adults and CKD patients have confirmed that, provided a sufficient energy intake (e.g., above 30 kcal/kg/d), the protein intake level can be safely decreased to 0.55-0.6 g protein/kg/d. A further reduction in protein intake to 0.3-0.4 g protein/kg/d can be achieved with the addition of pills of keto acid analogs to ensure a sufficient balance of the essential amino acids (EAAs) normally brought by animal proteins, which are essentially absent in these low protein vegan-like diets.

Protein Restriction Alone

In adults with CKD/kidney transplant, thirteen RCTs reported the effect of protein restriction only (no supplementation) on outcomes of interest (Cianciaruso et al, Cianciaruso et al, D'Amico et al 2004, Hansen 2002, Jesudason et al 2013, Kloppenburg et al 2004, Kuhlmann et al 1999, Locatelli et al 1989, Rosman et al 1989, Rosman et al 1985, Rosman et al 1989b, Sanchez et al 2010, Williams et al 1991). Duration of the follow-up in the included studies ranged from 3 months to 48 months.

Survival/Renal Death

Research reports a beneficial effect of protein restriction (0.55-0.6 g/kg/d) on ESRD/death in adults with CKD. In adults with CKD, 5 RCTs reported findings on effect of protein restriction on survival/deaths. Three studies clearly indicated a beneficial effect of moderate restriction in dietary protein on the development of ESRD/death (Hansen 2002, Rosman et al 1985, Locatelli et al 1989). Rosman et al indicated people consuming 0.6 g/kg/d of protein had better survival (55%) compared to patients consuming free protein intake (40%). Hansen et al 2002 also indicated that death or ESRD was significantly lower in low protein intake group (0.6g/kg/d) (10%) compared to usual protein intake (27%). Locatelli et al also indicated that low protein diet (LPD) (0.6 g/kg/d) had fewer events (27/192) compared to usual protein intake (1g/kg/d) (42/188), borderline significant ($p < 0.06$). Whereas, Cianciaruso et al indicated that cumulative incidences of death and dialysis therapy start were unaffected by the diet regimen, and low protein intake group (0.55 g/kg/d) does not seem to confer a survival advantage compared with a moderate protein intake group (0.80 g/kg/d) but may be explained by a relatively small sample size. Pooled together, results from the secondary analysis on the number of events of death/ESRD combined from the three studies indicated a beneficial effect of protein restriction on death/ESRD (OR 0.621 CI: 0.391, 0.985).

Quality of Life

Research reports an improved quality of life of a protein restricted diet in one study. In adults with CKD, one RCT examined the effect of protein restriction on quality of life (Sanchez et al 2010). QoL scores at the end of the study indicated that the protein restricted group had significantly higher scores for general health (MD 4.0, 95% CI: 3.1, 4.86) and physical status (MD 10.0, 95% CI: 9.1, 10.9) compared to the control group (0.6 g/kg/d vs 46.3g protein/day; $p < 0.05$).

There is seldom information on quality of life of patients with CKD undergoing a dietary protein restriction. The need for food information is important to obtain a good compliance to the restricted protein intake. However, therapeutic education can help patients to improve personal motivation, and can even become a personal goal to achieve. Getting more interested to food harvesting, preparation, cooking may improve quality of life. In addition, postponing dialysis start undoubtedly maintains a better quality of life rather than undergoing chronic dialysis (Neumann et al 2018).

Glomerular Filtration Rate

In adults with CKD, 5 RCTs reported on effect of protein-restricted diet on GFR. Results from the all the studies indicated that LPD (0.55-0.6 g/kg body weight) had no significant effect on GFR compared to the control group (0.8 g/kg protein). Hansen et al indicated that at a 6-month follow-up time, there was a comparable and significant decline in GFR in both groups (Hansen et al 2002). However, the difference between groups was insignificant ($p = 0.87$). Sanchez et al indicated that GFR rates decreased by 17.2% in the control group compared to only 6.9% in low protein group (NS between groups) (Sanchez et al 2010). Cianciaruso et al 2009 indicated no effect of diet assignments was noted on eGFR and proteinuria (0.55g/kg/d vs 0.80 g/kg/d). Juesudason et al reported that dietary treatment had no effect on changes in eGFR. Meloni et al 2002 (stage 3) also indicated no effect of protein restriction on GFR decline (0.6 g/kg/d). Decline in GFR was reported by three studies, a pooled analysis of these studies indicated no clear effect of protein restriction without supplementation on GFR (SMD -0.002, CI: -0.192, 0.188).

Electrolyte Levels

In adults with CKD, two RCTs reported mixed results regarding the effect of protein restriction on serum phosphate levels (Cianciaruso et al 2008, Rosman et al 1989). Rosman et al indicated that patients in the protein restriction group had significantly lower S. phosphate levels (used less phosphate binders) (0.4-0.6 vs 0.8 g) ($p < 0.05$). Whereas, Cianciaruso et al 2008, indicated that phosphate levels were similar in the two groups throughout the entire period of follow-up (0.55 g protein/kg/d group vs 0.8 g protein/kg/d).

Dietary Intake

Seven randomized controlled studies (D'Amico et al 1994, Hansen et al 2002, Jesudason et al 2013, Kloppenburg et al 2004, Sanchez et al 2010, Williams et al 1991, Meloni et al 2002) and 1 NRCT (Kuhlmann et al 1999) reported on dietary intake. These studies indicated that protein intake was lower in groups assigned to low-protein diet (0.6 g/kg/d) compared to control or standard groups (0.8-1.3 g/kg/d). Dietary intake was used as a compliance measure in most of the studies. In one study, the average protein intake during the entire duration of follow-up was higher than expected in both the groups (CPD=1.03 ± 0.18, LPD=0.78 ± 0.17 g protein/kg/d) (D'Amico et al 1994). Follow-up of at least 1.5 years indicated that compliance to diet did not change in time in either group. Hansen et al (2002) reported an estimated dietary protein intake at 4 years significantly lower in LPD compared to usual PD group (p=0.005) (Hansen et al 2002). Jesudason et al (2013) showed that the moderate protein intake group increased their protein intake (NS) and standard protein group decreased their protein intake. In the study by Kloppenburg et al the protein intake during the high protein diet was higher than during the regular protein diet. Kuhlmann et al reported that protein intake was not significantly different among the groups. However, total energy intake significantly differed among each other. In the Meloni et al study, patients in the low protein group were maintaining their intake at 0.68 g protein/kg/d level which was significantly lower than the free protein diet group. Phosphate intake was also significantly lower in the LPD group. Sanchez et al showed that protein intake in the LPD group decreased significantly from baseline to end of the study (p<0.05). Energy intake tended to decrease during the study duration in both the groups but it was non-significant. In Williams et al study, compared to control, only dietary protein and phosphate restriction group had significantly lower protein intake level. Finally, Cianciaruso et al reported that the 2 groups (LPD vs MPD) maintained significantly different protein intakes (p<0.05), with a difference between the 2 groups of 0.17 ± 0.05 g/d, which lasted from month 6 until the study end. Dietary intake can be used as a compliance index to the diet.

Nutritional Status

Research findings indicated that protein restriction did not affect serum albumin levels of anthropometrics in adult CKD patients. In adults with CKD, 2 RCTs reported no effect of protein restriction (0.55-0.9 g protein/kg/d) on serum albumin levels compared to control group (0.8-1.3 g protein/kg/d) (Cianciaruso et al 2009, Kloppenburg et al 2004). In adults with CKD, one RCT reported no effect of protein restriction (55-70 g/d) on anthropometrics compared to control group (90-120 g/d) (Jesudason et al 2013).

Blood Pressure

Two RCTs reported no effect of protein-restriction (0.6 g/kg bd wt vs usual) on BP values (Hansen et al 2002, Jesudason et al 2013). Hansen et al reported that BP changes were comparable in the two groups during follow-up period. BP was equally and significantly reduced during the study compared to baseline in both groups. Jesudason et al reported no overall changes in BP for both the groups. However, there was a time-by-treatment interaction (p<0.05) for diastolic BP. Diastolic BP was lower throughout the follow-up period in the moderate protein intake group.

Lipid Profile

Research reported an improvement in serum lipid profile during a low protein diet. Coggins et al intervention diet (Diet J/K) showed significant decreases in total cholesterol, HDL, and LDL between baseline and 6-month follow-up (p<0.05). Diet L reported trends for decreases in total and LDL-C between baseline and 6-month follow-up (p<0.10). Cianciaruso et al 2009 showed a significant decrease in LDL values in the low protein diet group, but not in the moderate protein intake group.

Protein Restriction plus Ketoacid Analogs Supplement

In international settings where ketoacid analogs are available, a very-low protein-controlled diet may be considered. For adults with CKD without diabetes, not on dialysis, with an eGFR below 20ml per minute per 1.73m², a very-low protein diet (VLPD) providing 0.28g to 0.43g protein/kg/d with addition of keto acid analogs to meet protein requirements may be recommended.

In adults with CKD/kidney transplant, fourteen studies reported the effect of protein restriction with KA supplementation on outcomes of interest. One non-randomized controlled trial (NRCT) (Bellizzi et al 2007), and 13 RCTs were included (Coggins et al 1994, Feiten et al 2005, Herselman et al 1995, Jian et al 2009, Junger et al 1987, Klahr et al 1994, Kopple et al 1997, Levey et al 1996, Li et al 2011, Malvy et al 1999, Menon et al 2005, Mircescu et al 2007, Prakash et al 2004).

Survival/Renal Death

Dietary Protein restriction with KA supplementation probably reduces RRT/renal survival in adults with CKD stages 3 to 5. In adults with CKD, 4 RCTs reported mixed effect of protein-restricted diet with KA on renal survival/RRT (Levey et al 1996, Malvy et al 1999, Mircescu et al 2007, Garneata et al 2016). Garneata and Mircescu et al indicated a significantly lower percentage of patients in the VLPD with KA group required RRT initiation throughout the therapeutic intervention. Whereas, Levey and Malvy et al indicated no effect, but Malvy study was underpowered. Pooled analysis of two studies that reported RRT incidence indicated that protein restricted diet with KA has a lower risk ratio for incidence of RRT (RR 0.412, CI: 0.219, 0.773). Levey et al indicated that after controlling for protein intake from food and supplement from the studies evaluated, assignment to the VLPD did not have a significant effect on renal failure/death risk. Malvy et al also indicated no effect of protein restriction +KA on renal survival. Whereas, Mircescu et al indicated a significantly lower percentage of patients in the VLPD+KA group required RRT initiation throughout the therapeutic intervention (4% vs. 27%); and Garneata et al also indicated a delay in dialysis initiation. Both Garneata and Mircescu are newer studies and shorter in duration (12 to 15 months) compared to Levey and Malvy (Levey-2.2 years). When pooled together, there is probably an overall benefit of dietary protein restriction + KA supplementation on RRT/renal survival in CKD stage 3 to 5 patients (RR 0.65, CI 0.49 to 0.85, p<0.001).

eGFR

A VLPD supplemented with keto-analogues (0.28-0.4 g protein/kg/d) could help preserve renal function in stage 3 to 5 CKD patients. One study was conducted in PD patients, and GFR was preserved. In adults with CKD, 1 NRCT (Bellizzi et al 2007) and 4 randomized controlled trial (Klahr et al 1994, Levey et al 1996, Mircescu et al 2007, Prakash et al 2004, Garneata et al 2016) reported on effect of protein-restricted diet+ KAA (0.28 - 0.4g/kg body weight) on estimated GFR. Results from all the 6 studies indicated that VLPD +KAA (0.3-0.4 g/kg body weight) supplementation helped preserve eGFR, whereas, subjects assigned to low protein diet only (0.58-0.68 g/kg protein) did indicate decline in GFR. All studies were conducted in subjects in stages 3 to 5. Pooled analysis for all five studies was not possible to conduct.

Bellizzi reported that GFR significantly decreased in the control group. Garneata et al indicated that the decrease in eGFR was less in keto-supplemented group compared with LPD. Klahr et al indicated that compared with usual protein group, the low-protein group had a more rapid GFR decline in the first four months (p=0.004) but slower decline from the first four months to the end (p=0.009). Among patients with GFR of 13-24 ml/min/1.73m² (study 2), there was a trend for slower GFR decline in the VLPD group when compared with the low-protein group (p=0.07). Levey et al (post-hoc analysis of MDRD) indicated that at a fixed level of protein intake from food only, assignment to a VLPD was associated with a decrease (trend) in the steepness of the mean GFR slope of 1.19 mL/min/yr (p=0.063). Similarly, after controlling for protein intake from food and supplement, assignment to the VLPD did not improve the rate of decline in GFR (p=0.71). Mircescu et al indicated that estimated GFR did not change significantly in patients receiving VLPD+KA but significantly decreased in the LPD group (p<0.05), suggesting renal protection for VLPD+KA. Prakesh et al also indicated that GFR stayed unchanged in the KA supplemented group, however, it significantly decreased in the placebo group (p=0.015). Keto-supplemented diet over the 9-month period helped preserve the GFR.

Electrolyte Levels

Research reports that a VLPD supplemented with keto-analogues (0.28-0.4 g protein/kg/d) could decrease serum phosphate and improve some markers of bone metabolism (calcium, parathyroid hormone). Four randomized controlled studies (stages 4-5) (Rosman et al 1989, Feiten et al 2005, Malvy et al 1999, Mircescu et al 2007) indicated a decrease in serum phosphate levels at the end of intervention among LPD+ KAA groups. One study with MHD patients also demonstrated a decrease in serum phosphate in the LPD +KAA group (Li et al 2011). Feiten et al indicated that serum phosphate did not change in the LPD group but tended to decrease in the VLPD + KA group (within VLPD, $p=0.07$). Serum PTH concentration did not significantly change in the VLPD with KA group; however, it increased significantly in the LPD group ($p=0.01$). Li et al in MHD patients indicated that in the LPD with KA group, no significant changes in serum calcium were observed, however, mean serum phosphate levels significantly fell at the end of the study ($p<0.001$) compared to the NPD group. Mircescu et al in stages 4 and patients indicated that in the VLPD+KAA group a significant increase was seen in serum calcium levels post intervention ($p<0.05$); serum phosphate levels decreased ($p<0.05$); whereas no statistical changes were observed in the LPD group. In Rosman et al study, patients in the low protein diet group showed significantly lower serum phosphate levels and used less phosphate binders ($p<0.05$). In a recent meta-analysis, it was reported that serum phosphate levels were lower in patients supplemented very low protein intake in two randomized studies from china (Jiang et al 2018).

Dietary Intake

Research findings indicate that a VLPD supplemented with keto-analogues (0.28-0.40 g protein/kg/d) can effectively be achieved. Dietary intake can be used as a compliance index to the diet. Five randomized controlled studies and 1 NRCT (4 studies with CKD stage 3-5 patients and 1 with PD patients) reported on dietary intake. These studies indicated that protein intake was lower in groups assigned to low-protein diet or very-low-protein diet groups compared to control or standard groups. Dietary intake was used as a compliance measure in most of the studies.

In Bellizzi (stage 4 and 5), at 6 months, protein intake and salt intake were significantly lower in VLPD than LPD ($p<0.0001$). Feiten et al (stage 4), reported a reduction in protein intake in the VLPD supplemented group; energy intake did not change in both groups during the whole study, and was low (approximately 23 kcal/g/d). Phosphorus intake decreased significantly only in the VLPD with KA group. Calcium intake was low and did not change during the intervention period for both groups. In Herselma et al study, protein intake during intervention was significantly reduced from baseline in both groups. In the study of Jian et al in PD, dietary protein intake between groups LP and HP was different in the 6th and 10th month ($p<0.05$). Kopple et al looked at both protein and energy intake (stage 3, 4), compared to usual protein diet, low-protein diet had significantly lower dietary protein intake in study A ($p\approx 0.001$). Compared to low protein diet, VLPD had significantly lower dietary protein intake in study B ($p\approx 0.001$). Dietary energy intake in low-protein diet was significantly lower in study A ($p\approx 0.001$) compared to usual protein diet, however, there was no significant difference between LPD and VLPD in study B ($p>0.05$). Mircescu et al (Stages 4 and 5) results indicated that compliance with prescribed diet was good throughout the study in both arms.

Nutritional Status

Research reports that a VLPD supplemented with keto-analogues (0.28-0.4 g protein/kg/d) had no significant effect on albumin levels and nutritional status as measured by SGA, and effects on anthropometry were inconclusive. In adults with CKD, 6 RCTs (Feiten et al 2005, Jiang et al 2009, Kopple et al 1997, Mircescu et al 2007, Prakash et al 2004, Garneata et al 2016) and 1 NRCT (Bellizzi et al 2007) reported no effect of protein + ketoanalogues intervention on serum albumin level. No significant effect of protein restriction with ketoanalogues supplementation was noticed in any of these studies on albumin levels. Jian et al and Garneata et al were the only studies that studied effect of protein restriction with ketoanalogues supplementation on SGA and no statistically significant effect was noticed. Both the studies indicated that nutritional status was maintained.

In the study by Kopple et al., (study B, stages 3 and 4), no significant differences in anthropometrics measurements were observed between groups ($p>0.05$). Malvy et al reported that for the patients in the VLPD group, a significant weight loss was observed at the end of the study ($p<0.01$) and lean and FM were reduced in this group at the end of study. Moderate protein group indicated no difference for weight variables. Garneata, in a larger and more recent study, reported no differences throughout the study period in both groups for BMI, MAMC, and TSF.

Blood Pressure

Research reports that the effects of a VLPD supplemented with keto-analogues (0.28-0.40 g protein/kg/d) on blood pressure are inconclusive. In adults with CKD, 1 NRCT (Bellizzi et al 2007) and 2 RCTs (Herselman et al 1995, Mircescu et al 2007) reported mixed effect of a protein-restricted diet (0.3-0.4 g/kg/d) + ketoanalogues supplements on BP. Only one study showed a significant reduction in systolic BP and diastolic BP (Bellizzi et al 2007). In this study, the VLPD had antihypertensive effect in response to the reduction of sodium intake, type of protein intake and ketoanalogue supplements, independent of actual protein intake. The other two studies reported no effect of protein-restricted diet with ketoanalogues on BP (Herselman et al 1995, Mircescu et al 2007).

Lipid Profile

Research indicates that a VLPD supplemented with ketoanalogues (0.28-0.40 g protein/kg/d) could improve serum lipid profile of CKD patients. In adults with CKD, 1 NRCT (Bellizzi et al 2007) and 4 RCTs reported on the effects of a protein-restricted diet (0.3-0.4 g/kg/d) + ketoanalogues on serum lipid profile (Coggins et al 1994, Feiten et al 2005, Malvy et al 1999, Garneata et al 2016). Feiten et al and Malvy et al reported no effect of VLPD + ketoanalogues on serum lipid profile, whereas, Bellizzi et al indicated a decrease in TC and TG only in the VLPD group. Coggins et al indicated a significant decrease in TC, HDL, LDL in the VLPD group. Garneata et al showed that cholesterol levels remained stable during the entire duration of the study however patients were taking statins/fibrates as standard therapy.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on protein restriction for pre-dialysis is based on Grade I/Grade A evidence. The evidence supporting the recommendation on protein restriction for MHD is based on Grade III/Grade C evidence. The evidence supporting the recommendation for PD is based on Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of protein restriction with a VLPD and KAA supplementation on ESRD in adults with CKD 3-5?](#)

[What is the effect of protein restriction with a VLPD and KAA supplementation on lipid profile in adults with CKD 3-5?](#)

[What is the effect of protein restriction with a VLPD and KAA supplementation on blood pressure in adults with CKD 3-5?](#)

[What is the effect of protein restriction with a VLPD and KAA supplementation on eGFR in adults with CKD 3-5?](#)

[What is the effect of protein restriction with a VLPD and KAA supplementation on phosphate levels in adults with CKD 4-5D?](#)

[What is the effect of protein restriction with a VLPD and KAA supplementation on anthropometric measures in adults with CKD 3-4?](#)

[What is the effect of protein restriction with a VLPD and KAA supplementation on albumin levels in adults with CKD 3-5D?](#)

[What is the effect of protein restriction with a VLPD and KAA supplementation on protein intake in adults with CKD 3-5D?](#)

[What is the effect of protein restriction with a VLPD and KAA supplementation on energy and electrolyte mineral intake in adults with CKD 3-5D?](#)

[What is the effect of protein restriction with a low protein diet on survival/mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of protein restriction with a low protein diet on quality of life in adults with CKD 3-5?](#)

[What is the effect of protein restriction with a low protein diet on LDL cholesterol levels in adults with CKD 1-5?](#)

[What is the effect of protein restriction with a low protein diet on blood pressure in adults with CKD 1-5?](#)

[What is the effect of protein restriction with a low protein diet on eGFR in adults with CKD 1-5?](#)

[What is the effect of protein restriction with a low protein diet on phosphate levels in adults with CKD 1-5D?](#)

[What is the effect of protein restriction with a low protein diet on anthropometric measures in adults with CKD 1-3?](#)

[What is the effect of protein restriction with a low protein diet on albumin levels in adults with CKD 4-5D?](#)

[What is the effect of protein restriction with a low protein diet on dietary intake in adults with CKD, non-dialyzed?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Macronutrients: Protein Type (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Macronutrients: Protein Type

In adults with CKD 1-5D, there is insufficient evidence to recommend a particular protein type (plant vs animal) in terms of the effects on nutritional status, calcium or phosphorus levels, or the blood lipid profile (1B).

Rating: Strong
Imperative

CKD: Macronutrients: Protein Type, Post-Transplant

In adults with CKD posttransplantation, there is insufficient evidence to recommend a particular protein type (plant vs animal) in terms of the effects on nutritional status, calcium or phosphorus levels, or the blood lipid profile (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms associated with this recommendation. Compliance to diets should be monitored frequently during the first year of dietary intervention by dietary interviews (3 is optimal) and urine collection for urea output measures. Then yearly follow-up may be recommended until start of maintenance dialysis

- [Conditions of Application](#)

So far, vegetarian protein diet (VPD) have not specifically been studied to test metabolic hypotheses in CKD. Particularly, the fact that phosphorus may be less absorbed during a VPD diet may benefit calcium and phosphate metabolism. This becomes more important since currently processed food contains much added inorganic phosphorus as compared with VPD. The fat content of VPD possesses a healthier profile and may benefit patients in long-term studies. Finally, toxic middle molecules such as P-cresyl sulfate, indoxyl-sulfate and trimethylamine oxide (TMAO), almost exclusively produced from animal source protein, have been reduced by VPD and this hypothesis should be tested in long-term clinical trials in CKD patients. As demonstrated in other subtopics of this guidelines, VPD has shown reduction in acid load, increase in dietary fiber intake, reduction of phosphorus, and body weight. There is increasing interest in the role of VPD in CKD due to the benefits of this dietary pattern on cardiovascular disease risk factors in the general population. However, current evidence from RCTs specifically comparing benefits of VPD vs APD in CKD patients is limited.

Implementation Considerations

- Work with you CKD patients to help them meet their individualized dietary protein intake needs.
- Based on CKD patients preference for animal or plant based protein ensure that they meet their dietary protein needs and that their diets provide adequate essential amino acids.

- [Potential Costs Associated with Application](#)

Although costs of MNT sessions and reimbursements vary, MNT sessions are essential for improving outcomes.

- [Recommendation Narrative](#)

Vegetable protein diets (VPD) may have beneficial effects on health. A recent population-based study suggested soy or soy isoflavones intake significantly reduced the risk of postmenopausal breast cancer (Wada et al 2013). Oxidative stress significantly decreased in postmenopausal women when treated with VPD (soy isoflavones) and in vitro experiments have shown that VPD protects against inflammation in vascular endothelial cells (Jing et al 2016). These findings lead to the development of preventive strategies for human health and disease. For example, the US Food and Drug Administration suggested that the intake of 25 g soy protein daily may prevent the risk of coronary heart disease because of reduced serum lipids and lipoproteins.

In patients with chronic kidney disease (CKD), VPD may have positive biological actions and possibly clinical benefits through a variety of mechanisms. In vitro studies showed that VPD reduce the expression of renin-angiotensin (Frigolet et al 2012). Studies in rodents demonstrated that VPD retard the development and progression of CKD, vs animal protein diets (APD) (Iwasaki et al 1988), presumably through favorable effects on glomerular filtration rate (GFR). In addition, in pre-dialysis CKD patients, Moe et al. reported that a vegetarian diet, although it may contain eggs and dairy products, was associated with a significant reduction in serum phosphate and FGF-23 levels and a decrease in 24-hour urinary phosphorus excretion, despite equivalent energy, protein and phosphorus intake, compared with a meat diet (Moe et al 2011). As a result, it was thought that VPD may be used in helping to reduce phosphorus load and CKD progression in this group of patients.

Detailed Justification

Three randomized controlled trials (HD & PD) and two randomized crossover (Stage 3-4 CKD) trials compared the impact of vegetable-based protein (VPD) vs animal -ased (APD) protein intake on biomarkers and health outcomes in patients with CKD.

Albumin

Protein type did not affect nutritional status as measured by serum albumin. In Soroka et al. serum albumin significantly increased after both VPD and APD, compared to the pre-study diet, but there was no significant difference on serum albumin between VPD and APD (Soroka et al 1998). Fanti et al. found no significant difference between VPD and APD on albumin levels. Tabibi et al., found a significant ($p < 0.05$) increase in serum albumin levels within both groups, but no significant difference found between groups. Finally, Chen et al. found no significant differences in albumin between groups. However, the power to discriminate might have been insufficient due to the small number of patients enrolled. In pooled analysis of four studies, there was no effect of protein type on albumin levels.

Protein Catabolic Rate (PCR)

VPD may be associated with a decrease in PCR after 6 months, but evidence was limited. In Saroka et al., PCR was significantly ($p < 0.05$) lower after 6 month of VPD compared to the pre-study diet, but there were no changes in the APD diet. In a secondary analysis, there was a mean difference (95% CI) of -0.10 (-0.17, -0.03) g/kg/d in PCR values in the VPD vs the APD. This might have been the consequence of a slightly reduced absorption of protein from vegetal source (estimated to be 90% of animal protein).

Pre-Albumin Levels

VPD did not affect serum pre-albumin levels compared to a control group, but evidence was limited. Fanti et al. found no significant difference between VPD and APD on serum albumin or pre-albumin levels after receiving soy protein for 8 weeks, compared to the control group.

Inflammatory Markers (CRP, IL-6, TNF- α)

Protein type did not affect inflammatory markers. Fanti et al. compared the impact of a soy protein vs a milk protein supplement on inflammation. No significant differences were found within or between groups for CRP, IL-6 or TNF- α levels.

Calcium and Phosphorus Levels

There was no effect of protein type on plasma/serum or urinary calcium levels. VPD for 7 days to 6 months did not affect plasma/serum phosphate levels, but did decrease 24-hour urinary phosphate levels by a mean difference of -126.6 (-200.4, -52.7) mg. Soroka et al. found no significant difference between VPD, APD, or pre-study diet on urinary sodium, potassium or calcium; or serum calcium or phosphate. Urinary phosphate was significantly lower after the VPD vs the APD and pre-study diet. In a small randomized crossover trial, Moe et al. demonstrated that plasma phosphate levels were significantly higher in the APD vs the VPD group at day 7 ($p = 0.02$), but there was no difference in urinary phosphorus excretion. There were no differences in plasma calcium or urinary calcium excretion levels between groups. In pooled analysis of these 2 studies, there was no effect of VPD, compared to APD, on serum/plasma phosphate. However, VPD did decrease 24-hour urinary phosphate levels by a mean difference (95% CI) of -126.6 (-200.4, -52.7) mg.

Total, LDL and HDL Cholesterol, Triglyceride Levels

Protein type did not affect lipid profile in MHD, PD or Stage 4 CKD patients. Three studies examined the effect of VPD vs APD on blood lipid panel. Chen et al. compared the impact of a soy protein vs a milk protein supplement on plasma lipids during 12 weeks in MHD patients with and without hyperlipidemia. In patients without hyperlipidemia, no significant differences were found in total cholesterol,

LDL-C, HDL-C and triglycerides levels within or between groups. In hyperlipidemic patients however, soy protein lead to a significant decrease in total cholesterol, LDL cholesterol and triglyceride levels, compared to milk protein, whereas HDL significantly increased. Tabibi et al. compared the impact of a soy protein supplement vs control in PD patients and found no significant impact on total cholesterol, LDL-C, HDL-C and triglycerides levels in the intervention group. Soroka et al found no significant differences after VPD, APD or pre-study diet on total cholesterol, LDL-C and triglycerides TG, in stage 4 CKD patients. HDL-C level was significantly lower after VPD compared to the pre-study diet. In pooled analysis of 3 studies, there was no mean difference in total, LDL or HDL-C levels or triglyceride levels between groups.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on protein type is based on Grade II /Grade B evidence for CKD 1-5D. The evidence supporting for PD is based on Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of protein type on total and LDL cholesterol and triglyceride levels in adults with CKD 4-5D?](#)

[What is the effect of protein type on HDL cholesterol levels in adults with CKD 4-5D?](#)

[What is the effect of protein type on hemoglobin levels in adults with CKD 4?](#)

[What is the effect of protein type on electrolyte levels in adults with CKD 3-4?](#)

[What is the effect of protein type on inflammatory markers in adults with CKD 5D on hemodialysis?](#)

[What is the effect of protein type on pre-albumin levels and PCR in adults with CKD 4-5D?](#)

[What is the effect of protein type on albumin levels in adults with CKD 1-5D?](#)

- [References](#)

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- [Chronic Kidney Disease](#)

- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD:Macronutrients: LC n-3 PUFA Nutritional Supplements for AV Graft and Fistula Patency and Kidney Allograft Survival (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: LC n-3 PUFA Nutritional Supplements for AV Graft and Fistula Patency for Maintenance Hemodialysis

In adults with CKD 5D on maintenance hemodialysis (MHD), we suggest not routinely prescribing fish oil to improve primary patency rates in patients with AV grafts (2B) or fistulas (2A).

Rating: Fair
Imperative

CKD: LC n-3 PUFA Nutritional Supplements for Kidney Allograft Survival

In posttransplantation adults, we suggest not routinely prescribing LC n-3 PUFA to reduce the number of rejection episodes or improve graft survival (2D).

Rating: Weak
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Adverse effects of capsule-based supplementation may lead to gastrointestinal side effects like stomach upset and eructation (though the latter can be masked by different formulations). Theoretical risks like bleeding have not been borne out in clinical trials.

- [Conditions of Application](#)

All treatment should be individualized according to a patient's preferences, health history and co-morbidities as well as clinical judgement.

- [Potential Costs Associated with Application](#)

- LC n-3 PUFA supplementation considerations will differ depending on whether the intervention is diet-based or capsule-based.
- For dietary interventions the goal of supplementation must be clearly defined. If it is to raise blood levels of $\pm$ -linolenic acid then supplementation should focus on soybean, flaxseed, and other oils as well as meat and dairy products. If it is to raise EPA or DHA blood/tissue levels then the primary dietary sources must be sardine, mackerel, salmon and other high-content marine-based foods. Potential limitations to dietary supplementation include their relatively high cost and difficulty in achieving high daily intake. In addition, the source and processing method will influence LC n-3 PUFA foodstuff content. For example, farmed fish typically (but not always) has lower LC n-3 PUFA compared to wild fish, while frying fish could alter the n-3/n-6 ratio which may be of clinical significance (Strobel, et al. 2012).
- Capsule-based supplementation involves a set of different considerations. While dozens of commercial LC n-3 PUFA supplements are available, quality control is often lacking (Kleiner, et al. 2015). This makes precise dosing recommendations difficult. An alternative route is to have the patient obtain supplements via physician prescription (e.g. icosapent ethyl, omega-3 ethyl esters). For either option cost could be an issue. Achieving high dose supplementation will be easier with capsules than through dietary consumption. Adverse effects of capsule-based supplementation may lead to gastrointestinal side effects like stomach upset and eructation (though the latter can be masked by different formulations). Theoretical risks like bleeding have not been borne out in clinical trials.
- LC n-3 PUFA content is listed on the website of the National Institutes of Health (NIH 2018) (<https://ods.od.nih.gov/factsheets/Omega3FattyAcids-HealthProfessional/>).

- [Recommendation Narrative](#)

Long chain omega-3 polyunsaturated fatty acids (LC n-3 PUFA) include eicosapentaenoic (EPA) and docosapentaenoic and docohexaenoic acids (DHA), both of which are obtained primarily from dietary sources like cold-water fish (i.e. fish oil), or linoleic acid, which is derived from flaxseed or certain other vegetable oils. In recent decades LC n-3 PUFA have demonstrated protean biologic effects that mediate eicosanoid production, cell membrane physiology, signal transduction, metabolism, apoptosis, oxidation, and inflammation. Accordingly, they have been tested in a variety of medical conditions. Of particular interest has been their putative effects on cardiac membrane stabilization leading to possible reduction of malignant arrhythmias and sudden cardiac death. Patients with CKD have been documented to have some of the lowest [IA1] [FAN2] blood levels of LC n-3 PUFA on record¹⁷⁸ thus making them potentially very suitable candidates for supplementation interventions. In fact, LC n-3 PUFA supplementation has been studied as possible therapy for a number of conditions commonly observed in patients with CKD including dyslipidemia, hemodialysis access failure, cardiovascular disease and death, as well as for their immunomodulatory effects in patients with kidney allografts.

Detailed Justification

Hemodialysis Access

Previous studies have suggested that LC n-3 PUFA, in particular those derived from fish oil, have anti-proliferative, anti-oxidant, and vasodilatory effects. This was the impetus for the four RCTs that examined whether LC n-3 PUFA supplementation could improve patency of arteriovenous grafts (AVG) or fistulas (AVF) in patients on MHD. Of the three RCTs (Lok et al 2012, Bowden et al 2007, Schmitz et al 2002) studying AV graft survival, the two smallest (using 0.96-1.76 g/d EPA and 0.6-0.96 g/d DHA) had mixed results with one showing no benefit at six months (n=29)(Bowden, et al. 2007) and the other (n=24) reporting higher primary patency rates compared to placebo group at 1 year (p<0.03) (Schmitz et al 2002). The third and much larger trial (n=201) noted a borderline statistically significant improvement in the loss of native patency at one year 0.78 [0.60-1.03](p=0.064) after providing 1.6 g/d EPA and 0.8g/d DHA (Lok, et al. 2012). While the overall results are not clearly positive they do suggest at a possible beneficial effect. However, by far the largest study in this field (n=567), which examined patency rates in new AVF at 12 months (Irish et al 2017), reported that fish oil 4 g/d (1.84 g/d EPA and 1.52 g/d DHA) had no benefit. When data was pooled from the three trials studying patency rates no benefit was observed heterogeneity of results was high ($I^2=72.19\%$).

Rejection Episodes and Graft Survival in Kidney Allografts

While LC n-3 PUFA have been reported to mediate the immunologic response they have not yet demonstrated any benefits on kidney allografts. Two RCTs (Maachi, et al. 1995; Bennett, et al. 1995) with differing study interventions (2.4 g/d EPA + DHA for one year, 9 vs 18 g/d EPA for 26 weeks) found no benefit on rejection episodes or a relationship between supplementation dose and rejection episodes (Bennett, et al. 1995). Supplementation using approximately 2.5 g/d EPA+DHA also did not influence graft survival (Maachi, et al. 1995; Berthouix, et al. 1992).

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on is based on Grade I to III /Grade A, B, D based on different outcomes and populations.

The clinical impact of LC n-3 PUFA supplementation in patients with CKD was challenging to assess due to short study durations, modest sample sizes, and broad heterogeneity in the composition of the supplements and the dosing strategies. Furthermore, baseline LC n-3 PUFA levels (either in blood or tissues) were not typically used to target populations that would most benefit. This is an important but often overlooked point because the putative benefits of LC n-3 PUFA supplementation may be inversely related to baseline blood or tissue concentrations.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of omega 3 fatty acid supplementation on hemodialysis access in adults with CKD 5D on hemodialysis?](#)

[What is the effect of omega 3 fatty acid supplementation on rejection episodes and graft survival in adults with CKD who are non-dialyzed and post-transplant?](#)

- [References](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Macronutrients: Oral, Enteral and Intradialytic Parenteral Nutrition Supplementation (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Oral Protein-Energy Supplementation for CKD 3-5D

In adults with CKD 3-5D at risk of or with protein-energy wasting, we suggest a minimum of a 3-month trial of oral nutritional supplements to improve nutritional status if dietary counselling alone does not achieve sufficient energy and protein intake to meet nutritional requirements

(2D).

Rating: Weak
Conditional

CKD: Oral Protein-Energy Supplementation for Post-Transplant

In adults with CKD posttransplantation at risk of or with protein-energy wasting, it is reasonable to consider a minimum of a 3-month trial of oral nutritional supplements to improve nutritional status if dietary counselling alone does not achieve sufficient energy and protein intake to meet nutritional requirements (OPINION).

Rating: Consensus
Conditional

CKD: Enteral Nutrition Supplementation

In adults with CKD 1-5D, with chronically inadequate intake and whose protein and energy requirements cannot be attained by dietary counselling and oral nutritional supplements, it is reasonable to consider a trial of enteral tube feeding (OPINION).

Rating: Consensus
Conditional

CKD: Total Parenteral Nutrition (TPN) and Intradialytic Parenteral Nutrition (IDPN) Protein-Energy Supplementation

In adults with CKD with protein-energy wasting, we suggest a trial of total parenteral nutrition (TPN) for CKD 1-5 patients and IDPN for CKD 5D on MHD patients, to improve and maintain nutritional status if nutritional requirements cannot be met with existing oral and enteral intake (2C).

Rating: Weak
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Energy-dense and low-electrolyte, renal-specific oral nutritional supplements (ONS) may be necessary to increase protein and energy intake and avoid fluid and electrolyte derangements. Gastrointestinal side effects can influence adherence to ONS and extended periods of monotonous supplementation can lead to flavor and taste fatigue as well as non-adherence to the prescribed ONS. Therefore, regular monitoring and evaluation during the supplementation period is crucial and adjustments to the ONS prescription may be necessary to improve adherence and optimize effectiveness.

- [Conditions of Application](#)

A complete nutritional assessment should be performed prior to considering ONS and should be repeated at regular intervals during the supplementation period.

IDPN therapy does not alter patient's eating behavior, nor does it encourage healthy eating habits. Patients on IDPN may suffer from time-limitation due to MHD frequency and duration. Because IDPN is usually given for 4 hours during dialysis thrice weekly, it may not provide sufficient calories and protein to meet long-term nutritional requirements. TPN is usually administered on a daily basis. The potential of IDPN to meet target protein and energy requirements in MHD patients mainly depends on the actual difference between these targets and spontaneous dietary intakes via ONS or dietary counselling. If the difference can be met by the IDPN regimen, the workgroup felt that IDPN should be considered in conjunction with ONS or dietary counselling.

This evidence review suggested that IDPN offers no further benefit over ONS. It was postulated that markers of nutritional status improved irrespective of the route of nutrient administration as long as dietary protein and energy targets are met (Cano et al 2007). However, a direct comparison between IDPN and ONS was lacking, this would only imply that ONS is equally effective as IDPN when oral intake is possible. Since ONS was included in the intervention arm as well, the inferiority of IDPN over ONS cannot be confirmed.

A recently published a RCT investigating the effect of IDPN therapy on pre-albumin and other biochemical and clinical nutritional markers in malnourished MHD patients (Marsen et al 2017), demonstrated that IDPN therapy increased pre-albumin levels and was superior to nutritional counselling after 16 weeks. This study was not included in this evidence review because the date of publication was beyond the cut-off time for study inclusion. In this study, patients randomized to the intervention group received standardized nutritional counselling plus IDPN three times weekly for 16 weeks. There were no within-group changes and between-group differences at week 16 in other clinical and biochemical nutritional markers (BMI, albumin, transferrin, PCR, phase angle alpha, and SGA).

Implementation Considerations

- ONS should be prescribed two to three times daily and patients should be advised to take ONS preferably 1 hour after meals rather than as a meal replacement in order to maximize benefit.
- Monitored in-center provision of high-protein meals or ONS during MHD may be a useful strategy to increase total protein and energy intake. Many of the perceived negative effects of intradialytic feeding such as postprandial hypotension, aspiration risk, infection control and hygiene, as well as diabetes and phosphorus control can be avoided with careful monitoring
- ONS prescription should take into account patient preference. The acceptability of ONS in terms of appearance, smell, taste, texture, and type of preparation (milkshake type, juice type, pudding type, protein/energy bar, or fortification powder) should be carefully considered.
- Energy-dense and low-electrolyte, renal-specific ONS may be necessary to increase protein and energy intake and avoid fluid and electrolyte derangements.
- Increased risk of infectious complications and the high cost of IDPN are the greatest barriers for regular use of IDPN.
- MHD patients meeting all of the following three criteria may benefit from IDPN therapy: 1) evidence of protein-energy malnutrition and inadequate dietary protein and/or energy intake; 2) inability to administer or tolerate adequate oral nutrition, including food supplements or enteral feeding; and 3) protein and energy requirements can be met when IDPN is used in conjunction with oral intake or enteral feeding.
- IDPN therapy should not be considered as a long-term approach of nutritional support. It should be discontinued and ONS should be attempted as soon as improvements in nutritional status are observed and patients are capable of using oral or enteral route.
- If IDPN therapy in conjunction with oral intake does not achieve the nutritional requirements of the patient, or the gastrointestinal tract is malfunctioned, then total parenteral nutrition (TPN) given on a daily basis should be considered.

Monitoring and Evaluation

Gastrointestinal side effects can influence adherence to ONS (Akpele et al 2004) and extended periods of monotonous supplementation can lead to flavor and taste fatigue as well as non-adherence to the prescribed ONS. Therefore, regular monitoring and evaluation

during the supplementation period is crucial and adjustments to the ONS prescription may be necessary to improve adherence and optimize effectiveness. Nutritional status should be monitored regularly throughout the supplementation period in order to evaluate effectiveness of ONS.

Ongoing monitoring and evaluation of nutritional status during IDPN therapy is necessary. Serum glucose should be closely monitored during and post MHD. In the case of insulin requirement, the use of subcutaneous short-acting insulin analogs should be chosen to avoid post-dialytic hypoglycemia. The ultrafiltration rate should be adjusted to remove the extra fluid provided by IDPN.

- **Potential Costs Associated with Application**

The cost of nutrition supplements should be considered before recommending to a patient.

- **Recommendation Narrative**

Protein-energy wasting (PEW) is common among patients with CKD, especially those undergoing maintenance dialysis therapy, and is associated with increased morbidity and mortality. The etiology of PEW in patients with CKD is complex and multifactorial, and includes reduced energy and protein intake resulting from anorexia and dietary restrictions, inflammation, hypercatabolism, protein losses during dialysis, metabolic acidosis, uremic toxicity, and the presence of comorbid conditions. As a result, patients with CKD may develop an imbalance between dietary intake and nutritional requirements. Indeed, many patients with CKD consume less protein and energy than their recommended intakes even when individualized dietary counselling is provided by a renal dietitian.

When dietary counselling alone proves insufficient to bridge the gap between protein-energy intake and the target requirements in patients with CKD, provision of oral nutritional supplements (ONS) is often the next appropriate step to prevent and treat PEW. Therefore, it is important to establish the effectiveness of ONS on nutritional status, clinical outcomes and quality of life in patients with CKD.

Although feeding through the gastrointestinal route should be the preferred choice of nutritional supplementation, feeding through the parenteral route (i.e. total parenteral nutrition), may be a safe and convenient approach for patients who cannot tolerate oral or enteral administration of nutrients (Ikizler et al 2013). In MHD patients, utilization of the hemodialysis access for TPN provides a significant advantage by eliminating the need for an additional permanent venous catheter placement. Since HD access is routinely utilized for the HD procedure, TPN can be conveniently administered during HD via the dialysis tubing. This type of TPN administration is called "intradialytic parenteral nutrition (IDPN)".

Detailed Justification

This evidence review included fifteen clinical trials, twelve of which were RCTs and three NRCTs. Most of the studies examined the effect of ONS in patients on maintenance hemodialysis (MHD). However, Moretti et al included patients on MHD and peritoneal dialysis (PD) with the results merged; Gonzalez-Espinoza et al and Teixido-Planas et al studied patients on PD only; and Wu et al studied patients with CKD, stages 3-4. No studies were performed in patients with CKD with kidney allografts. Most of the studies examined the effect of oral protein-energy or protein-based ONS using commercial products. However, Allman et al used a glucose-polymer ONS and Wu et al used a non-protein calorie ONS. Four studies used renal-specific protein-energy ONS. A major drawback of the literature was the limited use of a placebo group, though most studies did include a comparator group which was defined as participants not receiving ONS or receiving only nutritional counselling. Study durations ranged from 12 weeks to 13½ months. Seven of the RCTs included participants with some level of malnutrition at baseline. In contrast, five studies did not actively enroll malnourished patients. Of the NRCTs Sezer et al enrolled malnourished patients as defined by serum albumin or weight loss, Cheu et al enrolled patients with hypoalbuminemia, and Scott et al did not actively recruit patients with malnutrition.

Mortality, Hospitalizations, and Quality of Life

One NRCT examined the effect of ONS on mortality in 276 patients on MHD who received for ONS for a low serum albumin versus 194 similar patients who refused ONS or in whom treatment was deemed inappropriate (Cheu et al 2013). No difference in mortality [HR (95% CI); 0.70 (0.36, 1.35)] was noted over a median duration of 13.5 months.

Two RCTs (Wilson et al 2001, Moretti et al 2009) and one NRCT (Cheu et al 2013) evaluated the effect of ONS on hospitalization over a period of 6 to 13.5 months in patients on MHD or PD. A pooled analysis of the two RCTs (Moretti et al 2009, Cheu et al 2013) found no significant difference in odds of hospitalization by group assignment, but a NRCT (Cheu et al 2013) reported a 34% reduction in hospitalization risk [0.66 (0.50, 0.85)] by 12 months in patients receiving ONS compared to controls.

Three studies (two RCTs (Calegari et al 2011, Fouque et al 2008) and one NRCT (Scott et al 2009)) each of three months' duration examined the effect of ONS on quality of life (QOL) measures in patients on MHD. One RCT (Calegari et al 2011) and one NRCT (Scott et al 2009) reported that patients receiving general (Calegari et al 2011) or renal-specific (Scott et al 2009) protein-energy ONS had higher QOL scores in the domains of physical functioning (Calegari et al 2011, Scott et al 2009) and bodily pain (Calegari et al 2011) compared to receiving dietary advice only (Calegari et al 2011) or no supplementation (Scott et al 2009), but another RCT (Fouque et al 2008) reported that renal-specific protein-energy ONS did not influence QOL scores in any domain. A pooled analysis of the two RCTs (Calegari et al 2011, Scott et al 2009) found that ONS did not significantly influence bodily pain, physical functioning, or general health QOL domain scores.

CKD Progression

A RCT (Wu et al 2013) conducted for 24 weeks examined the effect of an energy-based ONS on progression of CKD in 109 patients with CKD 3-4 who were following a low-protein diet. While no difference in serum creatinine or eGFR was observed between ONS and controls, there was a comparative reduction in proteinuria in the ONS arm ($p < 0.05$).

Composite Nutritional Scores & Biochemical Markers of Nutritional Status

A 3-month RCT in 18 patients on MHD examined the effect of a food-based ONS on Subjective Global Assessment (SGA) scores (Calegari et al 2011). Authors describe a significantly greater SGA improvement in patients receiving ONS compared to patients receiving nutritional guidance only. One NRCT found that ONS over a six-month period did not influence the Malnutrition Inflammation Score (MIS) as compared to dietary advice (Sezer et al 2014).

Fifteen studies (twelve RCTs (Calegari et al 2011, Wilson et al 2001, Fouque et al 2008, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Hiroshige et al 2001, Moretti et al 2009, Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005, Wu et al 2013) and three NRCTs (Sezer et al 2014, Cheu et al 2013, Scott et al 2009) examined the effect of ONS on serum albumin in patients with 3-5D. These included eleven in patients on MHD 3 to 13.5 months' duration, one RCT (Moretti et al 2009) in patients on MHD and PD of 6 months' duration, two RCTs (Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005) in patients on PD of 6 months' duration, and one (Wu et al 2013) in patients with CKD 3-4 of 24 weeks' duration. Overall, the literature suggested that protein-energy ONS modestly improved serum albumin levels though the results should be interpreted with caution. A pooled analysis of 11 studies (Sezer et al 2014, Calegari et al 2011, Fouque et al 2008, Allman et al 1990, Bolasco et al 2011, Hung et al 2009, Moretti et al 2009, Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005, Wu et al 2013, Scott et al 2009) that included patients with CKD 3-5D found that ONS modestly improved serum albumin as compared to controls [mean difference (95% CI); 0.121 (0.006, 0.236) g/dL]. However, a subgroup analysis found the effect to be significant only when using protein-energy ONS (Sezer et al 2014, Calegari et al 2011, Fouque et al 2008, Hung et al 2009, Teixido-Planas et al 2005, Scott et al 2009) [mean difference (95% CI); 0.16 (0.08, 0.24)].

g/dl] and not energy (Allman et al 1990, Wu et al 2013) or protein-based (Bolasco et al 2011, Moretti et al 2009, Gonzalez-Espinoza et al 2005) supplements. Heterogeneity of results in the pooled analysis was high ($I^2=68.3\%$, $p<0.001$) so results should be interpreted cautiously.

One RCT (Fouque et al 2008) in 86 patients on MHD reported that ONS did not influence serum pre-albumin levels as compared to dietary advice. Two RCTs of 3-6 months' duration in patients on MHD reported conflicting effects of ONS on total protein, perhaps related to type of ONS (Allman et al 1990, Bolasco et al 2011). The first study of 30 patients reported a positive effect on total protein using an amino acid-based ONS (Bolasco et al 2011) while a second of 21 patients found no effect of a 6-month energy-based ONS intervention (Allman et al 1990). Two studies (an RCT (Allman et al 1990) and an NRCT (Scott et al 2009) in patients on MHD of 3-6 months' duration found no effect of ONS on serum transferrin, either individually or in a pooled analysis.

Anthropometric Measurements

The effect of ONS on anthropometric indices varied in large part according to the type of ONS used, with the greatest effects being seen in one study (Allman et al 1990) that used an energy based ONS.

Body Mass Index (BMI): Seven studies (six RCTs (Calegari et al 2011, Fouque et al 2008, Allman et al 1990, Bolasco et al 2011, Hung et al 2009, Gonzalez-Espinoza et al 2005) and one NRCT (Sezer et al 2014)) evaluated the effect of ONS on BMI over a 3-6-month period. Six of the studies were conducted in patients on MHD (Sezer et al 2014, Calegari et al 2011, Fouque et al 2008, Allman et al 1990, Bolasco et al 2011, Hung et al 2009) and one in patients on PD (Gonzalez-Espinoza et al 2005). A pooled analysis demonstrated no overall effect of ONS on BMI though the study using an energy-based ONS noted a rise in BMI (Allman et al 1990). Overall, the heterogeneity was moderate ($I^2=49.8\%$, $p=0.06$).

Body Weight: Six studies (5 RCTs and 1 NRCT) investigated the effect of ONS on body weight over 3 to 6 months in patients on MHD (Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Scott et al 2009), PD (Teixido-Planas et al 2005), and with CKD 3-4 (Wu et al 2013). Overall, ONS was linked to increased body weight but mainly in patients on MHD consuming an energy-based supplement. However, one RCT in patients on PD that used a protein-based ONS reported increased body weight (Teixido-Planas et al 2005). A pooled analysis of all six studies (Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Teixido-Planas et al 2005, Wu et al 2013, Scott et al 2009) found higher body weight in the ONS group compared to the control arm [mean (95% CI); 2.77 (1.19, 4.36) kg] in patients with CKD 3-5D. However, the difference was mainly driven by energy based ONS in patients on MHD.

Dialysis Target Weight: Four studies (3 RCTs (Hiroshige et al 1998, Calegari et al 2011, Fouque et al 2008) and 1 NRCT (Sezer et al 2014)) in patients on MHD (Sezer et al 2014, Hiroshige et al 1998, Fouque et al 2008), examined the effect of ONS on dialysis target weight over a 3 to 6-month period. Overall, no effect of ONS on target weight was observed, though one NRCT (Sezer et al 2014) reported an increase in target weight using a renal-specific protein-energy ONS (Sezer et al 2014) as did one RCT (Hiroshige et al 2001) using a protein-based ONS. A pooled analysis of three studies (Sezer et al 2014, Calegari et al 2011, Fouque et al 2008) found no overall effect. Hiroshige, et al. reported results in a figure and could not be included in pooled analysis.

Lean Body Mass/Fat Free Mass/Muscle Mass: Seven trials (six RCTs (Calegari et al 2011, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Hiroshige et al 2001, Teixido-Planas et al 2005) and NRCT (Sezer et al 2014)) in patients on MHD (Sezer et al 2014, Calegari et al 2011, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Hiroshige et al 2001) or PD (Teixido-Planas et al 2005) studied the effect of ONS on markers of lean mass over 3 to 6 months. Overall, ONS increased LBM or fat free mass only in patients on MHD who received an energy-based ONS. In patients on MHD, the effect of protein-based ONS on LBM was mixed. In a pooled analysis of 6 studies (Sezer et al 2014, Calegari et al 2011, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Teixido-Planas et al 2005) ONS was associated with a significant increase in lean body mass or fat free mass [mean difference (95% CI); 1.18 (0.16, 2.20) kg] compared to the control arm, but a subgroup analysis found the effect to be significant only in patients on MHD using energy-based ONS.

Body Fat: Seven studies (six RCTs (Calegari et al 2011, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Hung et al 2009, Hiroshige et al 2011) and one NRCT (Sezer et al 2014)) in patients on MHD evaluated the effect of ONS on body fat over a period of 3 to 6 months. A pooled analysis of six studies (Sezer et al 2014, Calegari et al 2011, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Hung et al 2009) reported no overall effect of ONS on body fat mass though subgroup analyses demonstrated that energy (Allman et al 1990) and protein-energy (Sezer et al 2014, Calegari et al 2011, Hung et al 2009) based ONS significantly increased body fat mass compared to controls, but protein-based ONS had no effect.

Skinfold Measurements: Five studies (four RCTs (Calegari et al 2011, Allman et al 1990, Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005) and one NRCT (Sezer et al 2014) in patients with CKD on MHD (Sezer et al 2014, Calegari et al 2011, Allman et al 1990) or PD (Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005) examined the effect of ONS on skinfold measurements over a 3 to 6-month period. A pooled analysis of 4 studies (Sezer et al 2014, Calegari et al 2011, Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005) reported that ONS significantly increased skinfold measurements [mean difference (95% CI); 3.91 (0.93, 6.90) mm] compared to dietary counselling or no supplementation, but this effect was significant only in patients on MHD using energy based ONS.

Arm or Muscle Circumference: Four RCTs in patients on MHD (Calegari et al 2011, Allman et al 1990) or PD (Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005) evaluated the effect of ONS on arm or muscle circumference over a 3 to 6-month period. None of the studies showed any effect.

Dietary Intake

Protein: Ten studies (nine RCTs (Calegari et al 2011, Fouque et al 2008, Allman et al 1990, Bolasco et al 2011, Moretti et al 2009, Wu et al 2013, Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005, Hiroshige et al 2001) and one NRCT (Sezer et al 2014)) examined the effect of ONS on protein intake as estimated by nPCR/nPNA, 24-hour dietary recall or multiple-day food records with study durations of three to six months. Overall, protein-based supplements (AA (Bolasco et al 2011) or BCAA (Hiroshige et al 2011)) increased reported protein intake and nPCR in patients on MHD and PD but energy (Allman et al 1990, Wu et al 2013) or protein-energy supplements did not influence either marker in patients with CKD 3-5D. A pooled analysis of seven studies (Sezer et al 2014, Calegari et al 2011, Fouque et al 2008, Bolasco et al 2011, Moretti et al 2009, Gonzalez-Espinoza et al 2005, Teixido-Planas et al 2005) found that ONS significantly increased nPCR in patients on dialysis [standardized mean difference (95% CI); 0.29 (0.04, 0.53)], suggesting a potentially clinically relevant effect. However a subgroup analysis found the effect to be significant only in persons receiving protein-based (Bolasco et al 2011, Moretti et al 2009, Gonzalez-Espinoza et al 2005) but not protein-energy based ONS (Sezer et al 2014, Calegari et al 2011, Fouque et al 2008, Teixido-Planas et al 2005). Similar results were noted in a pooled analysis of three studies (Fouque et al 2008, Gonzalez-Espinoza et al 2005, Wu et al 2013) examining the effects of ONS on reported protein intake where ONS increased reported protein intake only in one study that supplemented egg albumin protein (Gonzalez-Espinoza et al 2005).

Energy: Six RCTs 231, 232, 235, 237, 239, 242 with study duration of 3 to 6 months examined the effect of ONS on energy intake in patients on MHD, 231, 232, 235, 242 on PD, 237 and with CKD, stages 3-4. 239 Overall ONS raised energy intake though the effect was limited to patients on MHD receiving renal-specific protein-energy ONS. Four out of five studies in patients on dialysis reported that ONS increased energy intake. 231, 232, 235, 237, 242 However, a subgroup analysis found the effect to be significant only for patients on MHD receiving protein-energy ONS, 231, 235 but not receiving protein-237 or energy-based 239 ONS alone. The only study in patients with CKD 3-4 found no improvement in energy intake using a non-protein calorie ONS (Wue et al 2013).

Phosphorus and Calcium: An RCT of 3 monthsâ€™ duration in patients on MHD found no effect on phosphorus or calcium intake (Fouque et al 2008).

Other Biochemical Markers (CPR, anemia indices, electrolyte levels)

Seven studies (six RCTs (Calegari et al 2011, Fouque et al 2008, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Hung et al 2009, Wu et al 2013) and one NRCT (Sezer et al 2014) of 3-6 monthsâ€™ duration in patients on MHD (Calegari et al 2011, Calegari et al 2011, Fouque et al 2008, Bolasco et al 2011, Tomayko et al 2015, Hung et al 2009, Sezer et al 2014) and CKD 3-4 (Wu et al 2013) found no effect of ONS on CRP. Seven studies (five RCTs (Calegari et al 2011, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Gonzalez-Espinoza et al 2005) and two NRCT (Sezer et al 2014, Bolasco et al 2011)) in patients on MHD (Calegari et al 2011, Allman et al 1990, Bolasco et al 2011, Tomayko et al 2015, Sezer et al, Bolasco et al 2011) or PD (Gonzalez-Espinoza et al 2005) examined the effect of ONS on markers of anemia over a 3 to 6-month period. Overall, ONS had no effect on these markers. Five studies (four RCTs (Calegari et al 2011, Tomayko et al 2015, Gonzalez-Espinoza et al 2005, Wu et al 2013) and one NRCT (Scott et al 2009) examined the effect of ONS on serum calcium, phosphate, and potassium levels over 3 to 6 months. Three of the trials were in patients on MHD (Calegari et al 2011, Tomayko et al 2015, Scott et al 2009) one was in patients on PD (Gonzalez-Espinoza et al 2005), and one was in patients with CKD 3-4 (Wu et al 2013). None of the studies found any effect on ONS on these electrolytes. Five studies (four RCTs (Allman et al 1990, Hung et al 2009, Gonzalez-Espinoza et al 2005, Wu et al 2013) and one NRCT (Sezer et al 2014) examined the effect of ONS on plasma lipids over 3 to 6 months.

IDPN

This evidence review encompassed three studies that examined the effects of IDPN on nutritional status and clinical outcomes in MHD patients, including one NRCT (Hiroshige et al 1998) and two RCTs (Cano et al 2007, Toigo et al 1989). In all these studies, participants were malnourished. In Hiroshige, et al, participants in the intervention group received dietary counselling from an RDN and an IDPN infusion of 200 ml 50% dextrose, 200 ml 7.1% EAAs, and 200 ml 20% lipid emulsion, providing 2400 kcal and 42.3 g amino acid for one year. Results were compared to a group receiving dietary counselling only (control group). In Cano, et al, all participants were given an oral nutritional supplements (ONS) providing 25 g/protein/day and 500 kcal/day for one year, and the intervention group additionally received IDPN to meet target ranges of 30 to 35 kcal/day and 1.2 g/protein/kg/day; and included a standard lipid emulsion of 50% glucose, 50% non-protein energy supply, and a standard amino acid solution. In Toigo, et al, participants in the intervention group were given (EAAs) IV formula for 6 months. Results were compared to participants in the intervention group where they received an isonitrogenous standard formula containing both non-essential amino acids (NEAAs) and EAAs for 6 months. Both groups simultaneously received 500ml of 10% glucose. Participants were followed up for an additional 6 months.

Mortality and Hospitalization

Only one study examined and found no effect of IDPN on mortality and hospitalization. In Cano, et al, statistical comparisons were not provided but the authors described no significant differences in mortality or hospitalization events between ONS only and IDPN with ONS groups.

Anthropometric Measurements

The three studies examined the effect of IDPN therapy on anthropometric measurements in malnourished MHD patients (Hiroshige et al 1998, Cano et al 2007, Toigo et al 1989). The findings from these studies indicated that IDPN, in combination with dietary counselling (Hiroshige et al 1998) or ONS (Cano et al 2007), increased BMI (Hiroshige et al 1998, Cano et al 2007), dry body weight (Hiroshige et al 1998), skinfold measurements (Hiroshige et al 1998), and MAMC (Hiroshige et al 1998) compared to dietary counselling only. However, similar improvement in BMI was observed when adequate and comparable protein and energy were given to patients receiving ONS only (Cano et al 2007). Compared to a standard IDPN formulation of both EAAs and NEAAs, an IDPN formulation with EAAs did not affect % desirable body weight, skinfold measurements, and AMA (Toigo et al 1989).

Laboratory Markers of Nutritional Status (albumin, pre-albumin, transferrin, and nPCR)

Three studies (Hiroshige et al 1998, Cano et al 2007, Toigo et al 1989) examined the effect of IDPN on laboratory markers of nutritional status in malnourished MHD patients. The results from these studies concluded that IDPN in conjunction with dietary counselling or ONS increased albumin (Hiroshige et al 1998, Cano et al 2007), pre-albumin (Cano et al 2007) or transferrin levels (Hiroshige et al 1998), but similar improvements in albumin and pre-albumin levels were observed when adequate and comparable protein and energy were provided to patients receiving ONS only (Cano et al 2007). Compared to a standard IDPN formulation of both EAAs and NEAAs, an IDPN formulation with EAAs only did not affect albumin and transferrin levels (Toigo et al 1989).

Other Laboratory Markers (Inflammation (CRP); Hemoglobin, Lipid Profile)

One study evaluated and found no effect of IDPN on inflammation in malnourished hemodialysis patients. Cano, et al. reported no change in CRP levels in either ONS only or IDPN+ONS groups, although data was not provided. One study examined and found no effect of IDPN therapy with EAAs only vs standard IDPN formulation with both EAAs and NEAAs on hemoglobin levels in malnourished MHD patients after 6 months (Toigo et al 1989).

Two studies examined the effect of IDPN on lipid profile (Hiroshige et al 1998, Cano et al 2007). The results from these studies showed that combining IDPN with dietary counselling or ONS did not affect total cholesterol or triglyceride levels.

Dietary Intake (Energy and Protein Intake)

Two studies (Hiroshige et al 1998, Cano et al 2007) examined the effect of IDPN on dietary intake in malnourished MHD patients. The findings from these studies showed inconclusive effects of IDPN on dietary energy and protein intakes.

- [Recommendation Strength Rationale](#)

The evidence supporting these recommendations on Grade III /Grade D evidence and Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of oral nutrition supplementation on hard outcomes in adults with CKD 5D?](#)

[What is the effect of oral nutrition supplementation on lipid profile in adults with CKD 3-5D?](#)

[What is the effect of oral nutrition supplementation on eGFR and creatinine levels in adults with CKD 3-4?](#)

[What is the effect of oral nutrition supplementation on anthropometric measures in adults with CKD 3-5D?](#)

[What is the effect of oral nutrition supplementation on CRP levels in adults with CKD 3-5D?](#)

[What is the effect of oral nutrition supplementation on nutritional status indicators in adults with CKD 3-5D?](#)

[What is the effect of oral nutrition supplementation on protein and energy intake in adults with CKD 3-5D?](#)

[What is the effect of nutritional supplementation via IDPN on mortality and hospitalizations in adults with CKD on hemodialysis?](#)

[What is the effect of nutritional supplementation via IDPN on anthropometrics in adults with CKD on hemodialysis?](#)

[What is the effect of nutritional supplementation via IDPN on CRP levels in adults with CKD on hemodialysis?](#)

[What is the effect of nutritional supplementation via IDPN on hemoglobin levels in adults with CKD on hemodialysis?](#)

[What is the effect of nutritional supplementation via IDPN on triglyceride and total cholesterol levels in adults with CKD on hemodialysis?](#)

[What is the effect of nutritional supplementation via IDPN on nutritional status indicators in adults with CKD on hemodialysis?](#)

- [References](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Macronutrients: Dialysate (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Dialysate Protein-Energy Supplementation

In adults with CKD 5D on PD with protein-energy wasting, we suggest not substituting conventional dextrose dialysate with amino acid dialysate as a general strategy to improve nutritional status, although it is reasonable to consider a trial of amino acid dialysate to improve and maintain nutritional status if nutritional requirements cannot be met with existing oral and enteral intake (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Mild acidosis may occur in some patients, although it is readily treatable. In diabetic patients on peritoneal dialysis (PD) with uncontrolled hyperglycemia, substituting amino acid for glucose in PD solutions may serve as an immediate strategy for glycemic control.

- [Conditions of Application](#)

The recommendation statement is based on two randomized-controlled and one non-randomized crossover trials. The included studies only assessed intermediate nutrition-related outcome measures, including dietary intake (total energy and protein intakes, and oral energy intake); laboratory markers of nutritional status (albumin, pre-albumin, transferrin, and total protein levels); and anthropometry (MAMC, triceps skinfold and fat mass). The effects of substituting amino acid dialysate for conventional dextrose dialysate on patient survival, hospitalization, other clinical outcomes and quality of life have not been evaluated. The long-term effect of IPAA therapy remains unclear.

IPAA supplementation decreased bicarbonate levels, and mild acidosis may occur in some patients, although it is readily treatable. In diabetic patients on PD with uncontrolled hyperglycemia, substituting amino acid for glucose in PD solutions may serve as an immediate strategy for glycemic control.

- [Potential Costs Associated with Application](#)

The cost of dialysate protein-energy supplementation should be considered before recommending to a patient.

- [Recommendation Narrative](#)

Protein-energy wasting is common among patients on maintenance PD, and is associated with increased morbidity and mortality (Park et al 2006). Inflammation, acidosis, insulin resistance, insufficient dietary intakes of protein and energy as a result of anorexia, and peritoneal losses of proteins and amino acids contribute to protein-energy wasting (Tjiong et al 2008). Intraperitoneal amino acids (IPAA) supplementation was introduced to compensate for low protein intake and protein losses. Substituting amino acids for glucose in PD solutions should increase the amino acid intake and decrease the net amino acid losses of the patient, thereby increasing the net intake of protein precursors (Kopple et al 1995). IPAA supplementation may also reduce the infused carbohydrate load, thereby reducing the risk of hyperglycemia and the tendency to hypertriglyceridemia (Kopple et al 1995).

Detailed Justification

This evidence review included three studies that examined the effect of IPAA supplementation on nutritional status in malnourished PD patients, including two RCTs (Jones et al 1998, Li et al 2003) and one non-randomized crossover trial (Misra et al 1997). In the two RCTs (Jones et al 1998, Li et al 2003), results were compared between those receiving traditional 1.5% dextrose dialysate vs those who replaced one to two daily exchanges of 1.5% dextrose dialysate with 1.1% amino acid dialysate. Study durations ranged from 3 months (Jones et al 1998) to 3 years (Li et al 2003). In the non-randomized crossover trial, Misra, et al. utilized the same study design in which the participants were assigned to each exposure (amino acid dialysate for one exchange/day or dextrose dialysate only) for 6 months (Misra et al 1997). In all of these studies, PD patients demonstrated some level of malnutrition. In Misra, et al., the majority of patients were presented with hypoalbuminemia; in Li, et al., all patients were malnourished; and in Jones, et al., participants were mildly to moderately malnourished.

Anthropometric Measurements and Laboratory Measures of Nutritional Status

Two RCTs examined the effect of IPAA therapy on anthropometric measurements in malnourished PD patients (Jones et al 1998, Li et al 2003). MAMC, triceps skinfold measurements and FM were maintained at 3 months and 3 years in both IPAA and dextrose dialysate groups. The results from these studies indicated that substituting amino acid dialysate for dextrose dialysate had no effect on anthropometric measurements.

Two RCTs (Jones et al 1998, Li et al 2003) and one non-randomized crossover trial (Misra et al 1997) examined the effect IPAA supplementation on albumin, pre-albumin, and transferrin levels in malnourished PD patients. One randomized-controlled trial

evaluated the effect of IPAA supplementation on total protein level (Jones et al 1998). The findings from these studies concluded that substituting amino acid dialysate for dextrose dialysate in malnourished PD patients did not affect albumin, pre-albumin, transferrin, and total protein levels compared to those receiving dextrose dialysate only.

Electrolyte Levels (Phosphorus/Phosphate, Bicarbonate, and Potassium Levels)

One RCT (Jones et al 1998) and one non-randomized crossover trial (Misra et al 1997) examined the effect of IPAA supplementation on electrolyte levels in malnourished PD patients. The findings from these studies suggested that substituting amino acid dialysate for dextrose dialysate in malnourished PD patients decreased their phosphate and bicarbonate levels, but the effect on potassium levels was unclear.

Jones, et al. showed that serum potassium and phosphate levels decreased significantly in IPAA group and levels were different between groups at 3 months ($p < 0.05$ for each measure) (Jones et al 1998). In contrast, Misra, et al. showed no within-group changes in potassium, phosphate or bicarbonate levels in either IPAA or dextrose dialysate groups (Misra et al 1997). However, when averaged across time, patients receiving IPAA therapy had lower mean phosphate ($p = 0.018$) and bicarbonate levels ($p = 0.002$). In secondary analysis, the IPAA groups in Jones, et al. and Misra, et al. demonstrated a mean difference (95% CI) of -0.50 ($-0.87, -0.13$) mEq/L in potassium and -1.10 ($-1.43, -0.77$) mmol/L in bicarbonate levels respectively when compared to the dextrose dialysate group. In pooled analysis, there was a mean difference (95% CI) of -0.55 ($-0.70, -0.41$) mg/dL in phosphate levels in the IPAA group compared to the dextrose dialysate group.

Dietary Intake (Protein and Energy Intake)

One randomized-controlled trial examined the effect of IPAA supplementation on total and oral protein and energy intakes in malnourished PD patients (Li et al 2003). Compared to baseline intake levels, total protein intake increased in the IPAA group beginning at 6 months and continuing until 3 years ($p = 0.002$ for each measure), but there was no significant difference between IPAA and dextrose dialysate groups. Compared to baseline intake, total energy intake increased in the IPAA group at 6 months ($p < 0.001$) and 3 years ($p = 0.002$), but it decreased in the dextrose dialysate group ($p < 0.001$), though there were no significant differences between groups. Similar results were observed for oral and peritoneal energy intake only. nPNA (nPCR) increased in the IPAA group at 3 years, but decreased in the dextrose dialysate group, and values were significantly different between groups at 3 years ($p < 0.001$).

- [Recommendation Strength Rationale](#)

The evidence supporting this recommendation statement is based on Grade III/Grade C level of evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of nutritional supplementation via dialysate on lipid profile in adults with CKD on peritoneal dialysis?](#)

[What is the effect of nutritional supplementation via dialysate on anthropometric measurements in adults with CKD on peritoneal dialysis?](#)

[What is the effect of nutritional supplementation via dialysate on nutritional status indicators in adults with CKD on peritoneal dialysis?](#)

[What is the effect of nutritional supplementation via dialysate on electrolyte levels in adults with CKD on peritoneal dialysis?](#)

- [References](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Recommendations Summary

CKD: Macronutrients: Protein Amount, Diabetes Mellitus (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Protein Restriction, Non-Dialysis, Diabetic

In adults with CKD 3-5 and who have diabetes, it is reasonable to prescribe, under close supervision, a dietary protein intake of 0.6–0.8 g/kg body weight per day to maintain a stable nutritional status and optimize glycemic control (OPINION).

Rating: Consensus
Conditional

CKD: Dietary Protein Intake, Maintenance Hemodialysis and Peritoneal Dialysis, Diabetic

In adults with CKD 5D and who have diabetes, it is reasonable to prescribe a dietary protein intake of 1.0–1.2 g/kg body weight per day to maintain a stable nutritional status. For patients at risk of hyper and/or hypoglycemia, higher levels of dietary protein intake may need to be considered to maintain glycemic control (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

For all recommendations, nutrition status must be maintained, including adequate caloric intake and maintenance of lean body mass.

- [Conditions of Application](#)

Nutrition education and medical nutrition therapy are key in helping these patients manage their dietary intake. Patients should be monitored routinely to assess whether energy and protein requirements are being met satisfactorily. Changes in nutritional status should be treated and the protein prescription modified accordingly. Maintaining glycemic control is key in this population.

- [Potential Costs Associated with Application](#)

Although costs of medical nutrition therapy sessions and reimbursement vary, medical nutrition therapy sessions are essential for improved outcomes.

- [Recommendation Narrative](#)

Nutrition plays a significant role in the management of individuals with diabetic kidney disease (DKD) in conjunction with pharmacological interventions. The goal is to maintain optimal glycemic control and at the same time maintain adequate protein and energy intake to achieve optimal nutritional status. There are previous guidelines that suggest that 0.8 g/kg body weight/day among those with CKD stages 1-4 and also for CKD stage 5 (KDOQI 2007). However, KDIGO guidelines suggested that more liberalization with protein restriction and recommended that 0.8 g/kg body weight/day be maintained and avoiding levels above 1.3 g/kg/body weight (KDIGO 2012).

Evidence from controlled trials in this non-dialyzed DKD population has been conflicting (Walker et al 1989, Zeller et al 1991, Dullaart et al 1993, Raal et al 1994, Hansen et al 2002, Pijls et al 2002, Meloni et al 2002, Dussol et al 2005). Recent meta-analysis does show small beneficial impact of low protein diet on eGFR decline, however, the heterogeneity was really high (the type of diabetes, stages of CKD, types of interventions, duration, adherence to recommendations) (Robertson et al 2007, Nezu et al 2013).

For the DKD patients receiving dialysis, evidence from observational studies indicated low dietary protein intake is associated with higher hospitalization rates and higher risk of mortality (Kalantar-Zadeh et al 2003, Ravel et al 2013). The KDOQI guideline for dialysis patients suggests dietary protein intake of >1.2 g/kg body weight/day to manage the protein catabolism and losses of protein in dialysate.

Ko et al 2017, conducted an extensive review of existing guidelines and original research in patients with DKD and indicated that dietary protein intake of 0.8 g/kg body weight/day was advised for DKD not on dialysis and dietary protein intake >1.2 g/kg body weight/day was advised for DKD patients on dialysis.

- [Recommendation Strength Rationale](#)

These statements are based on workgroup expertise in the area.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

- [References](#)
- [References not graded in Academy of Nutrition and Dietetics Evidence Analysis Process](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Recommendations Summary

CKD: Macronutrients: Energy Intake (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Statement on Energy Intake in CKD 1-5D

In adults with CKD 1-5D (1C) and who are metabolically stable, we recommend prescribing an energy intake of 25-35 kcal/kg body weight per day based on age, gender, level of physical activity, body composition, weight status goals, CKD stage, and concurrent illness or presence of inflammation to maintain normal nutritional status.

Rating: Fair
Imperative

CKD: Statement on Energy Intake in Post-Transplant

In adults with CKD posttransplantation (OPINION) who are metabolically stable, we recommend prescribing an energy intake of 25-35 kcal/kg body weight per day based on age, gender, level of physical activity, body composition, weight status goals, CKD stage, and concurrent illness or presence of inflammation to maintain normal nutritional status.

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Individualization is the key. There is variation in the actual requirements between individuals. Hence, regular monitoring is important.

- [Conditions of Application](#)

Special Discussions

In the context of these recommendations, “metabolically stable” indicates absence of any active inflammatory or infectious diseases; no hospitalization within two weeks; absence of poorly controlled diabetes and consumptive diseases such as cancer; absence of antibiotics or immunosuppressive medications; and absence of significant short-term loss of body weight.

Implementation Considerations

- The registered dietitian nutritionist (RDN) should consider a number of factors when determining the energy requirements for adults diagnosed with CKD, and these include the patient’s overall health status, CKD diagnosis and associated therapies, level of physical activity, age, gender, weight status, metabolic stressors, and treatment goals.
- Patients should be monitored routinely to assess whether energy requirements are being met satisfactorily. Changes in nutritional status should be treated and the energy prescription modified accordingly.
- Among patients with CKD 5 on maintenance dialysis (hemodialysis or peritoneal dialysis), there are several factors that may influence energy expenditure, beyond the traditional determinants (age, sex, and fat-free mass), such as hyperparathyroidism, hyperglycemia, and chronic inflammation that should be considered into the overall energy prescription.
- Energy needs will be variable depending on the health status of the patient, e.g., acutely ill versus chronically managed and overall health goals, weight maintenance, repletion or loss.
- Energy needs may be different depending on the stage of CKD and its respective treatment (dialysis vs transplantation)
- Ideal body weight (IBW) is the body weight associated with the lowest mortality for a given height, age, sex and frame size and is based on the Metropolitan Life Insurance Height and Weight Tables and many other methods. [Caution: Not generalizable to the CKD population and data-gathering methods were not standardized.]. The IBW can also be estimated as follows: in males as 50.0 kg + 2.3 kg for each inch over 5 ft (each 2.5 cm over 152.4 cm) and in females as 45.5 kg + 2.3 kg for each inch over 5 ft.

- [Potential Costs Associated with Application](#)

There are no direct costs related to implementing these recommendation statements.

- [Recommendation Narrative](#)

Energy metabolism maybe impaired in patients with chronic kidney disease. Hence, maintaining adequate energy intake is necessary to prevent protein-energy wasting.

Evidence from ten controlled trials in pre-dialysis population and from 3 studies in maintenance hemodialysis (MHD) patients indicates that energy intake ranging from 30-35 kcal/kg/d helps maintain neutral nitrogen balance and nutritional status (Bellizi et al 2007; Herselma et al 1996, Mirescu et al 2007; Feiten et al 2005, Prakash et al 2004, Sanchez et al 2010, Williams et al 1991; Kopple et al 1997, Garneata et al 2016, Locatelli et al. 1991, Li et al 2011, Kloppenburg et al 2004, and Kuhlmann et al 1999). However, it is important to remember that many other factors may influence energy expenditure beyond traditional determinants like age, sex, and fat-free mass. Some of these factors include hyperparathyroidism, hyperglycemia, and chronic inflammation that should be considered into the overall energy prescription, health status (e.g., acutely ill versus chronically managed), overall health goals, and weight maintenance-repletion or loss.

There is still paucity of controlled metabolic studies, as well as long-term well-designed outpatient clinical trials studying energy intake in this population. Results from an old metabolic study examining energy requirements in MHD (sample size = 6) indicated that mean energy intake of 35 kcal/kg/d helped maintain neutral nitrogen balance and body composition (Kopple et al 1969). Another similar study in 6 individuals indicated that average intake of 38 kcal was desirable to maintain neutral nitrogen balance (Slomowitz et al 1989). Recent review articles not included in this evidence review, also suggest that energy intake in the range of 30-35 kcal/kg/d is appropriate to maintain maintains neutral nitrogen balance and nutritional status (Kalantar-Zadeh et al 2017, Fouque et al 2007).

- [Recommendation Strength Rationale](#)

The evidence supporting these recommendations is based on Grade III evidence and opinion of expert workgroup.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

- [References](#)

[Garneata L, Stancu A, Dragomir D, Stefan G, Mircescu G. Ketoanalogue-Supplemented Vegetarian Very Low-Protein Diet and CKD Progression. *Journal of the American Society of Nephrology* 2016; 27:2164-76](#)

[Mircescu G, Gârnea?? L, Stancu S, C?pu?? C. Effects of a supplemented hypoproteic diet in chronic kidney disease. *Journal of Renal Nutrition : the official journal of the Council on Renal Nutrition of the National Kidney Foundation* 2007; 17:179-88](#)

[Bellizzi V, Di Iorio B, De Nicola L, Minutolo R, Zamboli P, Trucillo P, Catapano F, Cristofano C, Scalfi L, Conte G. Very low protein diet supplemented with ketoanalogues improves blood pressure control in chronic kidney disease. *Kidney International* 2007; 71:245-51](#)

[Feiten S, Draibe S, Watanabe R, Duenhas M, Baxmann A, Nerbass F, Cuppari L. Short-term effects of a very-low-protein diet supplemented with ketoacids in nondialyzed chronic kidney disease patients. *European Journal of Clinical Nutrition* 2005; 59:129-36](#)

[Herselman M, Albertse E, Lombard C, Swanepoel C, Hough F. Supplemented low-protein diets--are they superior in chronic renal failure?. *South African Medical Journal - Suid-Afrikaanse tydskrif vir geneeskunde* 1995; 85:361-5](#)

[Mircescu G, Gârnea?? L, Stancu S, C?pu?? C. Effects of a supplemented hypoproteic diet in chronic kidney disease. *Journal of Renal Nutrition : the official journal of the Council on Renal Nutrition of the National Kidney Foundation* 2007; 17:179-88](#)

[Prakash S, Pande D, Sharma S, Sharma D, Bal C, Kulkarni H. Randomized, double-blind, placebo-controlled trial to evaluate efficacy of ketodiet in predialytic chronic renal failure. *Journal of Renal Nutrition : the official journal of the Council on Renal Nutrition of the National Kidney Foundation* 2004; 14:89-96](#)

[Li H, Long Q, Shao C, Fan H, Yuan L, Huang B, Gu Y, Lin S, Hao C, Chen J. Effect of short-term low-protein diet supplemented with keto acids on hyperphosphatemia in maintenance hemodialysis patients. *Blood Purification* 2011; 31:33-40](#)

[Kopple J, Levey A, Greene T, Chumlea W, Gassman J, Hollinger D, Maroni B, Merrill D, Scherch L, Schulman G, Wang S, Zimmer G. Effect of dietary protein restriction on nutritional status in the Modification of Diet in Renal Disease Study. *Kidney International* 1997; 52:778-91](#)

[Locatelli F, Alberti D, Graziani G, Bucciatti G, Redaelli B, Giangrande A. Prospective, randomised, multicentre trial of effect of protein restriction on progression of chronic renal insufficiency. Northern Italian Cooperative Study Group. *Lancet \(London, England\)* 1991; 337:1299-304](#)

[Sánchez C, Aranda P, Planells E, Galindo P, Pérez de la Cruz A, Larrubia M, Llopis J. Influence of low-protein dietetic foods consumption on quality of life and levels of B vitamins and homocysteine in patients with chronic renal failure. *Nutricion Hospitalaria* 2010; 25:238-44](#)

[Kloppenburg W, Stegeman C, Hovinga T, Vastenburg G, Vos P, de Jong P, Huisman R. Effect of prescribing a high protein diet and increasing the dose of dialysis on nutrition in stable chronic haemodialysis patients: a randomized, controlled trial. *Nephrology, Dialysis, Transplantation : official publication of the European Dialysis and Transplant Association - European Renal Association* 2004; 19:1212-23](#)

[Kuhlmann M, Schmidt F, Köhler H. High protein/energy vs. standard protein/energy nutritional regimen in the treatment of malnourished hemodialysis patients. *Mineral and Electrolyte Metabolism* 1999; 25:306-10](#)

- [References not graded in Academy of Nutrition and Dietetics Evidence Analysis Process](#)

[Ikizler TA, Wingard RL, Sun M, Harvell J, Parker RA, Hakim RM: Increased energy expenditure in hemodialysis patients. *J Am Soc Nephrol* 7:2646-2653, 1996](#)

Kopple JD, Shinaberger JH, Coburn JW, Sorensen MK, Rubini ME: Evaluating modified protein diets for uremia. *J Am Diet Assoc* 54:481-485, 1969

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Recommendations Summary

CKD: Micronutrients: General Guidance (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Dietary Micronutrient Intake

In adults with CKD 3-5D or posttransplantation, it is reasonable for the registered dietitian nutritionist (RDN) or an international equivalent, to encourage eating a diet that meets the recommended dietary allowance (RDA) for adequate intake for all vitamins and minerals (OPINION)

Rating: Consensus
Conditional

CKD: Micronutrient Assessment and Supplementation

In adults with CKD 3-5D or posttransplantation, it is reasonable for the registered dietitian nutritionist (RDN) or an international equivalent, in close collaboration with a physician or physician assistant, to assess dietary vitamin intake periodically and to consider multivitamin supplementation for individuals with inadequate vitamin intake (OPINION).

Rating: Consensus
Conditional

CKD: Micronutrient Supplementation, Dialysis

In adults with CKD 5D who exhibit inadequate dietary intake for sustained periods of time, it is reasonable to consider supplementation with multivitamins, including all the water-soluble vitamins, and essential trace elements to prevent or treat micronutrient deficiencies (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Supplementation must be considered on an individual basis. Individuals supplemented with micronutrients should be monitored regularly to determine continuing need for supplementation and to avoid toxicity.

- [Conditions of Application](#)

Special Discussions

Certain CKD population might be at higher risk of micronutrient deficiencies, and this must be taken into consideration. For example, pregnant women, gastric bypass surgery patients, patients with anorexia with poor intake, patients with malabsorption conditions, patients following vegetarian diets, and patients taking certain medications may have different micronutrient needs.

Nutrition Focused Physical Examination should be conducted with patients to identify if signs and symptoms of certain vitamin and mineral deficiency are present. These can be used in combination with lab measures to get a complete picture of problem.

If patients with CKD are meeting their recommended intake as assessed by 24-hr recall and have poor nutritional status, then it is likely that they might be at-risk for micronutrient deficiencies and appropriate intervention is required.

Implementation considerations

- Gather patient information on whether they are taking any micronutrient or multivitamin supplements.
 - Assess Dietary intake, including consideration of fortified foods.
 - Suggested vitamin intake should be based on recommendations for the general population (ex: Recommended Dietary Allowance) unless there are specific considerations requiring modification.
 - Supplementation dose should be individualized based on each patient's needs and risk profile.
- [Potential Costs Associated with Application](#)

The cost of nutrition supplements should be considered before recommending these to a patient.

- [Recommendation Narrative](#)

Micronutrients are essential for metabolic function and hence maintaining an adequate intake of these micronutrients is important. For healthy individuals, many countries have established dietary reference intakes (DRIs) for individual micronutrients. However, there is a paucity of guidance regarding appropriate intake for people with chronic diseases. There is some evidence to indicate that patients with chronic kidney disease are likely to be deficient in certain micronutrients. Some of the common reasons for this include insufficient dietary intake, dietary prescription may limit vitamin-rich foods (particularly water-soluble vitamins), dialysis procedure may contribute to micronutrient loss, improper absorption of vitamins, and certain medications and illness. Due to these concerns, there is a trend for routinely prescribing multivitamin supplements. Findings from the DOPPS study indicate that more than 70% of MHD patients in United States take supplements. However, there is insufficient evidence to indicate whether micronutrients or multivitamin supplementation is beneficial or detrimental in this population.

At present there is a paucity of good-quality evidence to either support or oppose routine supplementation on micronutrients, including multivitamins. There is some evidence to state that that patients with CKD might be deficient in thiamine (Frank et al 2000, Hung et al 2001, Ihara et al 1999), riboflavin (Porrini et al 1989), vitamin B-6 (Corken et al 2011, Kalantar-Zadeh et al 2003, Kopple et al 1981), vitamin C (Singer et al 2008), Vitamin K (Holden et al 2010) and/or vitamin D (Wolf et al 2007). However, most of the supporting evidence on deficiencies is for the MHD population and not much has been explored in other stages of CKD or for those on peritoneal dialysis or post-transplant.

This SR included a comprehensive search of controlled trials evaluating the effects of micronutrient supplementation (both water- and fat-soluble) in patients with CKD. A total of 80 controlled trials were included in the systematic review (Folic acid alone- 14 trials, folic acid + B vitamins- 13 trials, vitamin E- 8 trials, Vitamin K- 1 trial, vitamin D- 14 trials, vitamin B12- 4 trials, vitamin c- 8 trials, thiamine- 1, Zinc- 10 trials, Selenium- 7 trials). Some of the good quality evidence from these articles led to development of recommendation statements for specific micronutrients (see specific sections).

However, the current evidence in this field has significant limitations. A majority of the included studies in this SR did not report either baseline status of micronutrients examined or dietary intake during the trials. Moreover, the outcomes reported by these studies varied significantly across the studies, making it difficult to synthesize evidence. Also, the dosage of supplementation and duration of intervention varied across studies. Included studies primarily reported the effect of micronutrient supplementation on the serum level of the micronutrient being supplemented. The quality of evidence from these trials ranged from very low quality to moderate quality for a majority of the micronutrients. Due to these significant limitations, it is very difficult to provide recommendations regarding the exact levels of supplementation or routine supplementation for all patients with CKD. On the other hand, there is some evidence to support that there might be some individuals who are at higher risk of certain micronutrient deficiencies. Taking all these issues into consideration, the expert panel felt that it was important to draft expert opinion-based recommendations statements to guide practitioners and to emphasize the need for individualization of micronutrient use.

In recent years, there have been a few systematic or narrative reviews on the topic of micronutrient supplementation in patients with CKD. The findings from these SRs are in line with findings from the current SR. Tucker et al (2015), in a detailed review of micronutrients in patients on MHD, states that there is insufficient evidence to support routine supplementation and instead supplementation should be individualized and based on clinical judgement. Similarly, Jankowska et al (2017) and Kosmadakis et al 2014, also state that there is insufficient evidence to support or oppose supplementation and more good quality trials are needed to help clarify evidence in this area.

- [Recommendation Strength Rationale](#)

The recommendations regarding general guidance for micronutrient supplementation are based on Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

- [References](#)
- [References not graded in Academy of Nutrition and Dietetics Evidence Analysis Process](#)

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Holden RM, Morton AR, Garland JS, Pavlov A, Day AG, Booth SL. Vitamins K and D status in stages 3-5 chronic kidney disease. *Clin J Am Soc Nephrol*. 2010;5(4):590-7. PMID: 20167683

Hung SC, Hung SH, Tarng DC, Yang WC, Chen TW, Huang TP. Thiamine deficiency and unexplained encephalopathy in hemodialysis and peritoneal dialysis patients. *Am J Kidney Dis*. 2001;38(5):941-7. PMID: 11684545

Ihara M, Ito T, Yanagihara C, Nishimura Y. Wernicke's encephalopathy associated with hemodialysis: report of two cases and review of the literature. *Clin Neurol Neurosurg*. 1999;101(2):118-21. PMID: 10467908

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Kalantar-Zadeh K, Kopple JD. Trace elements and vitamins in maintenance dialysis patients. *Adv Ren Replace Ther*. 2003;10(3):170-82. PMID: 14708071.

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Tucker BM, Safadi S, Friedman AN. Is routine multivitamin supplementation necessary in US chronic adult hemodialysis patients? A systematic review. *J Ren Nutr*. 2015;25(3):257-64. PMID: 25446839

Wolf M, et al. Vitamin D levels and early mortality among incident hemodialysis patients. *Kidney Int*. 2007;72(8):1004-13. PMID: 17687259

- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Micronutrients: Folic Acid and Vitamin B12 Supplementation (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Folic Acid Supplementation for Hyperhomocysteinemia

In adults with CKD 3-5D or posttransplantation who have hyperhomocysteinemia associated with kidney disease, we recommend not to routinely supplementing folate with or without B-complex since there is no evidence demonstrating reduction in adverse cardiovascular outcomes (1A).

Rating: Strong
Conditional

CKD: Folic Acid Supplementation for Folic Acid Deficiency and Insufficiency, CKD 1-5 and Dialysis

In adults with CKD 1-5D, we suggest prescribing folate, Vitamin B12 and/or B-complex supplement to correct for folate or Vitamin B12 deficiency/insufficiency, based on clinical signs and symptoms (B2).

Rating: Fair
Conditional

CKD: Folic Acid Supplementation for Folic Acid Deficiency and Insufficiency, CKD Post-Transplant

In adults with CKD posttransplantation, we suggest prescribing folate, Vitamin B12 and/or B-complex supplement to correct for folate or Vitamin B12 deficiency/insufficiency based on clinical signs and symptoms (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Excessive folate intake inhibits zinc absorption in the gut by forming a complex with zinc in the intestinal lumen. High intake of folic acid may mask signs of pernicious anemia leading to undetected progression of neurological disease.

- [Conditions of Application](#)

Special Discussions

Folate status is most often assessed through measurement of folate levels in the plasma, serum, or red blood cells. Serum or plasma folate levels reflect recent dietary intake, so deficiency must be diagnosed by repeated measures of serum or plasma folate. In contrast, RBC folate levels are more reflective of folate tissue status than serum folate and represent vitamin status at the time the RBC was synthesized (i.e. longer-term folate status). Usually, RBC folate concentrations diminish after about 4 months of low folate intake reflecting the 120-day life span of RBC in healthy individuals. In patients with CKD such concentrations often decrease more rapidly reflecting the shorter RBC life span in CKD. Excessive folate intake inhibits zinc absorption in the gut by forming a complex with zinc in the intestinal lumen.

High intake of folic acid may mask signs of pernicious anemia leading to undetected progression of neurological disease. Based on the 2015 USRDS annual report, more than 2/3 (38.9%) of the patients who are on dialysis are 65 years or older. Older people have a higher risk of impaired gastrointestinal function. Since absorption of vitamin B12 is dependent on Intrinsic Factor and normal gut function and since the latter is often at least partially impaired in older individuals, assessment of serum vitamin B12 may be necessary if folate supplementation is considered.

Serum homocysteine levels, vitamin B12 and folate levels monitoring may be considered for patients who take certain medications such as methotrexate, nitrous oxide, 6-azaridine, phenytoin, carbamazepine, oral contraceptives, and excessive alcohol intake that can interfere with folate absorption.

Implementation Considerations

- Vitamin B deficiencies may be identified by clinical signs and symptoms. Assessment of serum vitamin B12 should be considered if folate supplementation is administered.
- High folic acid intake may mask signs of pernicious anemia and undetected progression of neurological disease, and thus levels of folate and vitamin B12 should be monitored if folate is being supplemented.
- Suggested vitamin intake should be based on recommendations for the general population (ex: Recommended Dietary Allowance) unless there are specific considerations requiring modification.
- Individualization of therapy, including supplementation dosage, is essential to the management of any comorbid condition.
- Individualization should include patient age since adults over 50 years may have increased needs due to the prevalence of atrophic gastritis in this population.

Monitoring and Evaluation

Serum/plasma/RBC folate level, serum vitamin B12 should be assessed as appropriate

- [Potential Costs Associated with Application](#)

The cost of nutrition supplements should be considered before recommending them to a patient.

- [Recommendation Narrative](#)

Folic acid is involved in the synthesis of several amino acids, including serine, glycine, methionine, and histidine. Folic acid can be provided by dietary sources as well as over-the-counter nutritional supplements. Over-the-counter supplements come in various forms, such as folic acid, methyl folate (also known as L-methyl folate, L-5-methyl folate or MTHF), and folinic acid, among others. Folic acid's primary mechanism of action is its role as a one-carbon donor. Folic acid is reduced to methyl folate which helps transfer single methyl groups in various metabolic reactions in the body. Folic acid also plays a role in the functioning of the nervous system, in DNA synthesis and in cell division. Food sources rich in folic acid include green leafy vegetables, fruits, yeast, and liver. Even though intake of food naturally rich in folic acid is limited in patients with CKD due to their high potassium content, folic acid deficiency among this patient population seems to be rare. This is especially true since 1996, when folic acid fortification of enriched cereal grain products was mandated in the United States and Canada. Because folate, vitamin B12 and vitamin B6 assist in the conversion of homocysteine to methionine (and therefore reduce serum homocysteine levels), they have received considerable attention as a putative treatment for cardiovascular disease in patients with CKD.

Mortality, Cardiovascular Outcomes and Vascular Function

Four RCTs did not show any effect of folic acid when taken with vitamins B6 and B12 on hard outcomes, including all-cause mortality and/or cardiovascular events in patients with stage 5 CKD, on MHD or PD, and post-transplant (Jamison et al 2007, Bostom et al 2011, Heinz et al 2012, Mann et al 2008). Folic acid and other B-vitamin supplementation ranged from 2.5-40 mg/d folic acid, 1.4-100 mg/d B6, 150 µg/week- 2 mg/d B12 for a duration of 2-5 years.

Folic acid (alone) intake of 1 to 5 mg/day for 4 to 40 weeks showed no effect on flow mediated dilation (van Guldener et al 1998 and Thambyrajah et al 2000). Additionally, folic acid supplementation did not alter the risk of cardiovascular outcomes in four RCTs (Righetti et al 2013, Wrone et al 2004, Vianna et al 2007 and Zoungas et al 2006). The 4 RCTs included patients with CKD, stage 5 non-dialyzed and on PD and MHD. The folic acid supplementation dose ranged from 1-15 mg/day and supplementation duration ranged from 1-3.6 years in these studies.

Supplementation with folic acid in combination with other B-vitamins did not improve total cholesterol levels, intima media thickness (IMT) or BP in MHD patients. Doses ranged from 5 mg to 15 mg folic acid and a B-complex vitamin for 3 to 6 months (Chang et al 2007 and Tungkasereerak et al 2006).

CKD Progression

One RCT examined the effect of folic acid supplementation on CKD progression (Xu et al 2016). In a sub-study of a larger primary stroke prevention trial including 15, 104 participants with CKD Stage 3 diagnosed with hypertension and taking the angiotensin converting enzyme inhibitor enalapril being randomized to receive 0.8 mg/day of folic acid or placebo for a median of 4.4 years. Compared to the group receiving enalapril and placebo, the enalapril + folic acid group significantly reduced the adjusted risk of CKD progression (Hazard Ratio (95% CI): 0.45 (0.27, 0.76); p=0.003), which was the sub-study's primary outcome. The limitation of this study was that a placebo alone group (without enalapril) was not included.

Two other RCTs showed no effect of supplementation with folic acid with vitamins B6 and B12 on the risk of dialysis initiation/kidney failure in participants with stages 3-5 chronic kidney disease and those post-transplant (Jamison et al 2007 and Bostom et al 2011).

Serum Homocysteine Levels

Fourteen studies examined the effect of folic acid supplementation alone on plasma homocysteine levels (Alvares et al 2007, Bernasconi et al 2006, Nafar et al 2009, Sunder-Plassmann et al 2000, Thambyrajah et al 2000, Vianna et al 2007, Wrone et al 2004, Xu et al 2016, Zoungas et al 2006, De Vecchi et al 2001, McGregor et al 2000, Ossareh et al 2009, Righetti et al 2013 and van Guldener et al 1998). Participants included were those with CKD, non-dialyzed (4 studies), on MHD (10 studies) and PD (4 studies), and post-transplant (1 study). In the ten RCTs, folic acid supplements ranged from 0.8-60 mg/day and duration varied from 4 weeks to 4.4 years in patients of various stages of CKD. All but one study (Tamadon et al 2011) concluded that folic acid supplementation significantly decreased homocysteine levels.

Thirteen RCTs examined the effect of supplementation with folate and other B-vitamins on homocysteine levels (Chang et al 2007, Azadibakhsh et al 2009, Chiu et al, Trimarchi et al 2002, Tamadon et al 2011, Bostom et al 1995, Bostom et al 2011, Jamison et al 2007, Heinz et al 2012, Mann et al 2008, Gonin et al 2003, Nakhoul et al 2004 and Tungkasereerak et al 2006). Serum homocysteine level was a primary outcome of interest in eight studies. Twelve out of 13 studies found that folic acid with other B vitamin supplementation decreased homocysteine levels in participants with CKD Stages 3-5, on MHD, PD and post-transplant. Supplementation doses in these studies ranged from 2.5-40 mg/d folic acid (one study utilized 3 mg IV folinic acid/week), 1 µg/d oral to 1000 mg/week IV B12, and 1.4-100 mg B6 and supplementation duration ranged from 8 weeks to 5 years.

CRP and IL-6 Levels

Daily oral folic acid (5 mg) with a B-complex vitamin for 3 months was associated with a decrease in CRP, but not IL-6, levels in a RCT that included 121 patients on MHD (Chang et al 2007).

Folic acid and B12 Levels

Six RCTs reported that supplementation of folic acid alone increased serum folic acid levels in participants with stages 3-5 CKD and those on MHD and PD (Sunder-Plassmann et al, Thambyrajah et al, Xu et al, Zoungas et al, De Vecchi et al and McGregor et al). When folic acid with vitamins B6 and B12 were provided, it is worth noting that serum folic acid level increased with a daily intake of 5 mg for 3 months, or a daily intake of 2.5 mg for a longer time frame. When folic acid with B6 and B12 was provided (Mann et al) to patients on MHD, serum folic acid significantly increased with an intake of 2.5 mg after 2 years of supplementation as compared to the control group. In Chiu, et al., a supplementation of 3 mg folinic acid weekly via IV for 3 months did not result in a significant increase in serum folic acid levels in participants with stage 3-5 CKD or were on MHD and PD.

Of the ten studies that examined the effect of supplementation of folic acid with B-complex, nine found a significant increase in serum folic acid levels (Chang et al 2007, Azadibakhsh et al 2009, Trimarchi et al 2002, Bostom et al 1995, Jamison et al 2007; Heinz et al 2012, Mann et al 2008, Gonin et al 2003 and Tungkasereerak et al 2006). Doses ranged from 2.5-60 mg folic acid and study duration ranged from 4 weeks to 5 years.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on folic acid and vitamin B12 supplementation is based on Grades I and II/Grades A and B evidence. The recommendation supplementation for people with CKD, post-transplant was based on Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of folate supplementation on cardiovascular disease outcomes in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of folate supplementation on intima media thickening in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of folate supplementation on endothelium-dependent flow mediated dilation in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of folate supplementation on folate levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of supplementation with folate combined with other B vitamins on all-cause and cardiovascular-related mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of supplementation with folate combined with other B vitamins on cardiovascular disease events in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of supplementation with folate combined with other B vitamins on folate and vitamin B12 levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of folate supplementation on homocysteine levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of supplementation with folate combined with other B vitamins on homocysteine levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of folate supplementation on mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of folate supplementation on end stage renal disease in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of folate supplementation on eGFR in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of supplementation with folate combined with other B vitamins on kidney failure/dialysis initiation in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of supplementation with folate combined with other B vitamins on blood pressure and intima media thickening in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of supplementation with folate combined with other B vitamins on total cholesterol in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of supplementation with folate combined with other B vitamins on inflammatory markers in adults with CKD 1-5D and post-transplant?](#)

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- [References not graded in Academy of Nutrition and Dietetics Evidence Analysis Process](#)

No additional references.

- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Micronutrients: Vitamin C Supplementation (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Vitamin C Supplementation

In adults with CKD 1-5D or posttransplantation who are at risk of Vitamin C deficiency it is reasonable to consider supplementation to meet the recommended intake of at least 90 mg/d for men and 75 mg/d for women (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

Toxicity is a possible concern for excessive vitamin C supplementation.

- [Conditions of Application](#)

Special Discussions

Current nutritional requirements or Recommended Dietary Intake of vitamin C for individuals with CKD stages 1-5D and post-transplant are not known and are based on those from the general population. The prevalence of vitamin C deficiency may vary according to the stage of CKD and dialysis modality. Toxicity is a possible concern for excessive vitamin C supplementation.

The above findings do not however preclude the importance of assessing for vitamin C supplementation or when to discontinuing supplementation. Ongoing monitoring of overall food intake and nutrition status is required to assess for vitamin C deficiency. An individualized approach to evaluation and monitoring of vitamin C status is ideally accomplished by the nephrology care team that includes Nephrologist, Nurse Practitioner, Physician Assistant, and RDN.

Implementation Considerations

- Initiation and cessation of vitamin C supplementation as well as supplementation dose should take into account of the subject's nutritional status, dietary intake, co-morbid conditions and dialysis modality.
- Suggested vitamin intake should be based on recommendations for the general population (ex: Recommended Dietary Allowance) unless there are specific considerations requiring modification.

Monitoring and Evaluation

Higher doses of Vitamin C supplementation (500 mg daily) have been shown to increase serum oxalate levels. Vitamin C is a potent physiologic antioxidant. Lipid metabolism may be affected by vitamin C supplementation and patients receiving vitamin C supplementation should have lipid fractions monitored. Vitamin C also affects immune function, and carnitine metabolism. Patients with any malabsorption or diseases of an inflammatory nature may be more prone to having lower plasma vitamin C levels than the general population. Therefore, supplementation dose should take into consideration of medical history, co-morbid conditions, and concomitant medications. Measurement of serum oxalate levels may be considered in patients prescribed high doses of vitamin C and/or who are susceptible to calcium oxalate stone formation.

- [Potential Costs Associated with Application](#)

The cost of nutrition supplements should be considered before recommending to a patient.

- [Recommendation Narrative](#)

There are currently a lack of studies identifying daily vitamin C requirements for individuals with CKD, including those on dialysis and post-transplant. Amount of daily intake and optimal serum levels of vitamin C required to maintain nutritional health, reverse deficiency, and to avoid toxicity are unclear. Studies included for this current review evaluated the effect of vitamin C supplementation on nutritional status, inflammation, anthropometrics, micronutrient levels, electrolytes levels, fluid status, serum uric acid levels, lipid levels, morbidity events, quality of life, mortality and hospitalizations. Limited data from a very small number of studies prohibit definitive evidence-based conclusions for all the above surrogate and hard outcomes. Therefore, Committee members suggest that individualized decision-making is the best clinical approach to determine if vitamin C supplementation, or termination of supplementation, is required for adults with CKD stages 1-5, non-dialyzed, on dialysis and post-transplant.

Nine studies examined the effect of vitamin C on nutrition-related outcomes in the CKD population, including five RCTs (Abdollahzad et al 2007, Fumeron et al 2005, Singer et al 2011, Biniiaz et al 2013, Khajehdehi et al 2000), one randomized crossover trial (Zhang et al 2013) and three comparative studies (Canavese et al 2005, De vriese et al 2007, Ono et al 1989) All studies examined MHD patients. Two studies (Canavese et al and Singer et al) also included PD patients and those with eGFR <20mL/min.

QOL (Quality of Life), Mortality, and Hospitalizations

In adults with CKD, one RCT (250 mg oral ascorbic acid 3x/week for 3 months) (Singer et al 2008) and one comparative study (Ono 1989) (500 mg oral vitamin C/day for 2 years) measured the effect of vitamin C supplementation, compared to either a placebo or control, on hard outcomes, including all-cause mortality, Quality of Life or hospitalizations events.

Singer et al reported no changes in symptom, cognitive, or nausea sub-scales of the KDQOL-SF in either the vitamin C supplemented or placebo groups in MHD/PD participants. QOL was the primary outcome of interest. Approximately 40% of participants were vitamin C deficient at baseline. In a comparative study by Ono et al., there were no differences in mortality rates or hospitalization events between vitamin C supplemented and non-supplemented periods in MHD participants. Mortality was a primary outcome measure. Baseline vitamin C status was not reported.

In summary, Vitamin C supplementation did not affect QOL, Mortality or hospitalizations in MHD patients, but evidence was extremely limited. Evidence based recommendations for the use of vitamin C in this patient population for these endpoints could not be provided.

Nutritional Status Parameters: serum Albumin, Pre-albumin, Transferrin, and Protein Nitrogen Appearance

Three studies examined the effect of vitamin C supplementation on nutritional status in MHD participants: one RCT (Fumeron et al 2005), one randomized crossover trial (Zhang et al 2013), and one comparative study (De Vriese et al 2008). . However, nutritional status was not the primary outcome of interest. In Zhang et al, all patients were vitamin C-deficient at baseline, while in DeVriese, et al. 44% of participants were deficient at baseline. In Fumeron et al, vitamin C deficiency status at baseline was unclear. All outcomes were reported as quantitative values but were not compared to a reference standard. Supplementation dosage and duration ranged from 750 mg/week for 2 months to 1500 mg/week for 3 months.

All three studies reported no effect of supplementation on albumin levels, as did pooled analysis of two of the RCTs. Zhang et al. measured the effect of vitamin C supplementation on pre-albumin levels in a randomized crossover trial with MHD participants. While one supplemented group experienced an increase in pre-albumin levels after three months of supplementation with 200 mg vitamin C, pre-albumin levels did not change in the other group after the same intervention. Therefore, the effect of vitamin C supplementation on pre-albumin levels is unclear. Fumeron, et al. supplemented MHD participants with 750 mg/week of vitamin C for 2 months. There were no significant changes in transferrin levels in either group. De Vriese et al measured nPNA (nPCR) in a NRCT and found no effect of vitamin C supplementation on nPNA (nPCR) following supplementation with 360mg/week or 1500mg/ week for 9 months in MHD patients.

CRP Levels

Three studies examined the effect of oral vitamin C supplementation on CRP levels in MHD participants (Fumeron et al 2005, Zhang et al 2013, De Vriese et al 2008) and found no significant effects compared to placebo or control groups, but evidence was limited.

Vitamin C Levels/Deficiency

Four RCTs (Abdollahzad et al 2009, Fumeron et al 2005, Singer et al 2011, Zhang et al 2013) and two comparative studies (De Vriese et al 2008, Ono et al 1989). examined the effect of Vitamin C supplementation in doses ranging from 360-3500 mg/week and duration ranging from 3 months to 2 years. In summary, oral vitamin C supplementation increased serum vitamin C levels in MHD patients and decreased the proportion of participants who were vitamin C deficient/insufficient (cut-offs were 11.44 and 23.0 µmol/L). However, in

pooled analysis of three RCTS, the increase in vitamin C levels may not be clinically significant. The quality of evidence in this regard remains low. Other CKD populations such as non-dialysis CKD 1-5, PD and post-transplant participants remain poorly studied.

These studies did not analyze the effects of vitamin C supplementation on optimal dosing or thresholds for toxicity. The potential for toxicity was acknowledged with dosage ranges maintained at 200-250 mg daily or three times weekly in most studies. The study by Ono that dosed MHD patients with daily 500 mg oral Vitamin C daily for 2 years reported an aggravation of hyperoxalemia (Ono et al 1989). De-Vriese et al had subjects dosed as 360 mg per week for 0-3 months followed by 1500mg/week dosing x 3-6 months and then no supplementations for 6-9 months in MHD patients. This study reported an increase in plasma malondialdehyde (De Vriese et al 2008). Supplementation with vitamin C increased the low levels but there is a potential risk of toxicity that requires monitoring.

Lipid Levels: Total Cholesterol, Triglycerides, LDL, HDL-C, LDL:HDL Ratio

In summary, the results of three trials (Abdollahzad et al 2009, Khajehdehi et al 2001, De Vriese et al 2008), demonstrated that vitamin C supplementation of 125-200 mg/d for 3 months may decrease total cholesterol and LDL cholesterol levels, but there was no effect on triglyceride or HDL cholesterol levels. Vitamin C supplementation of 125-200 mg/day decreased LDL:HDL ratio or prevented the increase seen in the placebo group.

There were several limitations to this evidence including a small number of studies, small sample sizes and low evidence quality. It is important to note that the study by Khajehdehi, et al included supplementing patients with potentially toxic doses of ergocalciferol, 50,000 IU daily x 3 months (Khajehdehi et al 2011). The impact of this amount of vitamin D on study outcome parameters, if any, cannot be ascertained.

Treatment of anemia with vitamin C supplementation was beyond the scope of this guideline.

- [Recommendation Strength Rationale](#)

The recommendation regarding vitamin C supplementation is based on Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of vitamin C supplementation on vitamin C levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin C supplementation on mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin C supplementation on quality of life in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin C supplementation on hospitalizations in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin C supplementation on lipid profile in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin C supplementation on CRP levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin C supplementation on nutritional status indicators in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin C supplementation on Uric Acid levels in adults with CKD 1-5D and post-transplant?](#)

- [References](#)

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- [References not graded in Academy of Nutrition and Dietetics Evidence Analysis Process](#)

No additional references

- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Micronutrients: Vitamin D Supplementation (2020)

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- [Recommendation\(s\)](#)

CKD: Vitamin D Supplementation for Vitamin D Deficiency and Insufficiency, CKD 1-5D

In adults with CKD 1-5D (2C), we suggest prescribing Vitamin D supplementation in the form of cholecalciferol or ergocalciferol to correct 25(OH)D deficiency/insufficiency.

Rating: Weak
Conditional

CKD: Vitamin D Supplementation with Proteinuria

In adults with CKD 1-5 with nephrotic-range proteinuria, it is reasonable to consider supplementation of cholecalciferol, ergocalciferol or other safe and effective 25(OH)D precursors (OPINION).

Rating: Consensus
Conditional

CKD: Vitamin D Supplementation for Vitamin D Deficiency and Insufficiency, CKD Post-Transplant

In adults with CKD posttransplantation, we suggest prescribing Vitamin D supplementation in the form of cholecalciferol or ergocalciferol to correct 25(OH)D deficiency/insufficiency (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There is no recommendation of safe dose of cholecalciferol or ergocalciferol supplementation to prevent for toxicity or adverse effects such as hypercalcemia or hyperphosphatemia in CKD.

- [Conditions of Application](#)

Special Discussions

There are potential benefits of vitamin D supplementation (cholecalciferol or ergocalciferol) in CKD. A systematic review with meta-analysis of observational and randomized studies showed a significant decline in PTH levels with cholecalciferol or ergocalciferol supplementation in patients who are non-dialyzed, on hemodialysis or peritoneal dialysis, and renal transplant recipients (Kandula et al 2011). However, whether such improvements translate into clinically significant outcomes is yet to be determined.

Cross-sectional analysis of Third National Health and Nutrition Examination Survey (NHANES III) showed progressively higher prevalence of albuminuria with decreasing 25(OH)D levels (de Boer et al 2007). In a prospective cohort study vitamin D deficiency was associated with a higher incidence of albuminuria (Damasiewicz et al 2013). There are limited randomized clinical trials investigating the effect of cholecalciferol or calcifediol on proteinuria in CKD and the results are not clear with both positive and null results (Susantitaphong et al 2017, Levin et al 2017).

Implementation Considerations

- The optimal serum 25(OH)D concentration for patients with CKD and the concentration at which patients with CKD are considered deficient/insufficient is not well defined but is generally considered to be the same as in the general population, although there is no absolute consensus about the definition of vitamin D sufficiency. For most experts, vitamin D insufficiency is defined as a serum 25(OH)D level between 20-29 ng/mL, deficiency is considered as 25(OH)D levels of less than 20 ng/mL and sufficiency serum 25(OH)D equal or greater than 30 ng/mL (Holick et al 2011).
 - Both the Kidney Disease Outcomes Quality Initiative (KDOQI) and Kidney Disease Improving Global Outcomes (KDIGO) experts recommend checking and supplementing low serum 25(OH)D levels in CKD and dialysis patients. In the most recent update of the KDIGO guidelines on bone mineral disorder, it is suggested based on low quality evidence that patients with CKD stage 1-5D have 25(OH)D levels measured, and repeated testing should be individualized according to baseline values and interventions. However, there was no clear suggestion on how frequently 25(OH)D levels should be reviewed.
 - With respect to vitamin D supplementation, current guidelines suggest that patients with CKD stages 1-5D and vitamin D insufficiency/deficiency should receive supplementation using the same strategies recommended for the general population. However, even for the general population, the optimal dosage of supplementation varies among the main guidelines. It has been recommended 1000-2000 IU/d of cholecalciferol for vitamin D repletion for the general population. However, KDOQI acknowledges that patients with CKD may require a more aggressive therapeutic plan.
 - There is also a debate regarding which form of vitamin D should be used, ergocalciferol or cholecalciferol. In the general population there appears to be some advantage of using cholecalciferol over ergocalciferol. PMID 22552031 Since in CKD there is no clear evidence about the superiority of chocalciferol, clinicians should use the form commercially available in the context of their clinical practice.
 - The tolerable upper intake levels (UL) proposed by the Institute of Medicine (IOM) for the general population is 4,000 IU/day. (IOM, Dietary Reference Intakes for Calcium and Vitamin D, 2011) There is no recommendation of safe dose of cholecalciferol or ergocalciferol supplementation to prevent for toxicity or adverse effects such as hypercalcemia or hyperphosphatemia in CKD. However, periodically measurement of serum calcium and phosphorus should be considered especially for patients who are on calcium-containing phosphate binders and/or on vitamin D active analogs.
- [Potential Costs Associated with Application](#)

The cost of nutrition supplements should be considered before recommending to a patient.

- [Recommendation Narrative](#)

Vitamin D2 (ergocalciferol) and Vitamin D3 (cholecalciferol) are recognized as a pro-hormones and comprise a group of fat-soluble secosteroids. A unique aspect of vitamin D as a nutrient is that it can be synthesized by the human body through the action of sunlight. These dual sources of vitamin D (diet and sunlight) make it challenging to develop dietary reference intake values (IOM, Dietary Reference Intakes for Calcium and Vitamin D, 2011). The classic actions of vitamin D are the regulation of calcium and phosphorus homeostasis contributing to bone health. More recently, there has been a growing interest in the potential pleiotropic actions of vitamin D on immune, cardiovascular and neurological systems and on antineoplastic activity since extra-renal organs possess the enzymatic capacity to convert 25 (OH)D to 1, 25(OH)₂D (Holick 2007)

Insufficiency/deficiency of vitamin D, assessed by serum concentration of calcidiol [25(OH)D], has been found to be common in the general population and even more prevalent in patients with CKD/ESRD (LaClair et al 2005, Wolf et al 2007, Taskapan et al 2006).

A number of factors or conditions are implicated in suboptimal vitamin D status in patients with CKD, including aging, diabetes mellitus, obesity, reduced sun exposure, loss of urinary/dialysate vitamin D binding protein (DBP), impaired tubular 25(OH) reabsorption and dietary restrictions (Cuppari et al 2008, Caravaca-Fontan et al 2016, Takemoto et al 2003, Barreto Silva et al 2017, Caravaco-fontan et al 2016). Considering the elevated prevalence of vitamin D deficiency/insufficiency in CKD/ESRD and the potential benefits of restoring the vitamin D status the K/DOQI and KDIGO Clinical Practice Guidelines for CKD-MBD have proposed ergocalciferol or cholecalciferol supplementation.

Due to the complex nature of vitamin D, the present guideline is focused on the effect of vitamin D supplementation, in the forms of cholecalciferol and ergocalciferol, on vitamin D insufficiency/deficiency in patients with CKD and not on outcomes related to CKD-MBD or other clinical disturbances. Supplementation of prehormone and activated forms of vitamin D, calcidiol and calcitriol, were not included in this guideline.

Vitamin D Levels and Deficiency

Despite differences in dosing regimens and vitamin D status at baseline, supplementation was effective in increasing 25(OH)D serum concentration in 14 RCTs, including in the form of ergocalciferol (Bhan et al and Miskulin et al) and cholecalciferol (Alvarez et al 2012 and 2013, Armas et al 2012, Chandra et al 2008, Hewitt et al 2013, Marckmann et al 2012, Meireles et al 2016, Massart et al 2014, Seibert et al 2013, Delanaye et al 2013, Mager et al 2016 and Tokmak et al 2008). This effect was demonstrated in HD patients (8 studies), HD and PD patients combined (1 study), stages 1-4 CKD patients (4 studies) and in 1 study with any CKD participants. Five studies reported that ergocalciferol using doses 50, 000 IU/week and dose dependent on status (Bhan and Miskulin) and cholecalciferol in doses ranging from 25, 000-50, 000 IU/week (Massart et al, Alvarez et al and Delanaye et al) improved vitamin D status. There were significant effects noted after three months of supplementation. However, there was no difference in vitamin D deficiency status between non-dialyzed groups receiving two different dosing regimens (Mager et al).

A meta-analysis was conducted to determine odds of vitamin D sufficiency according to vitamin D supplementation, which included Bhan, et al. (each group compared to the placebo group), Delanaye et al, Massart et al, and Alvarez et al 2012. Participants that were supplemented with vitamin D had an OR (95% CI) of 9.31 (3.38, 24.7) ($p < 0.001$) of being vitamin D sufficient (defined as either >30 or 32 ng/mL), though there was moderate heterogeneity in the data ($I^2=51.84$; $p=0.08$). Additionally, data from eight studies were pooled to determine mean difference (95% CI) in vitamin D levels according to vitamin D supplementation. There was a mean increase of 21.06 ($17.46, 24.66$) ng/mL in the vitamin D supplemented groups compared to the placebo groups, but heterogeneity was moderate ($I^2=67.3\%$; $p=0.003$), so results should be interpreted with caution.

Calcium and Phosphorus Levels

In adults with chronic kidney disease, twelve studies examined the effect of vitamin D intake on biomarkers and/or health outcomes (Alvarez et al 2012, Armas et al 2012, Bhan et al 2015, Chandra et al 2008, Delanaye et al 2013, Marckmann et al 2012, Massart et al 2014, Meireles et al 2016, Miskulin et al 2016 and Seibert et al 2013, Khajehdehi et al 2000 and Mager et al 2016). Moderate quality evidence demonstrated no effect of vitamin D supplementation on calcium or phosphorus levels.

In predominantly vitamin D deficient participants, there was no effect of ergocalciferol supplementation on effect of calcium levels (Bhan et al and Miskulin et al) in doses of 50, 000 IU/week or /month or in individualized doses. The effect of cholecalciferol on calcium levels was unclear with seven studies finding no effect on calcium levels and three studies determining supplementation increased calcium levels. In Massart et al., there was no effect of 25, 000 IU weekly cholecalciferol on proportion of HD participants reaching target levels at 3 months. There was no clear pattern of effect according to participant population, deficiency status or vitamin D dosage. In pooled analysis of four studies in which data could be combined (Delanaye et al 2013, Miskulin 2016, Khajehdehi 2000, Seibert et al 2013), there was no effect of vitamin D supplementation on calcium levels [MD (95% CI): 0.07 ($-0.18, 0.31$) mg/dL].

Vitamin D supplementation had no effect on phosphorus levels with ergocalciferol supplementation (2 studies with doses of 50, 000 IU/week or /month or in individualized doses) or cholecalciferol doses ranging from 50, 000 IU/day to 50, 000 IU/month (10 studies). In pooled analysis of five RCTs (Delanaye et al 2013, Khajehdehi et al 2000, Miskulin et al 2016, Seibert et al 2013, Meireles et al 2016), there was no effect of vitamin D supplementation on phosphorus levels [MD (95% CI): -0.15 ($-0.44, 0.15$) (mg/dL)].

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendations for vitamin D supplementation are based on Grade III /Grade C evidence as well as Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of vitamin D supplementation on vitamin D status and vitamin 25\(OH\)D levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on quality of life in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on hospitalizations in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on lipid profile in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on blood glucose parameters in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on arterial calcification in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on blood pressure in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on eGFR in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on creatinine levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on calcium and phosphorus levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on magnesium levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on sodium levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on vitamin D status and vitamin 25\(OH\)D levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on inflammatory marker levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin D supplementation on nutritional status indicators in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Micronutrients: Vitamins E and A Supplementation (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Vitamins E and A Supplementation and Toxicity

In adults with CKD 5D on MHD or CKD 5D on PD, it is reasonable to not routinely supplement Vitamin A or E because of the potential for vitamin toxicity. However, if supplementation is warranted, care should be taken to avoid excessive doses, and patients should be monitored for toxicity (OPINION).

Rating: Consensus
Imperative

- [Risks/Harms of Implementing This Recommendation](#)

There is a potential for toxicity in those patients who are being supplemented. High doses of vitamin E supplementation has the potential to increase risk of hemorrhagic stroke and impair platelet aggregation. Vitamin E interacts with anticoagulant and antiplatelet medications and therefore caution is advised on vitamin E supplementation for CKD patients already receiving these medications. The potential of vitamin A toxicity with supplementation is also a concern.

- [Conditions of Application](#)

Special Discussions

As a result of the limited number of high quality studies, (see study selection criteria) and the variability in the outcomes reported in these trials, there is insufficient evidence to make recommendations on vitamin E intake for CKD patients. The nutritional requirements or Recommended Dietary Intake of vitamin E for individuals with CKD 1-5, those undergoing chronic dialysis and post-transplant are unknown. Dose response studies identifying the relation between Vitamin E intake and serum levels of vitamin E are not available. The prevalence of vitamin E deficiency in CKD population is unclear. The potential of vitamin E toxicity with supplementation is a concern for this fat-soluble vitamin.

There is a potential for toxicity in those patients who are being supplemented. High doses of vitamin E supplementation has the potential to increase risk of hemorrhagic stroke and impair platelet aggregation. Vitamin E interacts with anticoagulant and antiplatelet medications and therefore caution is advised on vitamin E supplementation for CKD patients already receiving these medications.

Vitamin A was investigated in this SR, however there were no trials examining dietary intake of vitamin A, and supplementation trials included IV vitamin A, which the WG determined qualified it as a medication vs a nutritional supplement. However, the same concerns regarding toxicity of vitamin E supplementation apply to vitamin A supplementation.

Recommendations cannot be made with regard to vitamins A or E supplementation in CKD population. An individualized approach is required in considering the need to supplement vitamins A or E supplementation or terminate supplementation in adult CKD population and there is also a need to monitor for toxicity with supplementation.

Implementation considerations

- Suggested vitamin intake should be based on recommendations for the general population (ex: Recommended Dietary Allowance) unless there are specific considerations requiring modification.
- Vitamin E supplementation dose should consider individual patient's nutritional status, dietary intake, concomitant medications, co-morbid conditions particularly with regard to baseline cardiovascular disease, and lipid levels.
- Oral doses ≥ 400 IU of vitamin E are not recommended without at least intermittent monitoring of serum vitamin E levels.

Monitoring and Evaluation

Platelet count should be monitored as should any changes in medical status, medications, and nutritional status.

• [Potential Costs Associated with Application](#)

The cost of nutrition supplements should be considered before recommending to a patient.

• [Recommendation Narrative](#)

Vitamin E is a fat-soluble nutrient recognized for antioxidant properties. There are eight known naturally occurring forms of vitamin E (IOM et al 2000), but alpha-tocopherol is the only known form of vitamin E that meets human requirements and is the form found in plasma. Therefore, Dietary Reference Intake (DRI) for vitamin E is only available for alpha-tocopherol. The RDA for vitamin E was determined by identifying serum levels of vitamin E that provided protection to erythrocyte survival when exposed to hydrogen peroxide.

While vitamin E supplements are typically provided as alpha-tocopherol, products containing other tocopherols and tocotrienols have been reported. The potency of synthetic alpha-tocopherol (*RRR-alpha-tocopherol*, labeled as *D* or *d*) is not identical to the natural form. This is because synthetic alpha-tocopherol contains eight stereoisomers of which only 4 are found in tissues and serum. Synthetic alpha-tocopherol: *all rac-alpha-tocopherol*, labeled as *dl* or *DL* is therefore only half as active as the natural form, therefore requiring 50% more IU to receive a dose equivalent to the natural source. Most supplements provide vitamin E as alpha-tocopherol in a 100-400mg dose (IOM 2000).

Vitamin E is a fat-soluble vitamin. The potential risk of vitamin E toxicity is primarily related to the use of supplements (IOM 2000, Miller et al 2005). High doses of vitamin E supplements in the form of alpha-tocopherol have been reported to cause bleeding and/or disrupt blood coagulation *in vivo* and there are some *in vivo* data that suggest alpha-tocopherol inhibits platelet aggregation (Miller et al 2005). The RDA for vitamin E for normal adult men and woman is 15mg per day (22.4 IU). The Food and Nutrition Board has defined an upper level of intake for vitamin E in the form of alpha-tocopherol and the stereoisomer forms in synthetic vitamin E supplements as 1500 IU and 1100 IU/day respectively. While not definitive, these level of intake appear to be the safety limit with regard to the potential of vitamin E to confer bleeding risk.

Several studies evaluated the effects of Vitamin E coated dialyzer membranes on biocompatibility, blood pressure during dialysis and oxidative stress (Takouli et al 2010, Zweiger et al 2013). However, results were inconclusive. Data regarding the effect of vitamin E coated dialyzers on hemoglobin, lipid profile and nutritional status were inconclusive and study design for these trials and the meta-analyses were of low quality (Huang et al 2015).

Studies examining daily vitamin A requirements for individuals with various stages of CKD are lacking. Optimal serum levels of vitamin E are not defined for this population. Daily vitamin E required to maintain nutritional health, reverse deficiency, and to avoid toxicity in CKD population are unclear. Vitamin A was initially investigated in the systematic review, but there were no dietary trials available, only trials in which vitamin A was delivered intravenously, which was considered beyond the scope of this guideline.

The eight studies included for this review examined the effect of oral vitamin E supplementation in adults with CKD on serum indices and health outcomes (Ahmadi et al 2003, Boaz et al 2000, Daud et al 2013, Himmelfarb et al 2014, Mann et al 2004 and Ramos et al 2011, Hodkova et al 2006 and Khajehdehi et al 2000). In three of these studies, vitamin E supplementation was combined with lipoic acid supplementation (ALA) (Ahmadi et al 2013, Himmelfarb et al 2014, Ramos et al 2011). All studies examined MHD patients as the target population, except for Ramos et al, who examined Stages 3-5 CKD subjects. Subjects in Hodkova, et al. were vitamin E repleted, but baseline vitamin E status was not reported for any of the other studies.

All- Cause Mortality and Cardiovascular Disease Outcomes

Participants with CKD (serum creatinine $\approx$ 1.4 to 2.3 mg/dL) and high risk for cardiovascular events were given 400 IU daily oral vitamin E for a median of 4.5 years (Mann et al 2004). Compared to the placebo group, there was no difference in total mortality between groups. Additionally, there was no difference in the relative risk of myocardial infarction (MI), stroke, death from CV causes, unstable angina, heart failure hospitalizations, heart failure, TIA or composite of MI, stroke, or death from cardiovascular causes between groups.

Boaz et al 2000 examined the effect of vitamin E supplementation on CVD endpoints (primary outcome) and all-cause mortality. MHD subjects with pre-existing CVD were supplemented with daily oral 800 IU oral vitamin E for a median of 519 days. Risk of all-cause mortality was not significantly different between groups. The vitamin E group had a significantly decreased risk of experiencing a CVD endpoint compared to the control group, but the RR for fatal and non-fatal MIs, ischemic stroke and PVD were not significantly different between groups.

Based on these limited data, vitamin E supplementation did not affect all-cause mortality. Results regarding the effects of vitamin E supplementation on CVD outcomes were mixed. Differences may be due to the population studied or vitamin E dosage. In pooled analysis conducted in the current systematic review, there was no effect of vitamin E supplementation on CVD outcomes, though heterogeneity of results was high.

Anthropometric Measures

Two RCTs examined the effect of vitamin E supplementation on nutritional status in MHD participants (Ahmadi et al 2013, Daud et al 2013). Participants received either tocotrienols (90 mg) and tocopherols (20 mg) for 16 weeks or 400 IU oral vitamin E/day, 600 mg alpha lipoic acid (ALA)/day, or both for 2 months. While there were no changes in albumin levels between groups in the former study (Daud, et al.), in Ahmadi, et al., SGA score was improved in the vitamin E, ALA, and combined supplementation groups compared to placebo. SGA was the primary outcome of interest in this study. Vitamin E deficiency status at baseline was not described in either study.

Three RCTs examined the effect of oral vitamin E supplementation on anthropometric measures (Ahmadi et al 2013, Daud et al 2013, Ramos et al 2011). All studies reported no effect of vitamin E supplementation on BMI or body weight. Anthropometric measurements were not the primary outcomes of interest in any of these studies.

Inflammatory Markers: CRP and IL-6 (Interleukin-6)

Five studies examined the effect of vitamin E supplementation on inflammatory biomarkers, particularly CRP and IL-6 levels (Ahmadi et al 2013, Daud et al 2013, Himmelfarb et al 2014, Hodkova et al 2006, Ramos et al 2011). In three of the studies these inflammatory markers were the primary outcomes of interest (Ahmadi et al 2013, Hodkova et al 2006, Ramos et al 2011). Himmelfarb et al and Ramos et al gave vitamin E supplementation in combination with $\pm$ -lipoic acid, and Ahmadi, et al., examined vitamin E supplements alone and in combination with $\pm$ -lipoic acid. All studies assessed the effect of oral vitamin E supplementation on the CRP inflammatory marker levels in patients with CKD Stages 3-5 and MHD participants. None of them found any effect of vitamin E supplementation ranging from 400-800 IU oral vitamin E per day (with or without 600 mg $\pm$ -lipoic acid) for durations ranging from 5 weeks to 6 months on CRP levels.

Three of the studies also measured IL-6 and found no relationship between vitamin E supplementation and IL-6 levels (Daud et al 2013; Himmelfarb et al 2014, Ramos et al 2011). Ramos et al (Stages 3-5 CKD) and Himmelfarb et al (MHD patients) both supplemented with daily oral 666 IU mixed tocopherols (Vitamin E) + ALA 600 mg for 8 weeks and 6 months, respectively. Neither found an effect of supplementation on serum IL-6 levels. However, Ahmadi et al found that oral vitamin E alone (400 IU per day) or in combination with 600 mg $\pm$ -lipoic acid per day, reduced IL-6 cytokine levels in MHD participants. In pooled analysis of two RCTs that utilized vitamin E alone or with $\pm$ -lipoic acid, there was no effects on IL-6 levels compared to the placebo groups.

Serum Vitamin E Levels

Two RCTs examined the effect of daily oral vitamin E supplementation on vitamin E levels (Boaz et al 2011, Hodkova et al 2006). Both studies included MHD patients. Hodkova et al found that serum vitamin E levels increased in the vitamin E supplemented group ($\pm$ -tocopherol 400 mg/888 IU) after 5 weeks, but no change in the control. Between group differences were not reported. Boaz, et al. found that the vitamin E supplemented group had significantly higher vitamin E levels compared to the placebo group when MHD participants with pre-existing CVD were supplemented with 800 IU oral vitamin E/day for a median of 519 days, but between group differences were not reported. In pooled analysis of the two RCTs that examined vitamin E supplementation alone, there was no significant effect of supplementation compared to the placebo/control group. Therefore, available evidence indicates that vitamin E supplementation alone does not affect vitamin E levels.

Lipid Levels

Daily oral vitamin E supplementation of 110 mg for four months (Daud et al 2013) and 200 mg for three months (Khajehdehi et al.) did not change serum triglyceride, total cholesterol or LDL (low density lipoprotein) levels but demonstrated efficacy of increasing HDL-C levels.

- [Recommendation Strength Rationale](#)

The recommendation regarding supplementation of vitamins E and A is based on Consensus/expert opinion, since there was no literature to suggest a benefit of dietary or oral supplementation.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of vitamin E supplementation on all-cause mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin E supplementation on CVD outcome in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin E supplementation on lipid profile in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin E supplementation on vitamin E levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin E supplementation on CRP levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin E supplementation on IL-6 levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin E supplementation on SGA score in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Micronutrients: Vitamin K Supplementation (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Anticoagulant Medication and Vitamin K Supplementation

In adults with CKD 1-5D or posttransplantation, it is reasonable that patients receiving anticoagulant medicines known to inhibit vitamin K activity (e.g., warfarin compounds) do not receive vitamin K supplements (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks and harms associated with this recommendation.

- [Conditions of Application](#)

The United States [Institute of Medicine](#) states that the Adequate Intake of vitamin K is 120 and 90 micrograms per day for adult men and women, respectively (DRI 2001; Shearer et al 2014). These values are based on median vitamin K intakes reported in the NHANES III data. Globally, dietary recommendations for vitamin K usually vary from 50 to 120 micrograms/day (Fusaro et al 2017). These recommendations do not differentiate phyloquinone from menaquinone intake. At the time the US Institute of Medicine recommendations were set, the food composition databases on which these recommendations were made only contained the phyloquinone content of foods. Hence, these current recommendations are based on phyloquinone, which is the major form of vitamin K in Western diets.

A study of the NHANES data indicated that 72.1% of adults with mild-moderate CKD (eGFR-EPI 58 mL/min/1.73m²) had vitamin K intake below the recommended adequate intake (AI) level (mean, 97.5 µg/day; 95%CI, 89.7-105.3) (Cheung et al 2015). Studies in Italy confirmed that daily intake of vitamin K1 in MHD patients is commonly below recommended levels (Fusaro et al 2017). Several observational studies in advanced CKD (stages 3-5) or MHD patients indicated that serum vitamin K1 (phyloquinone) and vitamin K2 (menaquinone) concentrations were frequently low and that serum levels of other uncarboxylated compounds which, when elevated, indicated vitamin K deficiency were increased (Thamratnookoon et al 2017; Westenfeld et al 2012).

The recommended dietary vitamin K intake for patients with CKD 1-5, including those with the nephrotic syndrome, those who are undergoing MHD or PD or those who are post-transplant recipients were not defined and were based on that derived for the general population. In MHD patients, vitamin K intake and serum vitamin K levels are often low or undetectable, and serum uncarboxylated osteocalcin and PIVKA-II are commonly elevated (Westenfeld et al 2012, Elliott et al 2014).

Implementation Considerations

- Suggested vitamin intake should be based on recommendations for the general population (ex: Recommended Dietary Allowance) unless there are specific considerations requiring modification.
- Patients receiving antibiotics who have poor intake and at higher risk of bleeding (e.g., surgical patients) may be considered for vitamin K supplements, particularly if they have acute kidney injury or chronic kidney disease (Pineo et al 1973). However, the foregoing conclusions were essentially based on observational studies of small number of patients. Increasing age, platelet count and serum urea and creatinine and lower serum albumin concentrations were associated with more severe elevation in prothrombin time in patients taking antibiotics (Pineo et al 1973, Williams et al 1991). Vitamin K supplements may return prothrombin time to normal in such patients (Williams et al 1991). However, the foregoing conclusions were essentially based on observational studies of small number of patients.
- The RDN may provide dietary assessment/counseling related to excess dietary intake of Vitamin K or irregular excess intake of foods containing high vitamin K; and providing education regarding dietary sources of vitamin K.

◦ [Potential Costs Associated with Application](#)

There are no obvious costs associated with this recommendation.

◦ [Recommendation Narrative](#)

Vitamin K is a fat-soluble vitamin that acts as a cofactor for gamma-glutamyl carboxylase which enables the carboxylation of vitamin K-dependent proteins producing coagulation factors. Coagulation factors II, VII, IX and X are the most well-known vitamin K-dependent proteins, and deficiency in these factors can lead to impairment in blood clotting. Vitamin K also enables normal calcification processes to proceed in bone and soft tissues. Matrix Gla protein (MGP) is a vitamin K-dependent protein produced by vascular smooth muscle cells (VSMCs) that is a powerful inhibitor of vascular calcification in culture media and of intimal atherosclerotic plaque calcification. After carboxylation, MGP binds to calcium crystals, inhibiting further crystal growth. MGP binds to bone morphogenetic protein-2 (BMP-2) thereby blocking the differentiation of VSMCs towards osteochondrogenic type cells. Vitamin K participates in the enzymatic carboxylation of proteins controlling bone calcium deposition (e.g., osteocalcin) and also plays an important role in normal bone formation and structure.

Hence Vitamin K, by facilitating carboxylation of certain proteins, has major effects on blood clotting, preventing soft tissue calcification, including vascular calcification and controlling bone calcium crystal formation.

Two classes of vitamin K compounds are primarily responsible for vitamin K activity, phylloquinone (vitamin K1) and menaquinones (vitamin K2) (Harshman et al 2014). Phylloquinone is found primarily in foods, especially green and leafy vegetables (e.g., spinach, kale, cabbage, broccoli), plant based oils found in many food products, and cow's milk. There are more than 10 menaquinones which differ in the number of isoprenoid units in its side chain. Most menaquinones are produced by bacteria. Menaquinone 4 is different and appears to be produced in vivo from phylloquinone (Harshman et al 2014; Card et al 2014). Menaquinones are found in dairy products (yogurt) meats, and fermented foods, and also synthesized in the intestine by colonic bacteria. The intestinal absorption of vitamin K requires biliary and pancreatic secretions and occurs in the small intestine where vitamin K is incorporated into chylomicrons. The role of the menaquinones in vitamin K function and nutritional needs is still not completely understood. Large doses of vitamin E may induce vitamin K deficiency (Card et al 2014).

Vitamin K Levels

Only one short term randomized controlled study has been published that examined the effects of vitamin K supplements on vitamin K status in MHD patients (Westenfeld et al 2012). No such studies have been carried out in other stages of CKD or in PD patients or those post-transplant. The study involved small number of patients who received, by random assignment, supplements of 45, 135 or 360 micrograms per day of vitamin K2 (menaquinone-7) for only six weeks. In general, there was a dose dependent increase in serum vitamin K2 and decrease in serum $_{dpuc}$ MGP, $_{uc}$ osteocalcin and PIVKD-II. Mean serum vitamin K2 rose to previously reported normal values with the 45 µg/day dose and to modestly above normal values with the 135 and 360 µg/day doses. Serum $_{dpuc}$ MGP, $_{uc}$ osteocalcin and PIVKD-II decreased most with the 360 µg/day dose, but concentrations still tended to be above normal with this dose.

There are currently several clinical trials of vitamin K supplements in MHD patients, and more information regarding vitamin K supplementation should be available within the near future (Caluwe et al 2014; Holden et al 2015; Krueger et al 2014). (Clinical Trials Identifier: NCT01528800; NCT01742273; NCT2610933; NCT02870829; UMIN000011490; UMIN000017119). There is a paucity of data on the long-term safety of different vitamin K intakes and especially of vitamin K supplements and of the value, if any, of taking different vitamin K compounds. Individuals receiving vitamin K supplements should not receive anticoagulant medicines that inhibit vitamin K activity (e.g., warfarin compounds).

◦ [Recommendation Strength Rationale](#)

The recommendation regarding vitamin K supplementation is based on Consensus/expert opinion.

◦ [Minority Opinions](#)

Consensus reached.

• [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of vitamin K supplementation on vitamin K levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of vitamin K supplementation on dephosphorylated-uncarboxylated MGP levels in adults with CKD 1-5D and post-transplant?](#)

◦ [References](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Micronutrients: Trace Mineral Supplementation, Selenium and Zinc (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Selenium and Zinc Supplementation

In adults with CKD 1-5D, we suggest to not routinely supplementing selenium or zinc since there is little evidence that it improves nutritional, inflammatory or micronutrient status (2C).

Rating: Weak
Imperative

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms associated with this recommendation.

- [Conditions of Application](#)

Medical Nutrition Therapy requires individualization of nutrition care based on patient's CKD stage, comorbidities, diet and lifestyle as well as patient preferences. Recommendations are specific to CKD stage described.

Suggested intake should be based on recommendations for the general population (ex: Recommended Dietary Allowance) unless there are specific considerations requiring modification.

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with this recommendation.

- [Recommendation Narrative](#)

Selenium is a trace element that has known anti-oxidant properties and plays a role in enzymatic activities inside the body. It acts as a cofactor for the reduction in important antioxidant enzymes like glutathione peroxidase and thus protects against oxidation. Several studies have suggested that maintenance hemodialysis (MHD) patients have low levels of selenium compared with healthy controls, and deficiency of this trace element may contribute to increased oxidative stress and inflammation (Marti et al 2011, Chen et al 2009,

Chen et al 2013, Fujishima et al 2011). There is also some preliminary suggestion that low selenium levels may be associated with increased death risk in MHD patients, especially death due to infections (Fujishima et al 2011).

Zinc is an essential micronutrient and forms a component of bio-membranes. It functions not only as an antioxidant but also has anti-inflammatory effects and prevents free radicals-induced injury during inflammation. There is some suggestion that marginal zinc intake may be associated with an increased risk of cardiovascular disease in general population (Prasad et al 2008) and zinc has been shown to protect against atherosclerosis by inhibiting the oxidation of low-density lipoprotein cholesterol in animal studies (diSilvestro et al 1997). Zinc deficiency has been shown to increase oxidative stress and NF- κ B DNA-binding activity and induce inflammation in experimental models (Foster et al 2012, Prasad et al 2014, Shen et al 2008). Zinc is also essential for insulin synthesis and release and glucose homeostasis (Cooper-Capetini et al 2017) and zinc deficiency has been suggested to impair insulin secretion and decrease leptin levels (Ott et al 2001). Studies have reported a high prevalence of zinc deficiency in hemodialysis patients (Bozalioglu et al 2005, Hsieh et al 2006, Kiziltas et al 2008).

The current Recommended Dietary Allowance (RDA) for zinc is 8 mg/d for women and 11/d mg for men in the general population and for selenium is 55mcg/d for women and men. Whether similar amount of intake is recommended in various CKD stages and maintenance dialysis population is currently not known.

Selenium

In adults with chronic kidney disease (CKD), seven studies have examined the effect of selenium intake on biomarkers and other surrogate health outcomes. Most of the studies utilized oral selenium supplementation and all studies were performed in MHD patients. Koenig et al examined the effect of intravenous selenium supplementation and Stockler-Pinto examined the effect of selenium supplementation in the form of a Brazil nut. Selenium dosages generally ranged from 175-1400 μ g per week. The selenium dosage in Stockler-Pinto et al was not described (1 Brazil nut/day) and in Koenig et al, the parenteral dose of selenium used was much higher (400 mg 3 times a week) compared to other studies. Study duration ranged from 14 days to 6 months. In Temple et al, participants' selenium status at baseline was normal. In study by Tonelli et al, 28% of treatment group versus 15% of placebo group had low selenium levels after supplementation. Around 20% of participants were selenium deficient in Stockler-Pinto et al, and the remaining studies did not report selenium status at baseline.

Nutritional Status

Only one very short-term (12 weeks) randomized placebo-controlled study examined the effect of oral selenium supplementation of 200 μ g per day on nutritional status in 80 MHD patients (Salehi et al 2013). The study reported a significantly greater reduction in SGA and malnutrition-inflammation score in the selenium group compared to the placebo group. However, no significant difference was observed in serum albumin concentrations between the two groups (Salehi et al 2013). The same study by Salehi et al did not observe any difference in the median changes of CRP levels between selenium and placebo groups. Although a smaller increase in interleukin-6 levels was observed in selenium group compared to placebo group (Salehi et al 2013), this is the only study that examined inflammation as an outcome. Thus, there is not enough evidence to make recommendation of selenium supplementation for malnutrition-inflammation syndrome in MHD patients.

Selenium and Lipid Levels

Although two short-term small randomized controlled studies provided some evidence that selenium supplementation may be useful in increasing plasma and erythrocyte selenium levels (Temple et al 2000, Adamowicz et al 2002) it is not known if selenium supplementation may impact on any patient health-related or hard clinical outcomes. Only one short-term randomized study by Salehi et al examined the effects of oral selenium supplementation on lipid levels. The results showed no difference between selenium group and control group in any of the lipid parameters including triglyceride, total cholesterol, low density lipoprotein- and high density lipoprotein-cholesterol.

Zinc

Nutritional Status

Three small short-term RCTs examined the effects of zinc supplementation on nutrition status in MHD patients. The study duration ranged from 8 weeks to 90 days. The dose of zinc supplementation ranged from a daily dose of 11mg, 50mg to 100mg elemental zinc (Argani et al 2014, Guo et al 2013, Jern et al 2000). In the study by Argani et al, serum albumin levels increased in the zinc supplemented group but there was no change in the placebo group. Guo et al examined zinc supplementation of 11mg daily for 8 weeks in a cohort of 65 MHD patients with low baseline zinc level (<80mg/dL). Descriptive quantitative data was not provided but the authors concluded that protein nitrogen appearance and albumin levels significantly increased in zinc supplemented group but not in control group. Jern et al showed that protein catabolic rate increased with 50mg zinc supplementation for 90 days but no change in placebo group. Between group differences were not provided in these studies. The data from these three small low-quality trials were regarded as inconclusive and not enough to make recommendation.

Lipid Profile, Homocysteine Levels

Four short-term RCTs examined the effect of oral zinc supplementation on lipid levels (Argani et al 2014, Roozbeh et al 2009, Chevalier et al 2002, Rahimi-Ardabili et al 2012). The studies by Argani et al and Rahimi-Ardabili et al administered 100 mg oral zinc daily to MHD patients for two months. Argani et al showed no changes in cholesterol and triglyceride levels with zinc supplementation. Rahimi-Ardabili et al showed that cholesterol levels increased significantly in the placebo group but no change in the treatment group and total cholesterol levels were not different between the two groups after 2 months' study. In the other two studies, Roozbeh et al and Chevalier et al both supplemented MHD patients with 50 mg zinc daily for six weeks and 90 days, respectively. All patients in both these two studies were zinc deficient at baseline (<80 μ g/dL). Both studies showed that total cholesterol, LDL-C, HDL-C and serum triglyceride levels increased in zinc supplemented group but no change in the control group. The conclusions by the authors in these studies suggested that this increase in lipid parameters was desirable. Pakfetrat et al examined the effect of 50 mg oral zinc per day for 6 weeks in MHD patients, and found that significantly decreased homocysteine levels decreased in the zinc supplemented group compared to the placebo group. Two studies examined the effects of zinc supplementation on inflammatory parameters but results were inconclusive (Guo, et al. and Rashidi, et al). Data on the effects of zinc supplementation on body weight and BMI were mixed and limited (Argani et al 2014, Mazani et al 2013).

Zinc Levels

Six RCTs examined zinc supplementation in relation to serum zinc levels in MHD patients (Argani et al 2014, Guo et al 2013, Roozbeh et al 2009, Chevalier et al 2002, Tonelli et al 2015, Zachara et al 2001). All except Tonelli's study described zinc deficiency at baseline. The dosage of zinc supplementation used ranged from 11mg to 110mg. Study duration ranged from 5 weeks up to 6 months. In the study by Tonelli and co-workers, zinc levels in the medium dose (50mg per day) but not the low dose (25mg per day) group were significantly higher than the non-supplemented group at 90 days and 180 days after supplementation. A pooled analysis of these 6 studies showed a mean (95% confidence intervals) increase of 30.97 (17.45, 44.59) μ g/dL of serum zinc levels after supplementation compared to control group. However, heterogeneity was high. Furthermore, it is not known if zinc supplementation in deficient patients may impact on any health-related outcomes or clinical hard outcomes in CKD and dialysis patients. The long-term effects or any toxicity of zinc supplementation are also unclear at this stage.

There were no identified studies examining the effect of zinc supplementation on dysgeusia in patients with CKD, though this topic has been explored in other populations. .

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on zinc and selenium supplementation is based on Grade III /Grade C evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of selenium supplementation on comorbidities in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of selenium supplementation on selenium levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of selenium supplementation on inflammatory markers in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of selenium supplementation on nutritional status indicators in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of zinc supplementation on homocysteine levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of zinc supplementation on HDL cholesterol levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of zinc supplementation on total and LDL cholesterol and triglyceride levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of zinc supplementation on zinc levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of zinc supplementation on BMI in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of zinc supplementation on CRP levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of zinc supplementation on nutritional status indicators in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Electrolytes: Phosphorus (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Dietary Phosphorus Amount

In adults with CKD 3-5D, we recommend adjusting dietary phosphorus intake to maintain serum phosphate levels in the normal range (1B).

Rating: Strong
Imperative

CKD: Dietary Phosphorus Source

In adults with CKD 1-5D or posttransplantation, it is reasonable when making decisions about phosphorus restriction treatment to consider the bioavailability of phosphorus sources (e.g. animal, vegetable, additives) (OPINION).

Rating: Consensus
Conditional

CKD: Phosphorus Intake with Hypophosphatemia

For posttransplantation adults with hypophosphatemia, it is reasonable to consider prescribing high-phosphorus intake (diet or supplements) in order to replete serum phosphate (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

The serum phosphate level at which supplementation should be considered in these patients or the dose of replacement to be given is, however, not well studied, and should be decided based on patient needs and clinical judgment.

Recommendations to lower dietary phosphorus intake in patients with CKD have been met with concerns, often relating to the risk of limiting the intake of other nutrients, particularly protein, which is the main source of phosphate in the diet (Lynch et al 2011, Rufino et al 1998, Shinaberger et al 2008). These concerns are particularly relevant to patients treated with dialysis because of protein losses in dialysate and greater protein catabolism from hypermetabolic stress (Carrero et al 2013). Dietary counselling that includes information on not only the amount of phosphate but also on the source of protein from which the phosphate derives and suggestion on methods of cooking phosphate-rich foods can achieve phosphorus intake without compromising dietary quality or protein status (St. Jules et al 2016).

- [Conditions of Application](#)

Special Discussions

Hypophosphatemia in kidney transplant patients: Hypophosphatemia is a relatively common complication after kidney transplantation, especially during the first months, and possibly leading to osteomalacia and osteodystrophy. Its pathogenesis has been attributed to increased renal phosphate excretion due to elevated levels of phosphaturic hormones, the effect of glucocorticoid, persistent elevated PTH levels, suboptimal recovery of vitamin D activation, and imbalance in fibroblast growth factor 23 (FGF23) (Tomida et al 2012, Trombetti et al 2011, Sakhaee 2010). It has been proposed that dietary intensification of phosphorus can solve this complication; one small randomized controlled trial (Rosman et al 1989) examined the effects of 12-week dietary phosphorus supplementation by means of a neutral phosphate salt (disodium phosphate) in patients with early post-transplantation hypophosphatemia. The authors observed that, compared to sodium chloride, supplementation of phosphorus improved hypophosphatemia as well as adenosine triphosphate in the muscles and the acid excretion capacity of the kidney. No adverse effects on serum calcium and PTH concentrations were noted during intervention.

The serum phosphate level at which supplementation should be considered in these patients or the dose of replacement to be given is, however, not well studied, and should be decided based on patient needs and clinical judgment.

Implementation Considerations

Recommendations to lower dietary phosphorus intake in patients with CKD have been met with concerns, often relating to the risk of limiting the intake of other nutrients, particularly protein, which is the main source of phosphate in the diet (Lynch et al 2011, Rufino et al 1998, Shinaberger et al 2008). These concerns are particularly relevant to patients treated with dialysis because of protein losses in dialysate and greater protein catabolism from hypermetabolic stress (Carrero et al 2013). Dietary counselling that includes information on not only the amount of phosphate but also on the source of protein from which the phosphate derives and suggestion on methods of cooking phosphate-rich foods can achieve phosphorus intake without compromising dietary quality or protein status (St. Jules et al 2016).

- *Advise choosing natural foods that are lower in bioavailable phosphorus.* Animal- and plant-based foods contain the organic form of phosphate. While animal-based phosphate is absorbed in the GI tract by 40-60%, the absorption of plant-based phosphorus is lower (20-50%).³⁹⁴ In line with this, a small crossover trial including CKD patients³⁹⁵ stage 4 found that a 7-day vegetarian diet led to lower serum phosphate levels and decreased FGF23 levels than a 7-day meat-based diet (Moe et al 2011). Furthermore, foods with only organic phosphorus typically are more nutrient dense and have a higher nutritional value compared with processed foods containing phosphate additives, which tend to have a lower nutritional value, and are often paired with sodium and potassium additives (Gutierrez et al 2013).
- *Advise choosing commercial food items prepared without phosphorus-containing food additives.* Phosphorus additives are increasingly being added to processed and fast foods to preserve moisture or color, to emulsify ingredients and enhance flavor, and to stabilize foods. Phosphorus additives contain, however, inorganic phosphorus with a close to 100% intestinal absorption (St-Jules et al 2016, Kalantar-Zadeh et al 2010). Meat and poultry products that report the use of additives have an average phosphate-protein ratio much higher than additive-free products (Sherman et al 2009, Parpia et al 2017). The most-commonly used phosphorus additives in food industry can be found, for instance, in bakery products, enhanced meats, and processed cheeses (Karalis et al 2006).
- *Advise choosing natural foods that have low amount of organic phosphorus versus high amount of protein.* The content of organic phosphorus per gram of protein varies widely among foods. Nutrient composition tables reporting on phosphorus/protein ratio content can be used to recommend food substitutions that can considerably reduce the daily intake of organic phosphorus while ensuring adequate dietary protein intake (Cupisti et al 2003 and 2013, Barril-Cuadrado et al 2013, Kalantar-Zadeh et al 2010).
- *Advise preparing foods at home, using wet cooking methods such as boiling (and discard the water).* These methods are able to remove about 50% of phosphorus content from foods (Cupisti et al 2006, Ando et al 2015). Slicing the meat prior to boiling and the use of a pressure cooker has been shown more effective in terms of achieved protein to phosphorus content (Ando et al 2015). At the same time, these methods may remove other minerals (e.g. potassium) of concern for patients with CKD (Bethke et al 2008). Such practices, however, result in reduced palatability and texture of the food.

The work group emphasizes to individualize recommendations after appropriate evaluation of the patient daily intake. It requires nutrition expertise (preferably consultation with a renal dietitian) and should take into consideration culturally appropriate food substitutions. Nutritional counselling sessions should evolve, from the simple concept of phosphate restriction to opportunities of educating the patient on differentiation between organic and inorganic sources of phosphate and avoidance of phosphate additives (Karavetian et al 2014). Simple educational programs on how to read food labels and look for phosphate additives proved to be successful in helping dialysis patients reduce their serum phosphate levels (Caldeira et al 2011, Sullivan et al 2009). A meta-analysis suggested that nutritional counselling based on a structured behavioral change are, in general, successful in controlling hyperphosphatemia in these patients (Karavetian et al 2014). In this meta-analysis, however, only about half of the studies were randomized controlled interventions with a short duration ranging from 1 to 6 months, which calls for a need of more dedicated long-term interventional studies on this topic.

- [Potential Costs Associated with Application](#)

There are no obvious costs in implementing this recommendation.

- **Recommendation Narrative**

Phosphorus intake is necessary for bone growth and mineralization, as well as for regulation of acid-base homeostasis. Phosphorus is an essential nutrient, occurring in most foods both as a natural component and as an approved ingredient added during food processing. Because of difficulties of persons with chronic kidney disease (CKD) to clear excess phosphorus, additional means of serum phosphate control is necessary to avoid hyperphosphatemia, which could lead to bone and mineral metabolism disorders of CKD.

There are physiologic adaptations in the early stages of CKD that prevent excessive phosphorus retention, so the inability to promote phosphorus excretion to avoid phosphorus accumulation and hyperphosphatemia is generally seen when estimated glomerular filtration rate (eGFR) level decreases below 45 mL/min (Moranne et al 2009), being less common in earlier CKD stages. In the setting of anuria in patients on maintenance dialysis, hyperphosphatemia risks are particularly heightened (Block et al 1998), with a prevalence as high as 50% (Blayney et al 2009).

How much should dietary phosphorus/phosphate be restricted in adult patients with CKD is not well established. Traditionally, CKD-specific recommendations suggest maintaining phosphorus intake between 800-1000 mg/day in patients with CKD Stages 3-5 and those in maintenance dialysis in order to maintain serum phosphate in the normal range (Ketteler et al 2017, KDIGO 2017, Fouque et al 2007, KDOQI 2000 and 2003, Isakova et al 2017). The workgroup notes, however, that the efficacy of this recommendation has not been established. Further, such dietary phosphorus intake range is higher than current recommended dietary allowance for phosphorus in the adult general population (700 mg/d) (DRI 1997).

While dietary intake influences serum phosphate in CKD patients, factors other than intestinal phosphorus/phosphate absorption (namely exchange with bone and excretion by the kidneys in patients with residual renal function) may be major determinants of serum phosphate levels. Thus, the workgroup prefers not suggesting specific dietary phosphate ranges, and instead emphasize on the need to individualize treatments based on patient needs and clinical judgment, taking into consideration natural sources of organic phosphorus (animal vs. vegetal protein-based dietary phosphorus), or the use phosphorus additives in processed foods (Sherman et al 2009, Parpia et al 2011, Benini et al 2011).

With the goal to better understand the effect of dietary phosphate control, the workgroup decided in this evidence analysis to focus on reports that addressed dietary phosphorus intake/output/balance. This resulted in the exclusion of studies reporting solely on serum phosphorus levels.

Phosphorus Control

Limiting dietary phosphorus intake (*per se* or in combination with dietary protein restriction-the major source of dietary phosphorus) may be recommended to prevent / treat complications related to high phosphate load patients with CKD Stages 3-5 and maintenance dialysis. This can be achieved by intensified patient educational strategies or individualized dietary plans (Caldeira et al 2011). This evidence review included 5 short-term clinical trials that evaluated the effect of reduced dietary phosphorus on phosphorus intake, phosphate levels and urinary phosphorus excretion:

- *Phosphate restriction regimes in non-dialysis CKD:* Two randomized controlled trials (Williams et al 1991, Martinez, et al 1997) examined the effects of reduced dietary phosphorus in patients with CKD not undergoing dialysis. These studies evaluated the effect of a low phosphorus diet alone or in combination with a low protein diet, and observed significant reductions in serum phosphate, and urinary phosphorus excretion post-intervention.
- *Reducing phosphorus by limiting protein intake in non-dialysis CKD:* Five randomized controlled trials in CKD patients not undergoing dialysis stages 4-5 (Feiten et al 2005, Garneata et al 2016, Malvy et al 1999, Mircescu et al 2007, Rosman et al 1989) evaluated the effect of a low protein diet (LPD) or a very low protein diet (VLPD) supplemented with keto-analogs on serum phosphate levels. All five studies reported statistically significant (Garneata et al 2016, Malvy et al 1999, Mircescu et al 2007, Rosman et al 1989) or borderline-significant (Feiten et al 2005) reductions in serum phosphate levels at the end of intervention. *The interested reader can find more information on this topic in the evidence analysis of dietary protein restriction in these guidelines.*
- *Phosphate restriction regimes in maintenance dialysis:* Two randomized controlled trials (Lou et al 2012, Sullivan et al 2009) examined the effects of limiting dietary phosphorus in patients with CKD undergoing hemodialysis. Lou, et al. 2012 tested the effect of 3-month intensified dietary counseling in order to achieve 800 – 900 mg/d of dietary phosphorus, and observed a greater decrease in serum phosphorus concentration compared to standard care. Sullivan et al., tested the effect of patient education on identifying foods with phosphorus additives and observed, compared to standard care, a significant reduction in serum phosphorus levels after 3 months. No studies were identified that included peritoneal dialysis patients.

Although dietary phosphorus restriction may be a valid stand-alone strategy in patients with CKD-stage 3-4, the working group notes that, collectively, the serum phosphate reductions achieved solely by limiting dietary intake are modest (especially for dialysis patients) and recommend this strategy as one in the armamentarium of interventions to maintain serum phosphate levels in the normal range. For other non-dietary phosphate management strategies, the interested reader can consult recent guidelines on the management of mineral and bone disorders of CKD (Ketteler et al 2017, KDIGO 2017, Fouque et al 2007, KDOQI 2000 and 2003; Isakova et al 2017). Aligning with those guidelines, we recommend decisions to restrict dietary phosphorus be based on the presence of progressively or persistently elevated serum phosphate (that is, trends rather than a single laboratory value), and after consideration of concomitant calcium and PTH levels.

Clinical Consequences of Dietary Phosphorus Control

Whereas many studies have explored the outcome associations with serum phosphate levels throughout the spectrum of CKD, the clinical consequences of restricting dietary phosphorus are not well studied.

CKD Progression

Three observational studies evaluated the effects of dietary phosphate restriction on CKD progression. Results were mixed and evidence was limited. Williams et al 1991 studied impact of a dietary phosphorus restriction (alone or in combination with protein restriction) on creatinine clearance among 90 CKD patients of unreported aetiology or CKD stage over a median intervention time of 19 months. Compared to routine care, dietary protein and phosphate restriction or phosphate restriction only did not show any significant difference in the mean rate of fall of creatinine clearance. In an observational analysis from the Modification of Diet in Renal Disease (MDRD) study (Selamet et al 2016), greater 24-hr urinary phosphate excretion (taken in this study as an estimate of dietary phosphorus intake) was not associated with the future risk of ESRD. We note that in this study baseline phosphate levels were well controlled and normal on average, which may not be the case of real-world settings. A small retrospective observational analysis from Japan (Kawasaki et al 2015) including CKD patients stages 2-5 observed that higher phosphorus excretion per creatinine clearance was associated with a higher 3-year risk of CKD progression (defined as the composite of ESKD or 50 % reduction of eGFR).

It has been proposed that hyperphosphatemia in non-dialysis patients stages 2-5 may reduce the antiproteinuric effect of ACE inhibition (Zoccali et al 2011) or of very low protein diets (Di Iorio et al 2013). In a post hoc observational analysis from the Ramipril Efficacy In Nephropathy (REIN) trial, Zoccali et al 2011 evaluated the relationships between serum phosphate concentration at baseline, disease progression, and response to ACE inhibition among 331 patients with proteinuric nephropathies. Independent of treatment, patients with

higher phosphate progressed significantly faster either to ESRD or to a composite endpoint of doubling of serum creatinine or ESRD compared with patients with phosphate levels below the median, and the renoprotective effect of ramipril decreased as serum phosphate increased ($P = 0.008$ for interaction). In another post hoc study from a non-randomized, study in which 99 proteinuric CKD patients who sequentially underwent low-protein diet (LPD; 0.6 g/kg/day) and VLPD (0.3 g/kg/day) supplemented with keto-analogues, each for periods longer than 1 year, Di Lorio et al 2013 observed that 24-h proteinuria was reduced modestly in patients who maintained relatively higher serum phosphate levels or relatively higher phosphaturia to be maximal in those who achieved the lowest level of serum and urine phosphate.

Mortality

In observational studies involving CKD patients, the associations of dietary phosphorus intake on mortality are mixed, impacted by residual confounding and probably pointing to a null association. Three studies evaluated the cross-sectional association between measures of dietary phosphorus and mortality in individuals with non-dialysis CKD (Salamat, et al. 2016; Murtaugh, et al. 2012; Palomino, et al. 2013). Murtaugh et al. 2012 evaluated the association between 24-h dietary recall estimation of phosphorus intake in participants with eGFR < 60 ml/min/1.73m² from the community-based U.S survey National Health and Nutrition Examination Survey III, and observed no association between dietary phosphorus and mortality. Palomino et al. 2013 examined myocardial infarction patients from the Heart and Soul Study, the majority of which with normal kidney function, and observed no association between higher urinary phosphorus excretion and mortality, but noted an association with CVD-related mortality (P -trend across tertiles = 0.02). Selamet et al. 2016 involved nephrology referred patients with CKD from the MDRD study and failed to observe an association between 24-hr urinary phosphorus excretion and mortality.

One study in MHD patients that examined the association between dietary phosphate (as estimated from 3-day food recalls) and mortality (Noori, et al. 2010). Patients with higher dietary phosphorus intake were associated with greater 5-year mortality risk (p -trend across tertiles = 0.04). Lynch et al. 2011 explored the between prescribed dietary phosphorus restriction and mortality in a post hoc analysis of the Hemodialysis (HEMO) study, which included 1751 MHD patients. The study exposure was ascertained by the serum phosphate targets that the dietitians from the clinical dialysis centers settled annually to prescribe their dietary recommendations. A more restrictive prescribed dietary phosphate was associated with poorer indices of nutritional status on baseline analyses and a persistently greater need for nutritional supplementation but not longitudinal changes in caloric or protein intake. There was a stepwise trend toward greater survival with more liberal phosphate prescription, which reached statistical significance among subjects prescribed 1001 to 2000 mg/d and those with no specified phosphate restriction: hazard ratios (95% CIs) 0.73 (0.54 to 0.97) and 0.71 (0.55 to 0.92), respectively.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation phosphorus amount is based on Grade II /Grade B evidence. The remaining recommendations are based on consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of dietary phosphorus intake on mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of 24-hour urinary phosphorus/phosphate excretion on mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of 24-hour urinary phosphorus/phosphate excretion on cardiovascular events in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of urinary phosphorus/phosphate excretion per creatinine clearance on CKD progression in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of dietary phosphorus/phosphate restriction on CKD progression in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of 24-hour urinary phosphate excretion on CKD progression in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of dietary phosphorus/phosphate restriction on serum phosphorus and urinary phosphorus /phosphate excretion in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of dietary phosphorus/phosphate restriction on dietary phosphorus/phosphate intake in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Electrolytes: Sodium (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Sodium Intake and Blood Pressure, CKD 3-5, Non-Dialyzed

In adults with CKD 3-5 not on dialysis, we recommend limiting sodium intake to less than 100 mmol/day (or <2.3 g/day) to reduce blood pressure and improve volume control (1B).

Rating: Strong
Imperative

CKD: Sodium Intake and Blood Pressure, CKD 5D and Post-Transplant

In adults with CKD 5D or posttransplantation, we recommend limiting sodium intake to less than 100 mmol/day (or <2.3 g/day) to reduce blood pressure and improve volume control (1C).

Rating: Fair
Imperative

CKD: Sodium Intake and Proteinuria

In adults with CKD 3-5, we suggest limiting sodium intake to less than 100 mmol/day (or <2.3 g/day) to reduce proteinuria synergistically with available pharmacological intervention (2A).

Rating: Fair
Conditional

CKD: Sodium Intake and Dry Body Weight

In adults with CKD 3-5D, we suggest reduced dietary sodium intake as an adjunctive lifestyle modification strategy to achieve better volume control and a more desirable body weight (2B).

Rating: Fair
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

A nutrition prescription that is too high or too low in sodium may result in adverse outcomes.

- [Conditions of Application](#)

Implementation considerations

- Achieving a reduced sodium intake in CKD is recommended, however can be particularly challenging to achieve (Meuleman et al 2018). This is a result of the need to navigate a complex interplay between individual food choice and food supply, together with a range of other dietary recommendations that come with CKD. As sodium is consumed largely from processed foods, the World Health Organization (WHO) has initiatives for reducing sodium content in manufactured foods among the top priorities to combat non-communicable diseases (McMahon et al 2012). Consuming a low sodium diet generally requires education and skill development (cooking, label reading) and explicit choice to consume a low sodium diet. Therefore, a concerted and multi-faceted intervention strategy is required to support achieving this intake in clinical practice. This includes targeting individual behavior change for dietary choices, together with a wider public health strategy to reduce availability of sodium in the food supply (McMahon et al 2012).
- The interventions undertaken in clinical trials of sodium reduction have limited applicability when translating into practice. Many trials to date have used sodium supplementation or provided foods to enhance adherence in short-term effectiveness studies (McMahon et al 2012). Investigations of efficacy and behavioral interventions to adopt low sodium intakes in real-life settings are limited in the literature. Of those that exist, the evidence is either short term (< 6 months) or demonstrate that achieving a reduced sodium intake is only apparent whilst receiving active intervention (Meuleman et al 2017). The challenge for the future is to develop an evidence-base to inform successful strategies to support long-term adherence to dietary sodium reduction.
- Issues with sodium intake assessment: Measuring sodium intake and thereby accurately evaluating adherence to recommendations is extremely challenging in practice. Sodium intake can be measured in objective (urine collection over 24 hours or spot sample) and self-report (dietary recall) or a combination of methods. Urinary sodium excretion as a surrogate measure of intake assumes 1) a stable intake reflected in a single 24-hour collection, 2) sodium excretion is a direct reflection of intake. It is this latter assumption which has been recently challenged by Titze and colleagues, who have identified a sodium storage pool in the skin and a wide disparity between sodium intake and excretion day to day (Titze et al 2014). Increasing the number of 24-hour urinary collections may improve the accuracy to partially overcome these concerns, however it is not practical in clinical practice. Self-reported dietary assessment methods are prone to reporting bias and can be time consuming to collect and require technical expertise in the analysis. A panel of methods is therefore recommended, as no one method is ideal to adequately assess adherence (McMahon et al 2012).
- Sodium relative to potassium intake: Recent observational evidence suggests that the ratio of sodium-to-potassium intake may be as important, if not more important than lower sodium intake alone in CKD (He et al 2016). This is the premise of the DASH-Sodium trial, and has demonstrated benefits in the general population, with sodium reduction providing additive benefit in BP reduction to the DASH diet (Juraschek et al 2017). In hypertensive adults, post-hoc analysis of clinical trials indicate sodium-to-potassium ratio may be more effective in lowering BP than lowering sodium or increasing potassium as single interventions (Aaron et al 2013). However, there are unknown safety aspects in CKD, particularly with the risk of hyperkalemia. Investigating the relative benefit of sodium reduction compared to potassium intake is beyond the scope of the current guidelines however warrants further research. Evidence for *Potassium* recommendations is addressed within these guidelines.
- Currently, there is too much uncertainty in the evidence to advice on the effectiveness of sodium restriction based on specific thresholds of proteinuria. However, this intervention appears to be effective over a large range of proteinuria.

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with the implementation of these recommendations.

- [Recommendation Narrative](#)

Sodium is an extracellular cation responsible for fluid homeostasis in the body (Geerling et al 2008). Normovolemia is maintained through the action of the renin-angiotensin aldosterone system (RAAS). This system acts to adjust the quantity of sodium excreted by the body, and thereby ECF volume and arterial BP. Excess sodium intake is excreted in the urine and serum levels are tightly controlled, requiring normal kidney and blood vessel function (Schweda et al 2015). However, this system may be compromised with excessive sodium intake, and/or inadequate excretion, which may occur with chronic kidney disease.

Chronic high sodium intake may impact on a number of physiological functions relating to the vasculature, heart, kidneys and sympathetic nervous system (Kotchen et al 2013). Excessive sodium intake is thought to exert toxic effects on blood vessels through mediating factors such as oxidative stress, inflammation and endothelial dysfunction (Dinh et al 2014). Of particular interest in CKD is the role of sodium reduction in improving the pharmacological effect of antihypertensive medication thereby controlling hypertension.

In the general population, short-term intervention studies show significant reductions in BP (hypertensive subgroup, reductions of 5.8 mmHg systolic BP [SBP] and 2.82 mmHg diastolic BP [DBP]) with 100mmol/d reduction in sodium intake (He, et al. 2013). Indications from a small number of long-term studies (>6 months) suggest a benefit for CV-morbidity and mortality, although the studies were underpowered to adequately examine these outcomes (Adler et al 2014). The following will explore the evidence within CKD.

Overall, the evidence for reducing sodium intake comes from randomized controlled trials of short duration and typically small sample size. As a result, there is a focus on clinical markers such as BP, inflammation, body weight, fluid and proteinuria. There is limited evaluation of hard outcomes, which thereby rely upon observational evidence. In addition, the certainty of evidence for sodium reduction is limited by imprecision and risk of bias, particularly selection, attribution and performance bias.

Five randomised controlled trials (1 parallel (de Brito-Ashurst et al 2013) and 4 cross-over studies (McMahon et al 2013, Slagman et al 2011, Vogt et al 2008, Konishi et al 2001) examined the effects of reduced dietary sodium intake on blood pressure and a range of outcomes in CKD (Stage 2 to 5, non-dialysis). The cross-over studies utilised supplemental sodium (McMahon et al 2013, Slagman et al 2011, Vogt et al 2008) or provided meals (Konishi et al 2011) on the background of a low sodium diet to generate consistent intake in the high (180 mmol to 200mmol/d, with ~100-120mmol/d supplemented) vs low sodium intake group (placebo, total 50 to 0 mmol

sodium per day). The parallel RCT was the longest study duration (six months) conducted in a sample of Bangladeshi immigrants in the UK (n=48) (deBrito-Ashurst et al 2013). Participants were randomised to a tailored intervention including cooking classes modifying traditional cultural recipes together with regular telephone calls with a dietitian. From a baseline sodium intake of approximately 260mmol, the intervention group achieved 138mmol/day (a reduction of over 120mmol), whilst usual care stayed largely stable (to 247 mmol/d). Correspondingly mean SBP reduction of ~7mmHg and 1mmHg increase in the intervention and control group, respectively. Diastolic blood pressure (DBP) reduced by 5mmHg and 2mmHg the intervention and control group, respectively (de Brito-Ashurst et al 2013).

Two more recent studies build upon this evidence base and include a parallel (Meuleman et al 2017) and a cross-over trial (Saran et al 2017). Meuleman and colleagues conducted a 3 month open-label RCT, n= 138 adults with CKD, hypertension, and high urinary sodium excretion (≥ 120 mmol/day). The intervention focused on self-management advice to reduce sodium (goal <100mmol/day) and blood pressure monitoring, or usual care. The intervention led to significant decreases in 24-hour urinary sodium excretion and blood pressure at 3 months, as well as in the secondary outcomes of proteinuria and body weight. However, 3 months after the cessation of the 3 month coaching period, the sodium intake was no longer different between the groups, however clinic BP remained significantly lower at 6 months. In the most recent cross-over trial, Saran et al evaluated the effect of sodium restriction <2g/day vs usual diet for 4 weeks (with a 2 week washout in-between) in Stage3 and 4 CKD. This study improved upon previous cross-over trials as it used dietary counselling, rather than sodium supplementation, to achieve the difference between usual and sodium restricted intakes.

Four trials were conducted in the maintenance dialysis population. One RCT in peritoneal dialysis (PD) (Fine et al 1997), and two RCTs in maintenance hemodialysis (MHD) (Liang et al 2013, Telini et al 2014) and one non-controlled trial in both PD and MHD (Magden et al 2013). In the single study undertaken in MHD, there was no significant reduction in BP (Telini 2014). The difference with this study, compared to all others in dialysis, is that dietary prescription (rather than supplemental sodium) was used to achieve a modest reduction of intake (goal 34 mmol/d lower than usual intake). This compares to the other interventions in maintenance dialysis using sodium supplementation, which achieved a much larger gradient of difference in sodium intake between low and high intake groups (100mmol.d or 2.3g sodium difference).

One RCT was undertaken in patients post kidney transplantation (Keven et al 2006). This was a parallel RCT of a 12-week intervention that included counselling by a dietitian for a target intake of 80-100mmol/day compared with usual care. This trial demonstrated a significant reduction in sodium intake in the intervention group (from 190 ± 75 mmol/d to 106 ± 48 mmol/d) through dietary counselling, with no significant change in the usual care group (191 ± 117 mmol/d to 237 ± 113 mmol/d).

In the vast majority of trials, the target sodium restriction was 80-100 mmol/day (or 2-2.3g/day). However, there was a lack of consensus as to what constitutes a high sodium intake, which was either based on usual intake, or providing supplemental sodium to ensure a consistently high sodium intake, around 200mmol or 4g sodium per day.

Mortality, CKD Progression and Cardiovascular Events

There is insufficient evidence to make a statement on reduced sodium intake and kidney disease progression, mortality and cardiovascular events. The evidence for clinical endpoints is derived from observational studies as there were no RCTs in sodium reduction in CKD that reported CKD progression cardiovascular events, mortality outcomes. This is attributable to the small sample sizes and the longest trial duration only six months (de Brito-Ashurst et al 2013).

The post-hoc analysis of two observational cohort studies showed mixed results investigating the association between sodium intake (measured by dietary recall) and subsequent mortality in hemodialysis (McMahon et al 2015) and peritoneal dialysis (PD) patients (Dong et al 2010). The retrospective cohort study in 303 PD patients in Japan indicated that low sodium intake was significantly associated with higher overall and cardiovascular mortality. However, this study was open to indication bias as sodium intake was also associated with higher lean body mass, younger age and higher BMI. In contrast, in a post-hoc analysis of a prospective cohort of 1770 MHD patients, McCausland et al found higher dietary sodium intake associated with increased mortality (Mc Causland et al 2012).

More consistent results were demonstrated from a large high-quality prospective cohort (CRIC study) of predialysis Stage 2-4, using urinary sodium excretion. In He et al, 24-hour urinary sodium excretion was associated with greater all-cause mortality and CKD progression (defined as incident ESRD or halving of eGFR from baseline) (He et al 2016). Sodium excretion was also associated with composite CVD (heart failure, myocardial infarction, stroke) (Mills et al 2016).

Blood Pressure

Overall, sodium reduction probably reduces BP in kidney disease (moderate certainty evidence). This evidence review included 9 small (n=20 to n=52) randomized clinical trials (6 were cross-over trials) of short duration (1 week to 6 months), evaluating the effect on reducing sodium intake (typically to a level of <2g or 90mmol/d) on BP. In fact, lower sodium intake significantly decreased systolic BP in all but one study (Rodrigues et al 2014), which reduced intake by only 34mmol/d, compared with >90 mmol/d from the other trials. However, the certainty of evidence was limited by risk of bias, particularly risk of selection, attribution and performance bias. When evaluating the evidence across stages of CKD, the vast amount of evidence exists in pre-dialysis CKD, however the BP benefits were also apparent in trials in dialysis (Fine et al 1997, Rodrigues et al 2014, Magden et al 2013, Koomans et al 1985) and transplantation populations (Keven et al 2006).

Although this review was unable to derive a summary estimate, a Cochrane review on this topic published in 2015 showed dietary sodium reduction (MD -105.9, 95% CI -119.2 to -92.5mmol/day) resulted in significant reduction in systolic BP (MD -8.76, 95% CI -11.35 to -3.80 mm Hg). These short-term studies showing clinically meaningful systolic BP reductions ranging from 2-12mmHg systolic BP and 1-8mmHg diastolic BP in trials one week to six months in duration (McMahon et al 2015).

Inflammatory Markers

Sodium reduction may make little to no difference to inflammation (low certainty evidence). Two RCTs, a parallel RCT in MHD (Rodrigues et al 2014), and a crossover in Stage 3 and 4 (McMahon et al 2013), investigated the impact of sodium restriction on inflammation, measured by CRP, IL-6, TNF-alpha. In the Telini study there was a significant reduction in all inflammatory markers within the intervention group, however not reported between group differences (and no difference within control group) (Rodrigues et al 2014). The single crossover study in Stage 3-4 showed no difference in inflammation comparing high and low sodium intake (McMahon et al 2013).

Body Weight and Fluid

Sodium restriction may slightly reduce body weight and total body fluid in non-dialysis CKD (low certainty evidence). However, it is uncertain whether sodium restriction reduces body weight and body water in dialysis. The evidence from non-dialysis CKD comes from two randomized-cross-over trials, one using sodium supplementation to compare intake of 60-80mmol/d to 180-200mmol/d for 2 weeks (McMahon et al 2013) together with a more recent investigation by Saran et al evaluating the effect of sodium restriction <2g/day vs usual diet for 4 weeks (with a 2 week washout in-between). Both trials demonstrated a reduction in extracellular volume. Furthermore in maintenance dialysis, two RCTs demonstrated no significant difference in body weight with salt restriction in peritoneal (Fine et al 1997) or both hemodialysis and peritoneal dialysis. In one non-randomized study in hemodialysis, the group advised to restrict sodium intake (<3 g/day) and fluid (<1 L/d) demonstrated within group decrease in interdialytic fluid gain, but there was no change in the control group, and between group difference was not significant (Liang et al 2013).

Kidney Function (including Proteinuria)

Restriction of sodium intake may slightly reduce kidney function markers of creatinine clearance (Konishi et al 2001, Slagman et al 2011, Vogt et al 2008, Koomans et al 1985) and eGFR (Campbell et al 2014) demonstrated in short-term cross-over trials in the stage 1-5 non-dialysis population (low certainty evidence). In the single parallel RCT over 6 months of sodium restriction, deBrito-Ashurst found no difference in eGFR (de Brito-Ashurst et al 2013). The inconsistency in results may be due to the short-term cross-over trials demonstrating acute hyper filtration response to low sodium intake, compared with the longer-term parallel trial, reflecting a more clinically stable circumstance.

Restriction of sodium intake may reduce proteinuria as demonstrated in 3 randomized cross-over trials (McMahon et al 2013, Slagman et al 2011, Vogt et al 2008, Campbell et al 2014). This evidence is supported by further parallel RCTs and observational studies. Meuleman et al demonstrated a reduction in proteinuria over 3 months self-management intervention using a sodium intake <100mmol/d (Meuleman et al 2017). In addition, in a post-hoc analyses of clinical trials (REIN I & II) in proteinuric patients with established CKD have demonstrated participants consuming a higher sodium diet was associated with an increased risk of progressing to ESKD compared to a lower sodium diet <100mmol/d (Vegter et al 2012).

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on sodium and proteinuria in non-dialyzed patients is based on Grade I /Grade A evidence, while the remaining recommendations for sodium for non-dialyzed patients are based on Grade II/Grade B evidence. The recommendation for sodium and blood pressure for patients with CKD on MHD and post-transplant was based on Grade III/Grade C evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of sodium intake on blood pressure in adults with CKD, Stages 1-5?](#)

[What is the effect of sodium intake on blood pressure in adults with CKD, post-transplant?](#)

[What is the effect of sodium intake on proteinuria, albuminuria, protein:creatinine \(24 h urine\), and albumin:creatinine \(24 h urine\) levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of sodium intake on fluid status in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of sodium intake on weight in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of sodium intake on inter-dialytic weight gain in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of dietary sodium intake and mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of urinary sodium excretion and mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of urinary sodium excretion on CKD progression in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of urinary sodium excretion on composite CVD events in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of sodium intake on eGFR in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of sodium intake on creatinine clearance in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of sodium intake on urinary sodium excretion in adults with CKD, pre-dialysis?](#)

[What is the effect of sodium intake on urinary sodium excretion in adults with CKD, post-transplant?](#)

[What is the effect of sodium intake on inflammatory marker levels in adults with CKD 1-5D and post-transplant?](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Electrolytes: Potassium (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Dietary Potassium Amount

In adults with CKD 3-5D or posttransplantation, it is reasonable to adjust dietary potassium intake to maintain serum potassium within the normal range (OPINION).

Rating: Consensus
Conditional

CKD: Dietary and Supplemental Potassium Intake for Hyperkalemia or Hypokalemia, CKD Post-Transplant

In adults with CKD posttransplantation with either hyperkalemia or hypokalemia, we suggest that dietary or supplemental potassium intake be based on patient's individual needs and clinician judgement (OPINION).

Rating: Consensus
Conditional

CKD: Dietary and Supplemental Potassium Intake for Hyperkalemia or Hypokalemia, CKD 3-5D

In adults with CKD 3-5D with either hyperkalemia or hypokalemia, we suggest that dietary or supplemental potassium intake be based on a patient's individual needs and clinician judgment (2D).

Rating: Weak
Imperative

- [Risks/Harms of Implementing This Recommendation](#)

Potassium is widely distributed in foods but the main sources are fruits, vegetables, legumes and nuts. As these foods are major sources of fiber, vitamins, minerals and other important nutrients, efforts should be made to avoid restricting dietary potassium. In particular reduced fiber can lead to constipation which could lower potassium excretion. For these issues, consult the guideline statements on fiber and fruit/vegetable intake.

- [Conditions of Application](#)

Special Discussions

Research on this topic is complicated by the fact that potassium handling by the kidney will vary by disease state and CKD stage. In patients with pre-dialysis CKD the acute and chronic effects of dietary potassium loading are not consistently reflected in serum potassium levels due to compensatory mechanisms that are triggered to maintain homeostasis (Sterns et al 1979, Alvestrand et al 1984, Hayes et al 1967). Research and evidence on this area is also limited because of difficulties in obtaining reliable data on dietary potassium intake and absorption.

Potassium binders bind potassium in the gut and prevent hyperkalemia. In theory, these medications could lead to a more liberalized diet in terms of potassium (i.e. fruits and vegetables). However, none of the pivotal trials examining potassium binders evaluated dietary potassium intake, and currently there is no known study that investigates how potassium intake should be modified when taking potassium binders. Since the focus of this guideline was dietary intake, rather than pharmacological treatments, potassium binders were outside the scope of this guideline.

Implementation Considerations

- Potassium is widely distributed in foods but the main sources are fruits, vegetables, legumes and nuts. As these foods are major sources of fiber, vitamins, minerals and other important nutrients, efforts should be made to avoid restricting dietary potassium. In particular reduced fiber can lead to constipation which could lower potassium excretion. For these issues, consult the guideline statements on fiber and fruit/vegetable intake.
- When treating hyperkalemia clinicians are advised to first try and identify contributing factors that can be corrected such as a hypoinsulinemic state or certain medications. This is true in light of the physiological benefits high potassium intake may confer, such as putative antihypertensive effects (Adroque et al 2014). If hyperkalemia cannot be reversed the next step is to identify the most important dietary sources of potassium by interviewing the patient and dietary recalls. Clinicians preferably assisted by a renal dietitian should educate patients with hyperkalemia about fruits, vegetables, and other foods with low potassium content that ideally still contain higher levels of fiber and other micronutrients. Published food composition tables can be helpful in this regard (Cupisti et al 2018). In addition, potassium content in vegetables can be lowered by boiling and reductions in food taste and palatability associated with this strategy can be partially improved with the use of aromatic herbs (Burrowes et al 2006 and 2008).

- [Potential Costs Associated with Application](#)

There are no obvious costs to adjusting dietary potassium intake.

- [Recommendation Narrative](#)

As the main intracellular cation potassium plays a major role mediating cellular electrophysiology, vascular function and BP, and neuromuscular function. High or low serum potassium levels have been associated with muscular weakness, hypertension, ventricular arrhythmias, and death. The influence of dietary potassium consumption on serum potassium content is therefore of great clinical relevance. Because the mechanisms involved in potassium homeostasis and excretion (i.e. adrenergic system, insulin, aldosterone, and urinary clearance) are commonly impaired in patients with CKD and ESRD hyperkalemia is an especially salient concern. Dietary potassium is the focus of these recommendations (potassium binders were outside the scope of the current guideline).

There is a scarcity of studies on this topic and we found no data on how modifying diet can influence serum potassium in patients with CKD. The work group emphasizes that factors other than dietary intake influence serum potassium levels. These include medications, kidney function, hydration status, acid-base status, glycemic control, adrenal function, a catabolic state, or gastrointestinal (GI) problems like vomiting, diarrhea, constipation and bleeding. All these factors should be considered when formulating a strategy to keep the serum potassium within the normal range.

The consequences of dietary potassium intake in patients with CKD are not known. Indeed, no studies were identified that directly examined the relationship between dietary potassium consumption and either serum levels or clinical outcomes. However, several studies used urine potassium excretion or other surrogates for dietary intake to assess the following outcomes. While we acknowledge that urine potassium excretion may not necessarily represent dietary potassium in these patients, the studies showed:

Mortality

Data on the association between dietary and urinary potassium excretion and mortality in adults with CKD were mixed. A study in MHD (stage 5), found that compared to the lowest quartile of dietary potassium intake (879 mg or 22.5 mEq/24hr) as measured by the Block Food Frequency Questionnaire, higher quartiles of intake were associated with a stepwise increase in risk of 5-year mortality (p-trend=0.03) (Noori et al 2010). Another study in pre-dialysis (Stage 2-4) there was no significant association noted between quartiles of urinary potassium excretion and all-cause mortality (He et al 2016). Compared to the highest quartile of urinary potassium excretion (mean 3600 mg or 92.1 mEq/24hr) persons in the three lowest quartiles had higher all-cause mortality (hazard ratio (95% CI) of 1.53 (1.15-2.02), 1.7 (1.25-2.31), 1.71 (1.23-2.38) for quartiles 3, 2, and 1, respectively).. Results remained similar even after using time-updated average urine potassium excretion (Leonberg-Yoo et al 2017).

CKD Progression

Data on the association between urinary potassium excretion and CKD progression in adults with CKD were mixed. In Stage 2-4; pre-dialysis urinary potassium excretion in the highest quartile (≥ 67.1 mmol or 2617 mg/24 h) was significantly associated with CKD progression (defined as incident ESRD or halving of eGFR from baseline) (1.59, 95% CI: 1.25-2.03) compared to levels in the lowest quartile (<39.4 mmol or 1541 mg/24 h) (He et al 2016). In another study in Stage 2-4; pre-dialysis, baseline urinary potassium excretion was not significantly associated with kidney failure (defined as dialysis therapy or transplantation) even when using time-updated average urine potassium (Leonberg-Yoo et al 2017).

Nerve Function

One randomized study examined the effects of dietary potassium restriction on progression of peripheral neuropathy in CKD patients. In 42 patients randomized to either dietary potassium restriction vs. usual diet (change in dietary potassium -854 vs. -343, p=0.35), potassium restriction was associated with stabilization of a neuropathy score (difference 0.4 ± 2.2 , p<0.01) and several other nerve-related or general health scores over 24 months (Arnold et al 2017).

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on individualizing potassium based on needs is based on Grade III /Grade D evidence. The remaining recommendations are based on Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of dietary potassium intake on mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of urinary potassium excretion on mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of urinary potassium excretion on CKD progression in adults with CKD 1-5D and post-transplant?](#)

- [References](#)

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- [Chronic Kidney Disease](#)

Quick Links

Recommendations Summary

CKD: Electrolytes: Acid Load (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Dietary Management of Net Acid Production (NEAP)

In adults with CKD 1-4, we suggest reducing net acid production (NEAP) through increased dietary intake of fruits and vegetables (2C) in order to reduce the rate of decline of residual kidney function.

Rating: Weak
Conditional

CKD: Bicarbonate Supplementation

In adults with CKD 3-5D, we recommend reducing net acid production (NEAP) through increased bicarbonate or a citric acid/sodium citrate solution supplementation (1C) in order to reduce the rate of decline of residual kidney function.

Rating: Fair
Conditional

CKD: Bicarbonate Maintenance

In adults with CKD 3-5D, it is reasonable to maintain serum bicarbonate levels at 24-26 mmol/L (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

In CKD 5 maintenance hemodialysis (MHD) patients, higher bicarbonate in the dialysate bath is associated with increased mortality in epidemiological studies (Yamamoto et al 2015). In an analysis of the DOPPS data, it was reported MHD patients with either very low bicarbonate (≤ 17) or very high predialysis bicarbonate (>27) concentrations are at the greatest mortality risk (Bommer et al 2004).

- [Conditions of Application](#)

Special discussions

In CKD 5 MHD patients, higher bicarbonate in the dialysate bath is associated with increased mortality in epidemiological studies (Yamamoto et al 2015). In an analysis of the DOPPS data, it was reported MHD patients with either very low bicarbonate (≤ 17) or very high predialysis bicarbonate (>27) concentrations are at the greatest mortality risk (Bommer et al 2004). Paradigms that may apply to patients with residual renal function or those undergoing continuous therapy, such as peritoneal dialysis, do not directly apply to hemodialysis patients who are experiencing large changes in acid base equilibrium rapidly and/or discontinuously. Higher bicarbonate concentration in hemodialysis patients may also be reflective of lower protein intake.

Research on this topic is complicated by the fact that the effect of acidosis differs with the level of residual kidney function. With advanced CKD, net acid load has a higher potential to contribute to loss of kidney function. (Grade A evidence)

Dietary intervention is more complex, since the effects of specific amino acids or other dietary constituents on both renal outcomes as well as vascular and bone pathophysiology (Calcium/Phosphorous) may play a role that is independent from their effect on acid base physiology.

Implementation considerations

- Acid load is a consequence of protein load and is inversely associated with potassium intake. The estimation of net acid intake is (NEAP (mEq/d) = $-10.2 + 54.5 (\text{protein [g/d]}/\text{potassium [mEq/d]})$). NEAP can be reduced by administration of sodium bicarbonate or sodium or potassium citrate or by reduction in dietary acid content by changing the dietary pattern to increase fruits and vegetables. The latter can be accomplished by reduction in dietary protein intake and changing its composition. Low protein intake may have the added benefit of slowing the rate of progression of kidney disease through other mechanisms (See Protein Section). In the MDRD study, patients randomized to low protein intake exhibited a significant increase in serum bicarbonate (Gennari et al 2006), so that there is an interaction between intake of protein and net acid. Separating the effect of reduction in acid load and the effect of change in dietary protein amount and composition on outcomes is challenging.
- When increasing fruits and vegetables intake to correct acid load please use caution and monitor potassium levels.

Monitoring and Evaluation

Clinical trials have demonstrated compliance with expected changes in acid base status as evaluated by measurement of serum bicarbonate:

- Consuming fruit and vegetable in the amount that could reduce dietary acid by 50% generally had positive effects on acid-base biomarkers (Goraya et al 2012, 2013, 2014). Fruit and vegetable increased plasma total CO₂ (though not significant in one study) (Goraya et al 2012) and decreased potential renal acid load and 8h NAE. Except for Goraya et al, (0.5 mEq/kg/day) (Goraya et al 2012), oral bicarbonate supplements also had positive effects on acid-base biomarkers by increasing plasma total CO₂ or bicarbonate levels and decreasing potential renal acid load and 8h NAE in six studies with different supplement combinations and dosages.
- No hyperkalemia events were noted in the studies of Goraya et al. who provided a diet rich in fruits and vegetables to patients with advanced CKD. However, we note that inclusion criteria in those studies considered patients at low hyperkalemia risk not

consuming RASi. While no studies have formally evaluated the contribution of dietary potassium to hyperkalemia risk in these patients, we recommend caution if a fruit and vegetable rich diets is to be recommended to control metabolic acidosis. A close monitoring of serum/plasma potassium levels is encouraged, and fruit/vegetable consumption should be temporarily limited if the patient is considered at risk of hyperkalemia. Monitoring of circulating potassium is specially recommended in patients with CKD stage 4 or more, including those on dialysis, as this is the kidney function range where inability to compensate dietary potassium occur.

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with this recommendation.

- [Recommendation Narrative](#)

Acid base homeostasis is maintained by urinary acidification using titratable anions, such as phosphate, to trap proteins, and trapping ammonia that is generated as ammonium in an acid urine. As kidney function declines, the net acidification requirement by residual nephrons increases. This is accomplished by increasing ammonia production per residual nephron and requires delivery of glutamine to the residual nephrons. The increased per nephron need for increased acidification and ammonia genesis is in part endothelin controlled and may increase injury to residual nephrons. Acid retention also would have the potential to promote muscle wasting as part of the homeostatic processes of normalizing acid base status. Metabolic acidosis increases skeletal muscle proteolysis by a ubiquitin proteasome pathway that degrades actin potentially having adverse nutritional impact on the patient accompanied by an increase in protein catabolic rate.

Eleven studies examined the association between dietary acid load/oral bicarbonate supplements on health outcomes in the CKD population. Of the included studies, there were four RCTs (Goraya et al 2013, 2014, de Brito-Ashurst et al 2009; Szeto et al 2003), one NRCT (Goraya et al 2012), three non-controlled studies (Kooman et al 1997, Movilli et al 1998, Verove et al 2002), two prospective cohort studies (Banerjee et al 2015, Scialla et al 2012), and one retrospective cohort study (Kanda et al 2014).

CKD Progression: Effect of Reducing Net Acid Production

Studies aimed at evaluating the effect of reduction in net acid production (NEAP) have been two fold; either directly reducing NEAP by administration of sodium bicarbonate, or by dietary alteration using fruits and vegetables, which both decrease NEAP and alter the composition and quantity of dietary protein partially confounding the effect of reduction of NEAP alone. In adults with CKD, four RCTs (Goraya et al 2013, 2014, de Brito-Ashurst et al 2009; Szeto et al 2003), one non-RCT (Goraya et al 2012), two non-controlled studies (Movilli et al 1998, Verove et al 2002), two prospective cohort studies (Banerjee et al 2015, Scialla et al 2012), and one retrospective cohort study (Kanda et al 2014). examined the effects of dietary fruit and vegetable or oral bicarbonate supplements on CKD progression. In patients with CKD stages 2-4 (20-65 mL/min per 1.73 m²) higher quartiles of net endogenous acid production (NEAP) were associated with greater 125iothalamate glomerular filtration rate (iGFR) decline (p-trend=0.02) (Scialla et al 2012). In CKD 3-5 (≤60 mL/min per 1.73 m²) higher NEAP is associated with CKD progression (p<0.05 for all quartile groups) (Kanda et al 2014). In CKD stages 3-4 (≤60 mL/min per 1.73 m²) compared to lowest dietary acid load tertile, highest dietary acid load had greater relative hazard of ESRD (p=0.05) (Banerjee et al 2015).

Studies reducing NEAP by the use of administration of oral sodium bicarbonate are not confounded by alteration in dietary protein composition and easier to study in randomized controlled prospective manners. In studies involving patients with CKD 4-5, the oral sodium bicarbonate group had significant greater creatinine clearance after 18 and 24 months (p<0.05). Rapid CKD progression (creatinine clearance loss of >3ml/min per 1.73m²/yr) was lower in the oral sodium bicarbonate group (RR: 0.15; 95% CI: 0.06-0.40). Development of ESRD was lower in the oral sodium bicarbonate group (RR: 0.13; 95% CI: 0.04-0.40) (de Brito-Ashurst et al 2009). In another study of CKD 4-5; not on dialysis, there was no significant difference in creatinine clearance between before and after intervention (p>0.05) (Verove et al 2002). In patients with less impaired renal function at baseline (CKD 3), there was a reduction in eGFR in all groups, however, at 3 years, lesser reduction in eGFR was observed with HCO₃ group or fruits and vegetables than usual care group (Goraya et al 2014).

In a study by Goraya et al, 2013 in patients with CKD 4 using either fruits and vegetables or NaHCO₃ as the intervention, plasma creatinine levels were comparable between the subjects treated either with HCO₃ or fruits and vegetables at baseline and 1-year follow-up (p=0.99, 0.49, respectively), eGFR were comparable between the two groups at baseline and 1-year follow-up (p=0.84, 0.32, respectively) (Goraya et al 2013). This study does not isolate the effects of alteration in dietary composition and NEAP sufficiently to establish which intervention is associated with any biological change observed.

The outcome of studies in patients with CKD1-2 are less clear and the outcomes not as definitive. This may in part be due to the fact that the per nephron stress of maintaining acid/base balance is reduced, either decreasing the renal risk of acidification below a critical threshold, or by reducing the power necessary to measure an effect. Additionally studies that alter NEAP by changing dietary composition are confounded by other variables, such as amino acid load and quality. One of the outcome variables measured was urinary albumin excretion.

Net urine albumin excretion was not different among the three groups in CKD 1 patients (p>0.05). However, in CKD 2 patients, fruits and vegetables had greater decrease in net urine albumin excretion than both HCO₃ and control (p<0.05) and HCO₃ group had greater decrease in net urine albumin excretion than control (p<0.05) (Goraya et al 2012). It should be noted that a change in diet towards higher intake of fruit and vegetables is a different and more complex intervention than change in NEAP since the amino acid load and composition is changed. This may affect urinary protein loss and have an effect on progression that is independent of NEAP if the patient population has significant proteinuria.

Hospitalization

The effects of oral bicarbonate supplements on hospitalization in CKD patients were mixed, though evidence is limited. In adults with chronic kidney disease, two RCTs (de Brito-Ashurst et al 2009, Szeto et al 2003) examined the effects of oral bicarbonate supplements on hospitalization. Among CKD Stage 5 patient; peritoneal dialysis, compared with placebo group, intervention group had lower hospital admission (trend) and hospital length of stay (p=0.07 and 0.02, respectively) (Szeto et al 2013). In CKD Stages 4-5; pre-dialysis, there was no significance difference in hospitalization for heart failure between the two groups (p=N/A) (de Brito-Ashurst et al 2009).

Nutritional Status

In CKD 3-5 patients including ones on maintenance dialysis, oral bicarbonate supplements improved nutritional status (e.g., SGA scores, nPCR, albumin, and prealbumin) in most studies. Oral bicarbonate supplements increased overall SGA scores (Szeto et al 2003-2.7 g/day) and lowered nPNA (nPCR) (de Brito-Ashurst ~1800 mg/day) (de Brito-Ashurst). Except for Kooman et al, (dialysate bicarbonate and oral sodium bicarbonate (1500-3000 mg) if pre-dialytic bicarbonate did not reach desired level), the other three studies observed positive effects of oral bicarbonate supplements on serum albumin or prealbumin levels (de Brito-Ashurst ~1800 mg/day; Movilli et al, - mean dose 2.7±0.94 g/day; 1.4 g/day; Verove et al, 2002 - mean dose 4.5±1.5g/d. Oral bicarbonate supplements also had no effects on TSF measurements (Kooman et al 1997). de Brito-Ashurst et al, (~1800 mg/day) noted significant increases in mid arm muscle circumference (MAMC) measurements with oral sodium bicarbonate, while Kooman et al, 1997 did not.

Two RCTs (de Brito-Ashurst et al 2009, Szeto et al 2003) and three non-controlled studies (Kooman et al 1997, Movilli et al 1998, Verove et al 2002) examined the effects of oral bicarbonate supplements on nutritional status in adults with CKD. In CKD Stage 5; peritoneal

dialysis, the oral bicarbonate group had higher overall SGA scores starting at 24 weeks (p-value <0.0003) (Szeto et al 2013). In CKD 4-5; pre-dialysis, the oral sodium bicarbonate group had significant lower nPNA (nPCR) at 12 and 24 months (p<0.05) and the oral sodium bicarbonate group had significant higher serum albumin at 12 and 24 months (p<0.05) (de Brito-Ashurst et al 2009).

In contrast, in a group of CKD 5; hemodialysis patients, there was no significant difference in serum albumin among time points (p>0.05) (Kooman et al 1997). In CKD 5; hemodialysis, oral sodium bicarbonate increased serum albumin level (p=0.01) (Movilli et al 1998).

It should also be noted that high dialysate bicarbonate concentrations are associated with increased mortality in observational studies (Yamamoto et al 2015). Among CKD 4-5; pre-dialysis, oral sodium bicarbonate increased both serum albumin and prealbumin levels between before and after intervention (p<0.05) (Verove et al 2002). Among CKD 1-2 compared to control and HCO₃, fruit and vegetable group had significantly greater decrease in body weight at the end of the intervention for both individuals with CKD 1-2 (p<0.05 for both). No difference between HCO₃ and control (Goraya et al 2012). Thus there does not appear to be a significant effect of reduction in NEAP on nutritional status in patients with CKD 1-2. In CKD 4 compared to HCO₃ group, FV group had lower weight at 1-year follow up (p<0.01) but baseline weight did not differ between the two groups (p=0.24) (Goraya et al 2013). In CKD 3, fruits and vegetables had greater net body weight loss than both HCO₃ and control (p<0.05) and control group had greater net body weight loss than HCO₃ group (p<0.05) (Goraya et al 2014).

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendations on dietary management of NEAP with fruits and vegetables and bicarbonate supplementation is based on Grade III /Grade C evidence. The recommendation regarding bicarbonate maintenance based on Consensus/expert opinion.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of acid-base interventions on hospitalizations and mortality in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on blood pressure in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions or NEAP/acid load levels on CKD progression in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on acid-base levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on fluid status in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on body weight/BMI in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on MAMC/TSF in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on CRP levels in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on nutritional status indicators in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on nPNA in adults with CKD 1-5D and post-transplant?](#)

[What is the effect of acid-base interventions on caloric and protein intakes in adults with CKD 1-5D and post-transplant?](#)

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- [References not graded in Academy of Nutrition and Dietetics Evidence Analysis Process](#)

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- [Chronic Kidney Disease](#)
- [EAL-KDOQI \(CKD\) Guideline \(2020\)](#)

Quick Links

Recommendations Summary

CKD: Electrolytes: Calcium (2020)

[Click here](#) to see the explanation of recommendation ratings (Strong, Fair, Weak, Consensus, Insufficient Evidence) and labels (Imperative or Conditional). To see more detail on the evidence from which the following recommendations were drawn, use the hyperlinks in the [Supporting Evidence Section](#) below.

- [Recommendation\(s\)](#)

CKD: Total Calcium Intake, CKD 3-4

In adults with CKD 3-4 not taking active Vitamin D analogs, we suggest that a total elemental calcium intake of 800-1,000 mg/d (including dietary calcium, calcium supplementation and calcium-based phosphate binders) be prescribed to maintain a neutral calcium balance (2B).

Rating: Fair
Conditional

CKD: Total Calcium Intake, CKD 5D

In adults with CKD 5D, it is reasonable to adjust calcium intake (dietary calcium, calcium supplements or calcium-based binders) with consideration of concurrent use of vitamin D analogs and calcimimetics in order to avoid hypercalcemia or calcium overload (OPINION).

Rating: Consensus
Conditional

- [Risks/Harms of Implementing This Recommendation](#)

There are no obvious risks or harms associated with this recommendation.

- [Conditions of Application](#)

Special Discussions

In maintenance dialysis patients, calcium balance is more complex. In addition to dietary calcium load and use of vitamin D analogs, calcium concentration in the dialysate and mode of dialysis also determine the mass balance of calcium. Studies using a mathematical modeling have shown a positive calcium balance mass in patients on MHD (Bushinsky et al 2010, Gotch et al 2010). According to estimates and assumptions made, extracellular fluid calcium increased with an elemental daily calcium intake >1.5 g and was numerically more positive when patients are given active vitamin D analogs (Bushinsky et al 2010). The excess of extracellular calcium is deposited in either osseous or extraosseous sites. The extensive soft tissue calcification highly prevalent in MHD patients suggests that extraosseous sites seem to be the repository for this calcium (London et al 2003).

Although calcium balance studies are demanding, they are essential to provide data to make conclusive recommendation for calcium intake from diet or supplements for patients on maintenance dialysis. In KDIGO (CKD-MBD) 2009 and 2017 there is no recommendation regarding calcium intake for patients on maintenance dialysis or with kidney transplantation.

Implementation Considerations

Hypercalcemia is relatively common in patients on maintenance dialysis. Evidence has been accumulated linking higher serum calcium concentrations to increased nonfatal cardiovascular events (Gallieni et al 2012) and mortality (Coen et al 2010, Floege et al 2011, Fukagawa et al 2014, Markaki et al 2012). In the event of hypercalcemia the following adjustments are recommended (K/DoQI et al 2003):

- In patients taking calcium-based phosphate binders the dose should be reduced or therapy switched to a non-calcium phosphate binding.
- In patients taking active vitamin D analogs the dose should be reduced or therapy discontinued until serum concentration of calcium return to normal.
- If hypercalcemia persists, consider using low dialysate calcium (1.5 to 2.0 mEq/L). This should be done with caution because observational studies have linked this approach with increased risk for arrhythmia and heart failure (Pun et al 2013, Brunelli et al 2015).

- [Potential Costs Associated with Application](#)

There are no obvious costs associated with modifying dietary calcium intake. The cost of nutrition supplements should be considered before recommending to a patient.

- [Recommendation Narrative](#)

Calcium is a multivalent cation important for many biologic and cellular functions. Approximately 99% of total body calcium is found in the skeleton and the remainder is present in the extracellular and intracellular spaces. In addition to its role in maintenance of bone health, calcium serves a vital role in nerve impulse transmission, muscular contraction, blood coagulation, hormone secretion, and intercellular adhesion.

Calcium balance is tightly regulated by the concerted action of calcium absorption in the intestine, reabsorption in the kidney, and exchange from bone, which are all under the control of calciotropic hormones triggered by demand for calcium.

Serum calcium concentrations are maintained in the normal range until very late in CKD (CKD) when it decreases slightly (Levin et al 2007). However, calcium balance in CKD is poorly understood. Calcium deficiency due to decreased intestinal calcium absorption is a stimulus for the development of secondary hyperparathyroidism and resultant bone disorders. On the other hand, calcium excess may promote extra-osseous calcification contributing to increasing the risk of cardiovascular disease and mortality of these patients (Sigrist et al 2007). In kidney transplant, calcium balance is complex and depends on several factors such as the post-transplant renal function, the persistence of hyperparathyroidism, the previous bone disease and the immunosuppressive therapy (Hirukawa et al 2015).

Serum calcium levels do not reflect the overall body calcium balance and may not be very informative except at extremes. The maintenance of serum calcium in the normal range in CKD depends on several factors such as bone turnover, mineral regulating hormones, degree of kidney function, use of vitamin D analogues, dialysate calcium concentration and calcium intake especially from supplements. A careful medical and nutritional history may provide some insight into the adequacy of calcium intake. However, due to the multifactorial causes of altered calcium metabolism in CKD, the establishment of adequate amount of dietary calcium is challenging and depends on the investigation of calcium balance.

The evidence review included three small short-term clinical trials in pre-dialysis CKD patients that investigated the effect of calcium intake in food or supplements on mineral bone biomarkers and on calcium balance (Martinez et al 1997, Spiegel et al 2012, Hill et al 2013). No other outcomes were investigated in these studies.

Calcium Balance and other Lab Measures

In an NRCT, 51 patients in the early stage of CKD (creatinine clearance: 66 to 82 mL/min) were placed in a low protein (40g/d) and low phosphorus (600 mg/d) diet supplemented with or without 0.5 g/d of elemental calcium for 10 days (Martinez et al 1997). A decrease in intact parathyroid hormone (iPTH) was observed only in the group receiving calcium supplementation and no changes in serum calcium, phosphorus and calcitriol were found in the other groups.

In a crossover study, six patients with CKD stages 3 and 4 consumed controlled high (2,000 mg/d) and low calcium diets (800 mg/d) for 9 days (Spiegel et al 2012). Calcium balance was slightly negative to neutral in both patients and healthy controls on the low calcium diet (-91 ± 113 and -144 ± 174 mg/d respectively, $p > 0.05$) and more positive in patients than in controls on the high calcium diet (759 ± 120 and 464 ± 225 mg/d respectively, $p < 0.05$). Serum calcium and phosphate concentrations were unchanged and iPTH and 1,25-dihydroxyvitamin D decreased in the high calcium diet.

In a 3-week randomized cross-over balance study, eight patients with CKD stages 3 and 4 were randomized to a controlled calcium intake of 2457 mg/day (1,500 mg of elemental calcium from calcium carbonate used as phosphate binder + 957 mg/day of dietary calcium) versus placebo (957 mg/day of dietary calcium) (Hill et al 2013). The calcium balance was neutral in the placebo and positive in the calcium carbonate groups (508 vs. 61 mg/d, respectively, $p = 0.002$). Serum calcium, phosphate and iPTH concentrations were unchanged in both groups.

Despite the small number of patients investigated, these well performed balance studies showed that a dietary calcium intake of approximately 800 to 1,000 mg/d may be adequate to maintain calcium balance in patients CKD stages 3 and 4 who are not receiving active vitamin D analogs, at least at short term (DRI et al 2011). These values are close to the current estimated average requirement (EAR-800-1000 mg/d) and the recommended dietary allowance (RDA- 1000-1200 mg/d) for healthy individuals proposed by the Institute of Medicine.

- [Recommendation Strength Rationale](#)

The evidence supporting the recommendation on calcium is based on Grade II/Grade B evidence.

- [Minority Opinions](#)

Consensus reached.

- [Supporting Evidence](#)

The recommendations were created from the evidence analysis on the following questions. To see detail of the evidence analysis, click the blue hyperlinks below (recommendations rated consensus will not have supporting evidence linked).

[What is the effect of calcium intake on electrolyte levels in adults with CKD 1-5D and post-transplant?](#)

- [References](#)

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